Battery Manager

Manual

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1 Welcome to Battery Manager

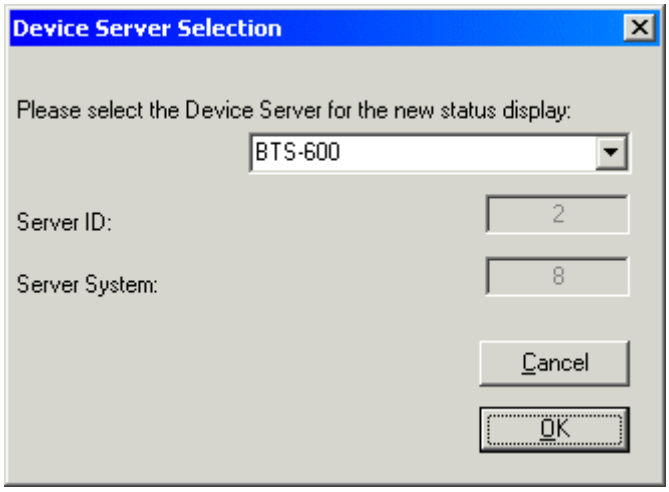
Battery Manager is a database-oriented system for controlling and monitoring test-, formation- and production processes in connection with various controllers.

Battery Manager contains all features found in existing Digatron / Firing Circuits Battery Testing and Formation systems under one common user interface. Thereby Battery Manager connects all Digatron/Firing Circuits device types to one database system.

Battery Manager integrates itself into Windows networks and SQL database arrangement. Inside a Windows network Battery Manager can be operated from every workstation. Remote access via Internet is also possible.

1.1 To start Battery Manager

Prior to starting Battery Manager, the servers have to be activated. Then the actual Battery Manager program can be started on the Workstation. Depending on the system's configuration (number of PCs, number and location of Communication Servers and Workstations) the start sequence may vary. The Battery Manager system will provide a service to deal with this automatically.



1.2 Establishing the Data Communication

Battery Manager consists basically of four software components:

- BM Workstation
- BM Server Manager
- BM Communication Server
- BM Database and Database Server

BM Workstation

The Workstation is the actual user interface for controlling circuits and graphical evaluation of measuring data. The battery-related program maintenance and other functions are started here, too.

BM Communication Server

The Communication Server is an independent application that works as communication interface towards the physical circuits. In accordance with various interfaces the Communication Server transports data to and from the respective circuit. One system can have several Communication Servers. Each server provides one Partyline. The number of circuits on a Partyline depends on the existing installation.

BM Server Manager

The Server Manager coordinates and prepares the data between the Communication Servers and the Database Servers.

BM Database Server

The Database Server links the system to the database and coordinates all access to the database.

1.3 Database Architecture

Battery Manager can be linked to either Microsoft Access or for large scale installations to Microsoft SQL Server database. This manual does not describe the handling of databases. It is assumed that the BM system administrator knows the basics of operating and maintaining the selected database.

Circuit Database

The Circuit database contains tables that include information about:

- all available circuits (physically available circuits)
- Communication Server information
- program information
- batteries
- the name of the session

User Database

The User database stores all information about the Users and groups of users (departments). This database is always password protected. The password is shipped with the initial system as part of the license agreement.

1.4 Database Maintenance

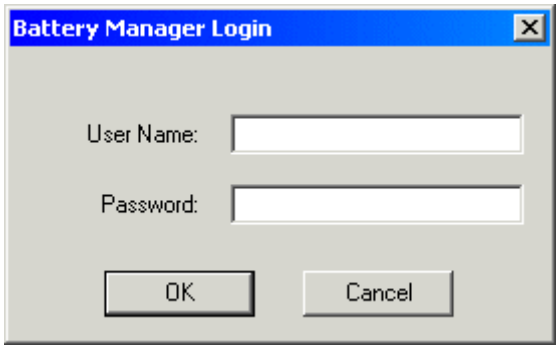
Some basic database maintenance is required for the BM system to run at optimum performance. While adding and deleting permanent records to the database, the database performance will decrease over time. To compact the used databases periodically all connections to the database should be disconnected and then the database should be compacted.

If data registration is used to store all registration data that occurred during the program, the database will grow quickly. To maintain a reasonably sized database, older data should be deleted periodically.

To back up the system, regular database backups are necessary. These backups then contain all information needed to restore the system in case of e.g. a Hard Drive crash. A weekly backup is a recommended method of long term data archiving. Battery Manager therefore delivers several automatic scripts that will be installed if required.

1.5 Logging in

Press F5 or choose **Log In** under the **Control** item on the main menu to invoke the Login dialog. Enter user name and corresponding password in the places provided. Press **OK**.



When first logging in, these standard entries are available:

User name: digatron

Password: bm

1.6 Logging out

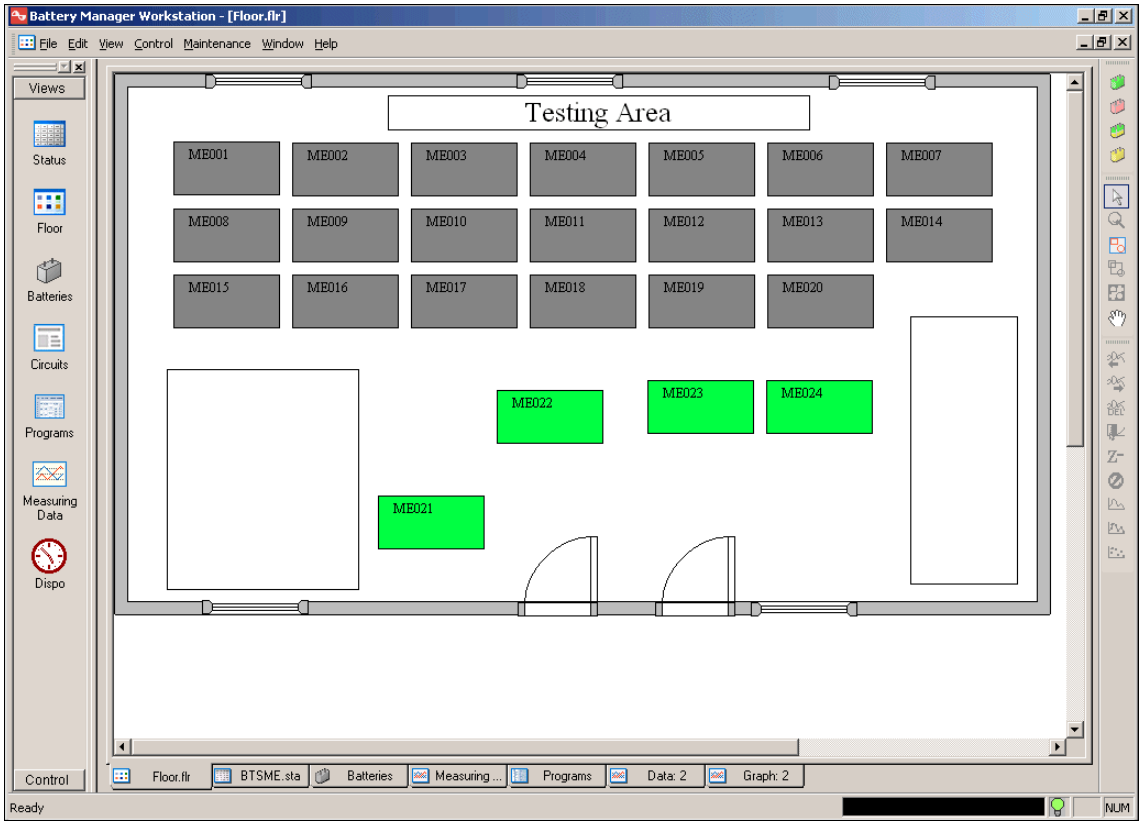
Press F4 or choose **Log Out** under the **Control** item on the main menu to log out. Prior user access is immediately lost.

2 Floor Display

The Floor Display gives the user a broad view of a large number of circuits. The Floor Display consists of multiple color blocks and, optionally, graphic elements. Each block represents a given circuit. The circuit name is in the upper left corner of the block. The group name is in the lower right corner of the block. The color of the block represents the status of the circuit. This allows the user to determine the status of many circuits with a quick scan.

Various graphic elements, including walls, doors, boxes and windows, can be added to the Floor Display layout to enhance realism. Text blocks can also be added to provide orientation information. The user determines the size of the blocks, their positions, and adds structural elements during the Floor Display design process.

To open the floor view click **View|FloorView** or click the **Floor** entry from the Shortcut Bar. Multiple Floor views can be opened at the same time.



Moving within the Floor Display

The Floor Display shows one portion of the entire floor. The user can change the viewed area by using the page-up and page-down keys or the scroll bars. This will alter the currently displayed set of circuits.

2.1 Zooming In and Out

The Floor Display can be zoomed to adjust the number of displayed circuits. There are several ways to zoom:

Use the Floor Display Zoom/Pan toolbar:



 Sets the cursor to normal point and click mode.

 Each click of the right mouse key zooms out and increases the number of circuits displayed. Pressing the left mouse key and dragging the pointer allows you to box the circuits that you want to zoom in and decreases the number of circuits displayed.

 Zooms out so that all circuits are displayed.

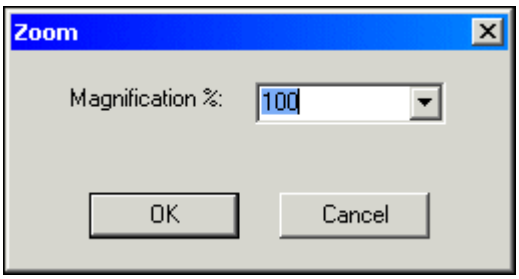
 Sets the magnification to 100%.

 Zooms into the selected items.

 Allows you to grip the display and drag it up/down and left/right.

Use the **View|Floor View** functions:

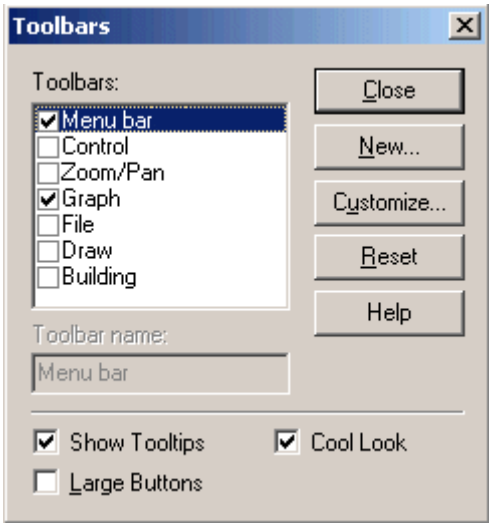
- Zoom – Enables cursor/mouse base zoom.
- Zoom Normal – Sets the magnification to 100%.
- Zoom Percent – Provides pre-selected magnifications of 50% and 75% (zoom out), 100% (zoom normal) and 200% (zoom in).
- Zoom Custom – Allows you to specify the magnification percent.



- Zoom to Fit – Zooms out so that all circuits are displayed.
- Zoom to Selection – Zooms the selected element(s).

2.2 Adjust Toolbars

If a toolbar is not visible you can activate it by selecting the entry **Toolbars** from the **View** menu:



Click an entry from the list to make the corresponding toolbar visible. You can also hide toolbars by deactivating entries.

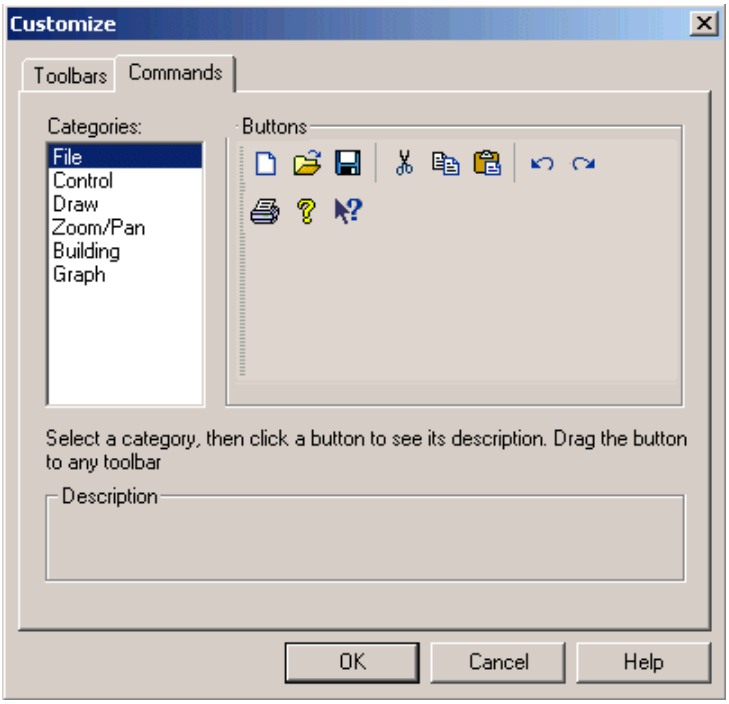
The option **Show Tooltips** determines if a short description appears in a yellow popup window on a screen when you point on a toolbar button.

The deactivation of the option **Cool Look** makes the toolbars appear more old-fashioned.

The option **Large Buttons** allows to enlarge the size of all toolbar buttons for better differentiation.

Use the **New** button for creating your own toolbars. Self-created toolbars can be deleted by highlighting and then clicking the **Delete** button.

The button **Adjust** makes it possible to pick toolbar buttons from the **Commands** tab to your self-created toolbar via drag and drop:

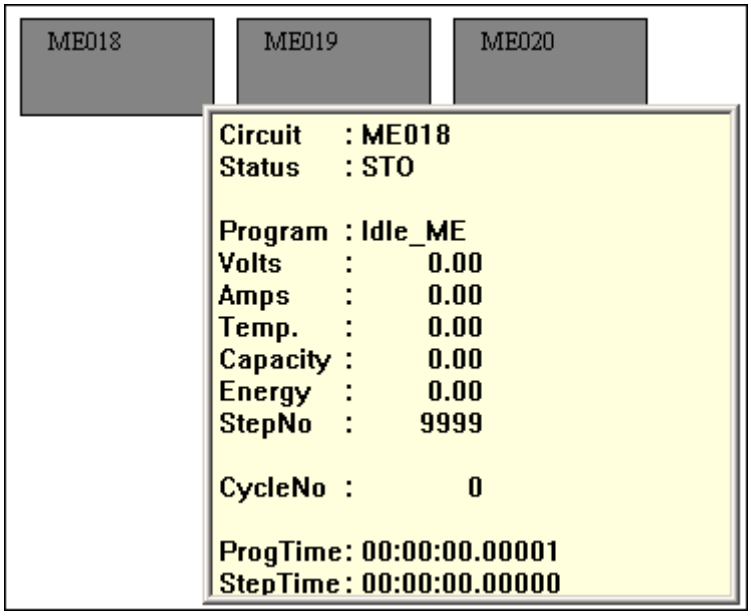


When clicking a symbol here you get a small description.

When the dialog **Adjust** is open you can remove symbols from all toolbars by pulling them away with the left mouse button pressed. The button **Reset** resets all toolbars to their initial state.

2.3 QuickView

Placing the mouse pointer over a circuit box will activate the QuickView status display. Limited status information, such as the circuit name, circuit status, battery type, etc. will be displayed in a small window. This information will be automatically updated.



Move the mouse pointer away from a circuit box and the QuickView window will disappear.

Enabling and Disabling QuickView

The QuickView feature can be disabled if desired. Click **View|Properties** and uncheck the **QuickView** option to disable.

2.4 Controlling Circuits

It is possible to control one or more circuits from the Floor Display.

Select the circuits that you wish to control:

- to select one circuit, click on it with the left mouse button
- to add to the selection set, hold down the **<Ctrl>** key and click on the additional circuits with the left mouse button
- to select multiple circuits that are near each other on the floor display layout, press the left mouse key and drag the pointer to outline the circuits that you want to control
- grab handles will appear identifying the selection set

Control the circuits in one of the following ways:

- position the mouse pointer inside one of the selected circuits, click the right mouse button and select the desired control function from the menu
- by using the associated control function key, such as F9 to start
- select the control function on the **Control** toolbar or from the menu **Control**

Elements on the toolbar **Control**:

 **Start**

 **Stop**

 **Continue**

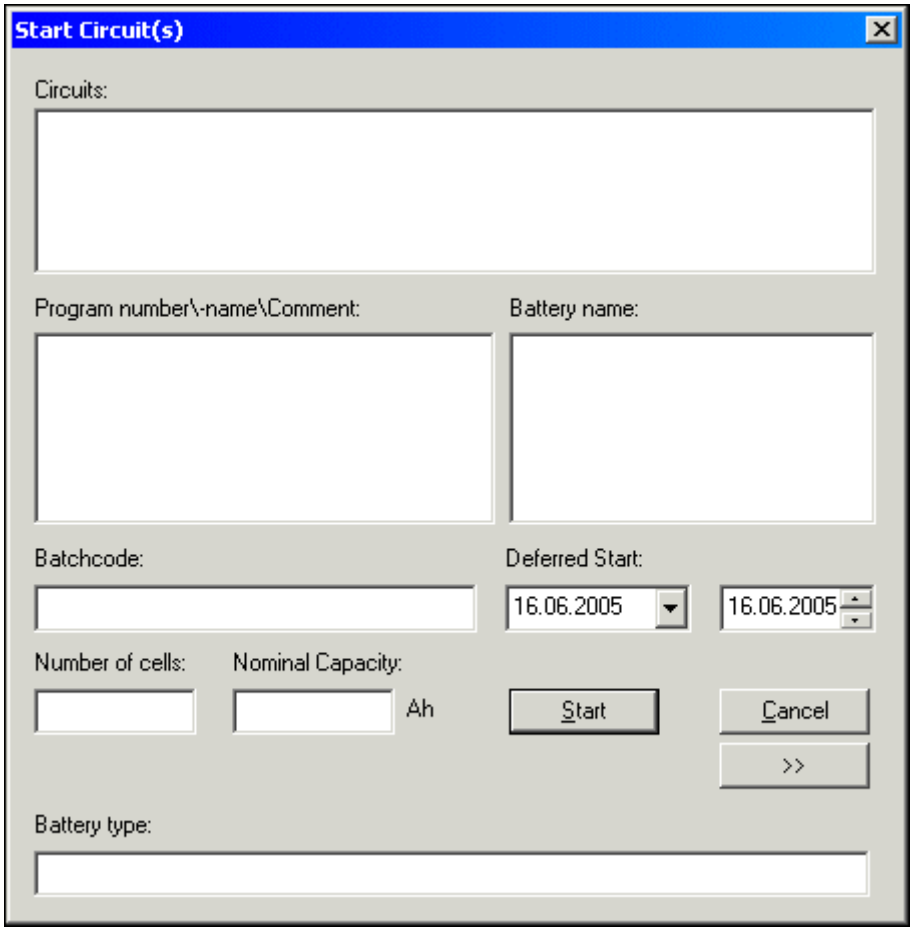
 **Interrupt**

2.5 Start Circuits

To start a circuit, highlight it in the Floor View and then click **Start** from the **Control** menu or click the symbol  from the **Control** toolbar. In the status view you may as well right-click an entry in the column **Circuit Name** and select the desired function from the appearing menu.

Depending on the installation there appears a dialog for the selection of various start parameters. When starting a **BTS-600** circuit after clicking **Start** the Dispo list appears instantly.

With the installation of a formation system the following dialog appears:



Perform the following steps here:

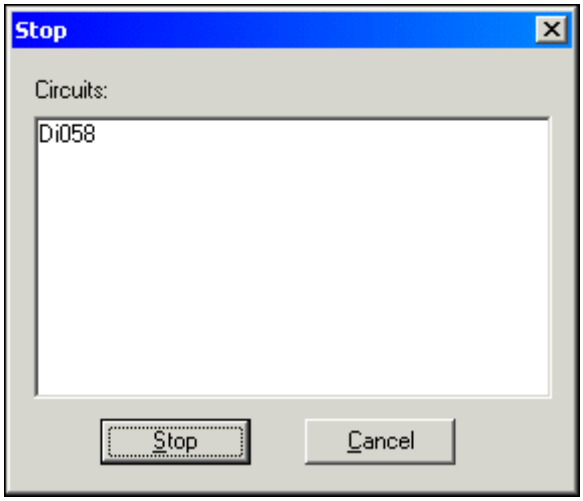
Click a program from the list. When starting a circuit, a deferred start is also possible. Thereby you can determine date and time of a circuit that is about to be started. If you want to run a circuit immediately, leave date and time fields unchanged. If you enter an order number/Batchcode, it will be displayed in the measuring data overview later.

Click the button **Start** to perform the process you have chosen or click **Cancel** to abort. Afterwards you can recognize the new state of the circuit by its new color in the **Circuit Overview**.

2.6 Stop Circuits

To stop a circuit, highlight it in the Floor View and then click **Stop** from the **Control** menu or click the symbol  from the **Control** toolbar. In the Status view you may as well right-click an entry in the column **Circuit Name** and select the desired function from the appearing menu.

A box appears where you can pick the circuits to terminate. To highlight several circuits, keep the **Ctrl** key pressed while clicking on circuits.

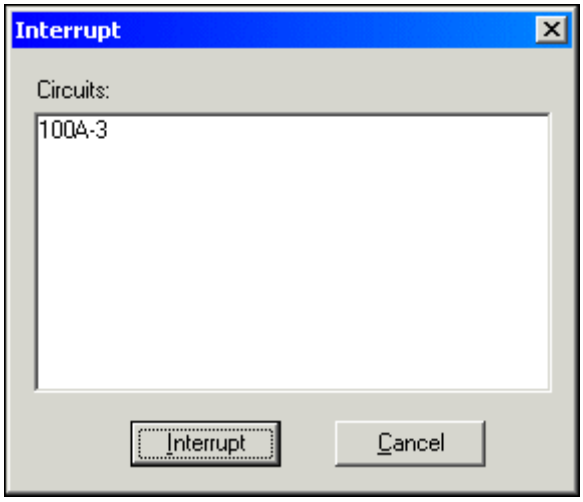


Finally, click the button **Stop** to confirm your selection.

2.7 Interrupt Circuits

To stop a circuit, highlight it in the Floor View and then click **Interrupt** from the **Control** menu or click the symbol  from the **Control** toolbar. In the Status view you may as well right-click an entry in the column **Circuit Name** and select the desired function from the appearing menu.

A box appears where you can pick the circuits to interrupt. To highlight several circuits, keep the **Ctrl** key pressed while clicking on circuits.

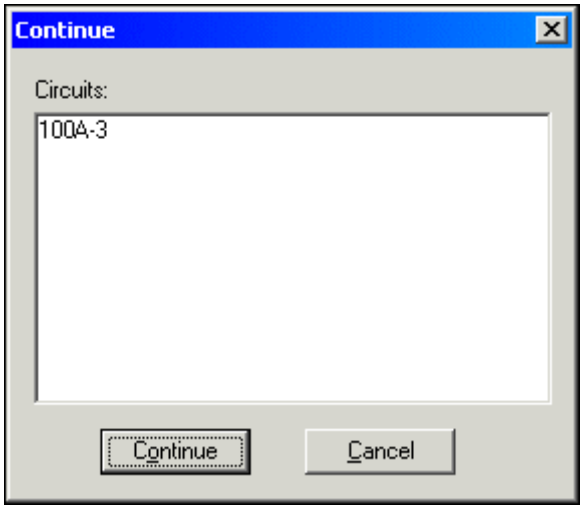


Finally, click the button **Interrupt** to confirm your selection.

2.8 Continue Circuits

To continue a circuit, highlight it in the Floor View and then click **Continue** from the **Control** menu or click the symbol from the **Control** toolbar. In the Status view you may as well right-click an entry in the column **Circuit Name** and select the desired function from the appearing menu.

A box appears where you can pick the circuits to continue. To highlight several circuits, keep the **Ctrl** key pressed while clicking on circuits.

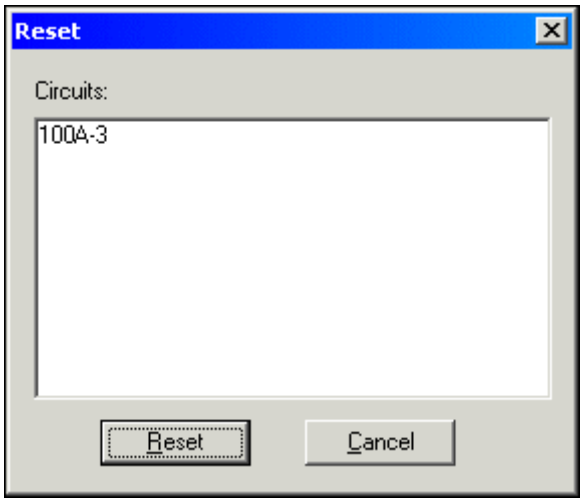


Finally, click the button **Continue** to confirm your selection.

2.9 Reset Circuits

To reset a circuit, right-click it in the Floor View and then click **Reset circuit(s)** from the appearing menu. You may as well right-click an entry in the column **Circuit Name** of the Status view and select the desired function.

A box appears where you can pick the circuits to reset. To highlight several circuits, keep the **Ctrl** key pressed while clicking on circuits.



Finally, click the button **Reset** to confirm your selection.

Note:

This procedure only resets the arrangement inside the database. A direct circuit reset is not possible by this means.

2.10 Circuit Status and Circuit Color

Every circuit of **Battery Manager** has a defined status. This status is represented by the color of the circuit in the Floor Display.

Circuits that are **offline** are displayed in White.

Circuits in **STOP** mode are displayed in **Darkgray**.

Circuits in **Interrupt** mode are displayed in **Lightgrey**.

Circuits in **Pause** mode are displayed in **Green** color.

Circuits in **Charge** mode are displayed **Yellow** color.

Circuits in **Recharge** mode are displayed in **Lightyellow** color.

Circuits in **Discharge** mode are displayed in **Blue** color.

Circuits in **Error** mode are displayed in **Red** color.

2.11 Saving and Recalling the Current Floor Display Configuration

The current Floor Display configuration can be saved for future reuse. If you have disabled QuickView, for example, then you can save this configuration and recall it next time it is needed. The configuration is saved to a file.

To save the configuration:

- Click File|Save As
- Be sure that **Save in:** says **data**
- Enter a file name. The file will automatically be saved with the required **.flr** extension.

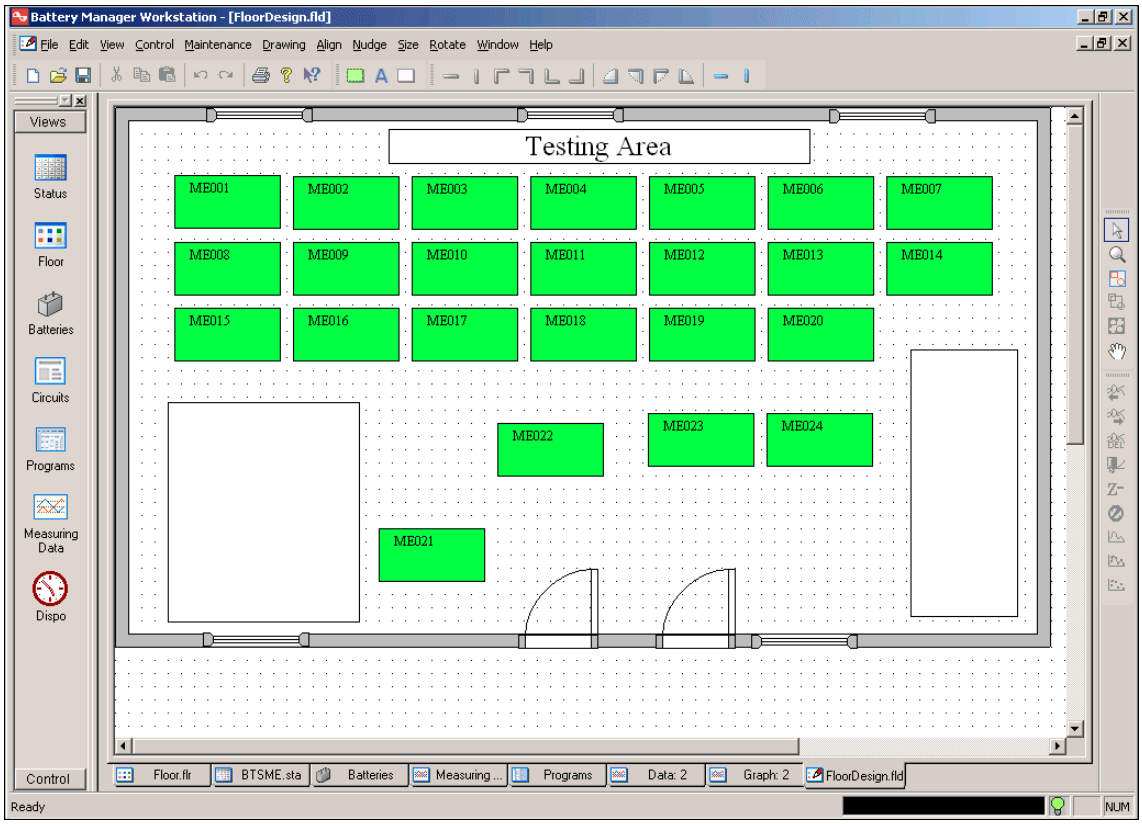
To recall the configuration and start the Floor Display:

- Click File|Open
- Be sure that **Look in:** says **data**
- Use the **Files of type:** drop down list to select **.flr**
- Select a file name from the list.

3 Floor Designer

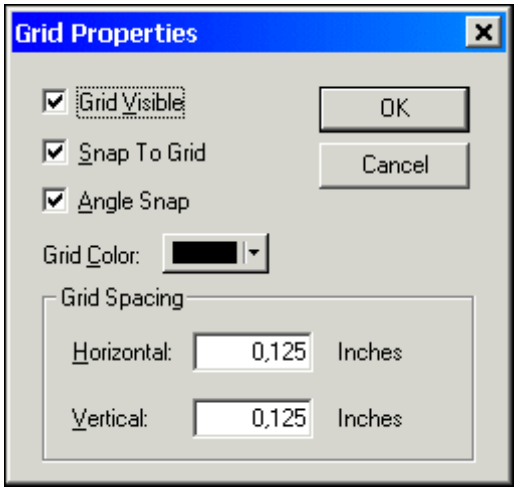
The Floor Designer will be used during initial system installation and configuration, and whenever changes are made to the floor. The floor designer allows you to create a floor display layout that closely matches the actual installation. This graphic representation can be taken over for the Floor Display.

To start the Floor Designer, point at **Configuration** in the menu **Maintenance** and click **Floor Designer**. If the entry is disabled, you either have a Floor View or Floor Designer already open or you don't have the necessary rights to make changes in the design.



During a design process you can allocate different properties like **Name** and **Device server** to a circuit. More information can be found in the chapter Adding Circuits and Modifying Configuration Information.

You can change the grid structure of the Floor Designer by right-clicking a free piece of floor. Now select **Grid Properties** from the appearing menu. The following dialog allows to switch the grid on and off and change several more features:



When the floor design process is complete, the resulting floor display layout will be saved to a file. For information about using the design in the Floor View, see Saving the Floor Display Layout.

3.1 Adding Circuits and Modifying Configuration Information

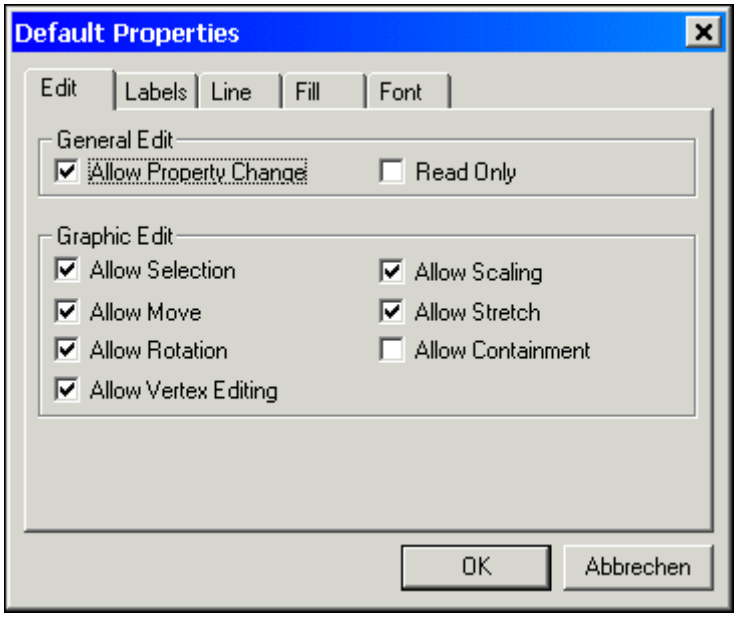
Circuits can be added to a new or existing floor display layout. Once the circuit is added, it must be configured. Existing circuits can be reconfigured at any time.

Add a circuit box by using the **Draw** toolbar:



- Click on the green controller icon. A shaped icon will be added to the mouse pointer.
- Move the mouse pointer to the desired spot and click the left mouse button. A standard size circuit box will be inserted.

In the menu **Edit** you can click the entry **Default Properties** to determine the properties for future circuit boxes:

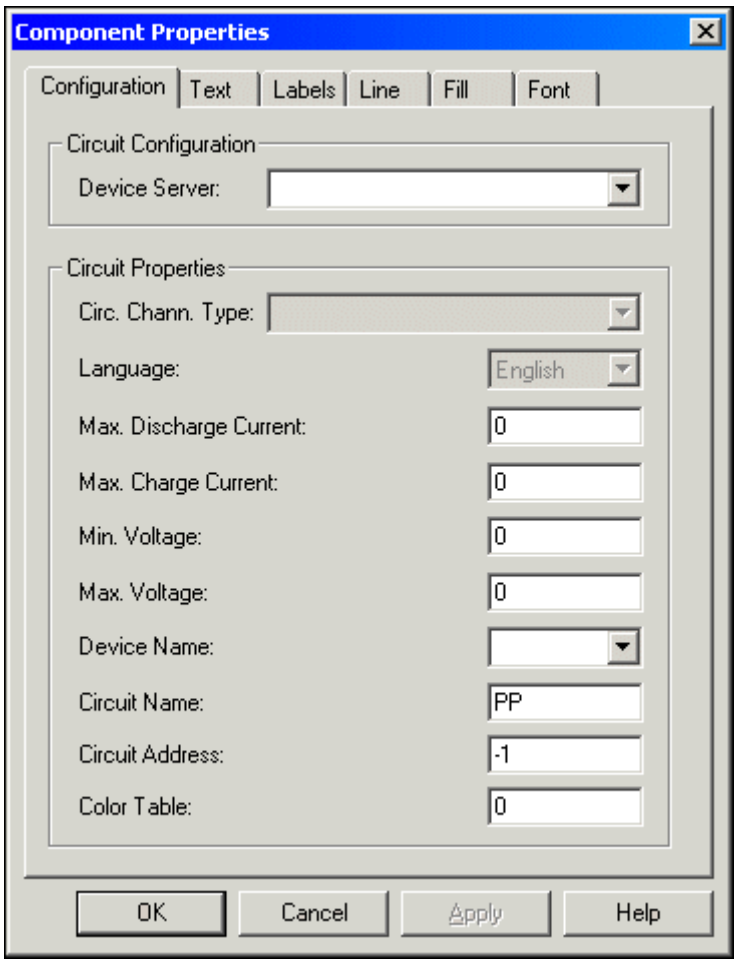


With this dialog you can define how boxes should look standard-wise (e.g. color, labeling) and what kind of properties they possess. You can thereby achieve that an element e.g. can not be moved or changed in size. However these properties can be adapted individually after an element has been created.

Configuring the circuit information:

- Move the mouse pointer to the desired circuit and double-click the left mouse button. The Component Properties dialog box will appear opened to the Configuration tab. You may as well right-click an element an select Properties from the appearing menu.

The dialog **Component Properties** is now shown with the tab **Configuration**:



- Use the list under **Configuration** to select a device server. You can then enter further circuit properties.
- With the tab **Text** you can alter the labeling of an element.
- The tab **Labels** allows to determine where the labeling appears on an element.
- With the **Line** tab you can determine the border of an element.
- The tab **Fill** influences the look of an element (color, hatch etc.).
- With the **Font** tab you can determine font size and type.

When you completed your settings, click **OK**.

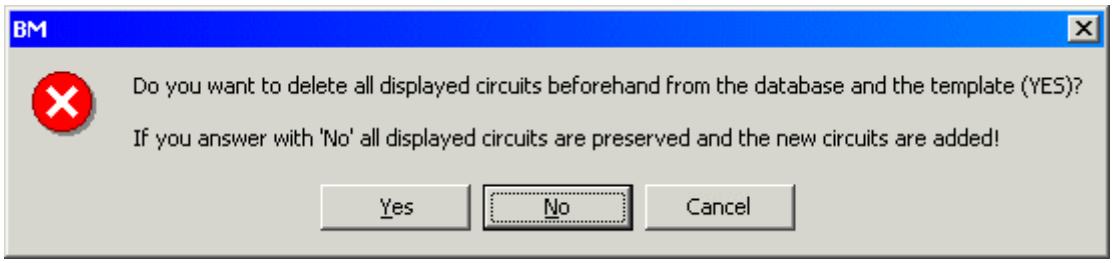
The settings described here may vary or be limited depending on the kind of object you selected.

Within the **Floor Designer** you can start a Wizard to create and define similar circuits quickly. Click **Floor Designer Wizard** from the **Edit** menu.

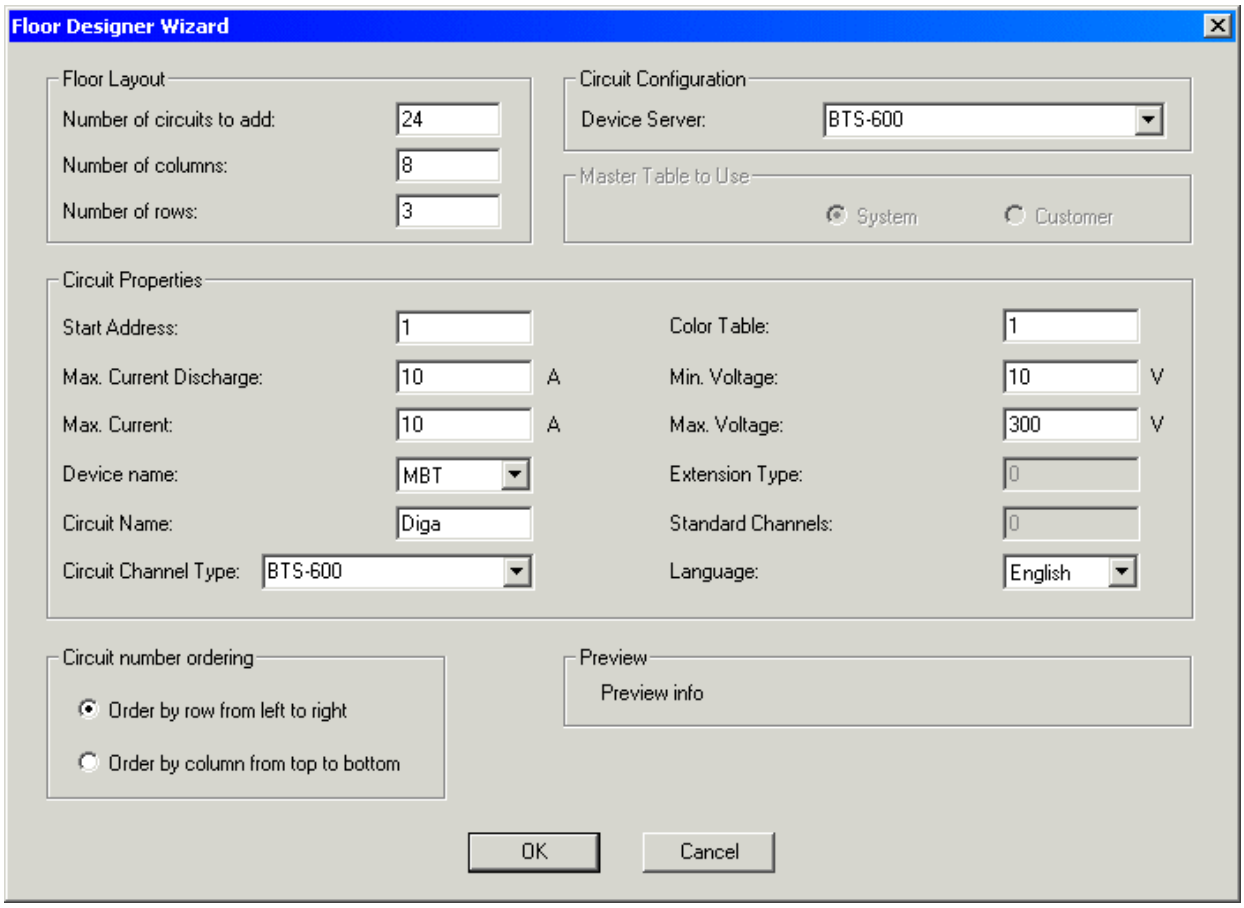
Note:

This entry is only active if the user has the appropriate rights for this process (login).

A request appears asking if you want to replace all displayed circuits or add the new circuits to the existing ones.



After clicking **Yes** or **No** the Wizard appears where you can define the number of new circuits and their common parameters:



When creating several similar circuits, the **Floor Designer Wizard** saves time and effort.

3.2 Moving and Resizing Circuits

To **move** a circuit box:

- Move the mouse pointer over the box until a four-way arrow appears.
- Hold down the left mouse button and move the box. The box will snap to the grid allowing easy alignment of the boxes.

To **resize** a circuit box:

- Move the mouse pointer to the desired circuit and click the left mouse button. Grab handles will appear.
- Move the mouse pointer over the desired grab handle until an arrow appears.
- Hold down the left mouse button and resize the box. The box will snap to the grid allowing easy alignment of the boxes.

3.3 Grouping Circuits

The menu item **Grouping** allows to summarize various elements on the screen and treat them as one. Grouped elements can e.g. be moved together. Select elements for the grouping by either

- pressing down the left mouse button and drawing a rectangle around the elements to be grouped

or

- clicking individual elements while pressing the **Ctrl** key

Perform a right-click on one of the elements and select **Grouping** from the context menu it is possible to change the look of an element or group.

3.4 Copying Circuits

One or more circuits can be copied to a different spot in the floor display layout. This makes it easy to duplicate a section of the floor display layout.

To copy one or more circuits:

- Select the circuit(s) that you wish to copy:
- To select one circuit, click on it with the left mouse button.
- To add to the selection set, hold down the Ctrl key and click on the additional circuits with the left mouse button.
- To select multiple circuits that are near each other on the floor display layout, press the left mouse key and drag the pointer to box the circuits that you want to copy.
- Grab handles will appear identifying the selection set.
- Move the mouse pointer over one of the boxes until a four-way arrow appears, click the right mouse button and select Copy from the menu.

- Move the mouse pointer to the desired spot, click the right mouse button and select Paste from the menu.
- Move the mouse pointer over one of the boxes until a four-way arrow appears, hold down the left mouse button, and jog the selection set to force it to snap to the grid allowing easy alignment of the boxes.

Note

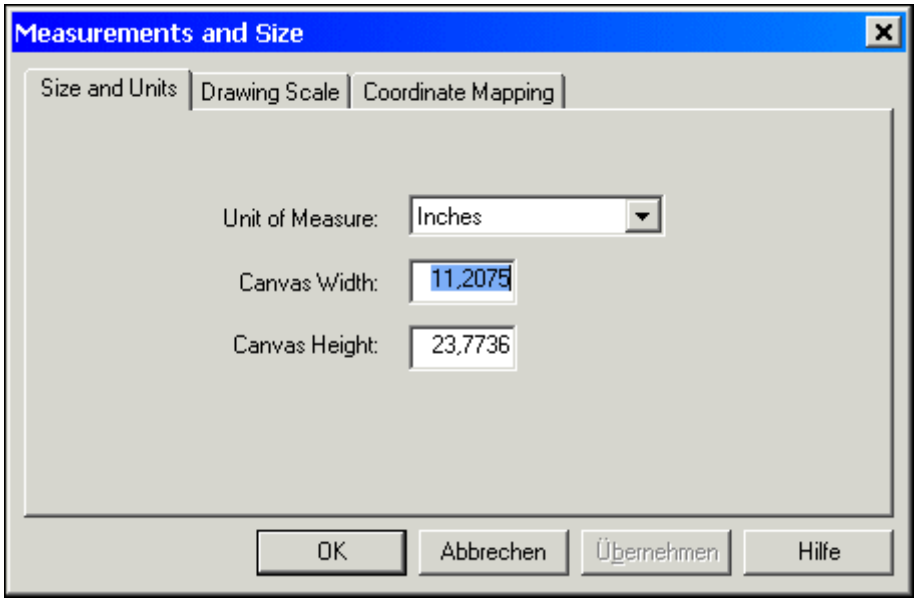
After the copy is complete, each of the copied circuits will have the configuration information of the circuit that it was copied from, and will therefore need to be reconfigured.

3.5 Adjusting Floor Size

The size of the floor display layout can be modified to fit the desired number of circuit boxes. Ideally, this size should be only slightly larger than necessary to fully contain all circuit boxes. Remember, the floor display layout will typically be larger than what can be viewed with the floor display view, and you will have to scroll to the desired area or zoom out to fit more of the floor display layout into the floor display view.

The size of the floor display layout is in inches. Note that the Undo/Redo feature is not available when adjusting the floor size.

- Click the entry **Measurements and Size** from the **Edit** menu. The dialog box **Measurements and Size** appears with the tab **Size and Units**.



- Be sure that **Unit of Measure** says **Inches** or **Centimeters**.
- Set the **Canvas Width** and **Canvas Height** as necessary. Experiment with these numbers.

The tab **Drawing Scale** enables you to determine the scale of the display on the screen (e.g. when drawing rooms).

The tab **Coordinate Mapping** defines the mapping mode and the range of values to display.

3.6 Adding Graphic Elements

You can create a true bird's eye view of the floor by adding graphic elements to the Floor Display Layout. Walls, doorways, windows, support beams, computer rooms, and signs can be added where needed. In addition, text boxes can be added for orientation purposes, such as **North West Entrance**, etc.

Add building elements by using the **Building** toolbar:



- Click on any of the building structure icons. A shapes icon will be added to the mouse pointer.
- Move the mouse pointer to the desired spot and click the left mouse button. The building structure will be inserted.
- These elements can be used to construct walls and insert doors and windows. Stretch the wall element to connect two 90° corner elements.

Add a text box by using the **Draw** toolbar:



- Click on the text icon  and position and insert a text box.
- Change the text box size by selecting it and using the grab handles to stretch it.
- Change the text and font as necessary. Move the mouse pointer to the text bock, click the right mouse button and select **Properties**. The **Component Properties** dialog box will appear opened to the **General** tab.
- To modify the text, select the **Text** tab. Click the option box **Multiple Lines** to create a multiline text. With the option **Word Break** you can create a wordwrap.
- To modify the font, select the **Font** tab.
- To determine the graphical styling, select the **Fill** tab.
- To change the border around the box, select the **Line** tab.
- To change the properties of a text box, select the **Edit** tab.

You may as well click an element once and then select **Properties** from the **Edit** menu to see the **Component Properties** dialog.

Add a box by using the **Draw** toolbar:



- Click on the right box icon and position and insert a box.
- Change the text box size by selecting it and using the grab handles to stretch it.

Additional capabilities:

- From the menu **Drawing** you can then select from a number of graphic elements, including line, rectangle, closed curve, ellipse, etc.
- From the menu you can select **Align**, **Nudge** and **Rotate** to change the position and orientation of a graphic element.

Stacking graphic elements and adjusting the stacking order:

- You can stack graphic elements on top of each other. For example, you could create an ellipse and then place a text box on top of it. In this case, it may be necessary to adjust the **order** of the items as they appear, e.g., you want the text box on top of the ellipse.
- To adjust the order, move the mouse pointer to the stack, click the right mouse button and select **Order…**. Use **Bring to Front**, **Send to Back**, **Bring Forward** and **Send Backward** as necessary. Experiment for best results. Note that the **Undo/Redo** feature is not available when ordering graphic elements.

3.7 Saving the Floor Display Layout

The floor display layout must be saved upon completion of the design process. The results are saved to **..\data\FloorDesign.fld**, which is the default file path and name. This is the default file used by the Floor Display.

To save the configuration:

- Click File|Save

If you make any changes to the floor display layout and attempt to exit the Floor Designer without saving, you will be prompted to save the file.

To use a changed or newly created Floor Design as new standard Floor View, store the layout under the name **FloorDesign.fld** into the **data** folder. When calling the Floor View for the next time, this Layout will be used as a template.

4 Workspace Manager

The Workspace is the appearance or layout of Battery Manager as seen on the screen. This layout consists of the position and size of the Battery Manager application itself, each of its open windows, and the position, size and docking status of each dockable toolbar.

The current workspace is saved on system exit. This workspace will automatically be restored during system restart. The workspace is saved to the default workspace name **..\data\STA-[Computer name]-UNI.wsp**.

4.1 Saving and recalling the Workspace

It is possible to save the current workspace to a different file name for future use.

To save the workspace:

- Click File|Save Workspace As
- Be sure that **Save in:** says **data**
- Enter a file name. The file will automatically be saved with the required **.wsp** extension.

To recall the configuration and start the Floor Display:

- Click **File** and select one of the listed workspace names.

or

- Click File|Open Workspace
- Be sure that **Look in:** says **data**
- Use the **Files of type:** drop down list to select (***.wsp)**.
- Select a file name from the list.

4.2 Care When Using a Recalled Workspace

If you recall a specific workspace, remember to save it before exiting the system to preserve any changes that you make. Click **File|Save Workspace**. When you restart the system, the workspace that was visible when you exited the system will be restored. If you then recall that specific workspace without having saved it, it will not have the latest changes in it.

5 Measuring Data

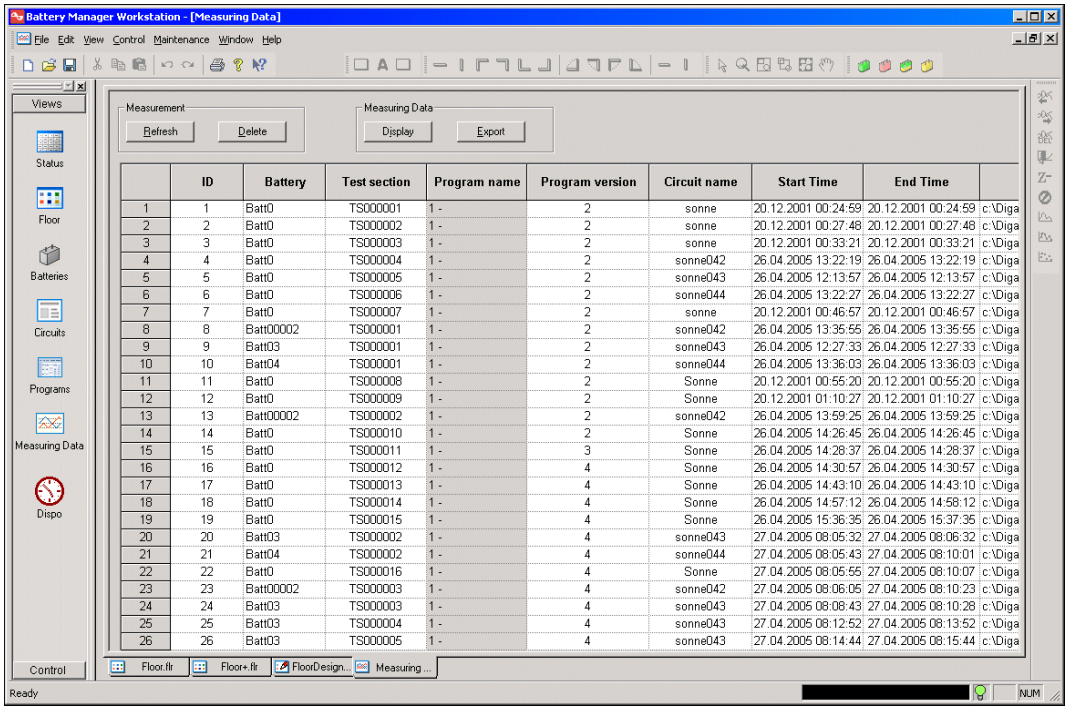
The **Measuring Data** environment allows the user to view data collected by the system during the execution of a program. Viewing is allowed for both completed sessions and sessions still in progress.

Specifically, the registration data viewer:

- Allows the user to display/analyze one or multiple sessions of collected data at a time.
- Displays data in a tabular worksheet (spreadsheet).
- Allows the user to view the data both numerically and graphically.
- Allows graphic representation of data to be manipulated by changing x and y-axis values.
- Allows the user to print both the numerical and graphical views of the registration data.

5.1 Using Measuring Data

At the top-level menu, select **Measuring Data** from the **View** menu. The Measuring data page shows all session data in spreadsheet form:



	ID	Battery	Test section	Program name	Program version	Circuit name	Start Time	End Time	
1	1	Batt0	TS000001	1 -	2	sonne	20.12.2001 00:24:59	20.12.2001 00:24:59	c:\Diga
2	2	Batt0	TS000002	1 -	2	sonne	20.12.2001 00:27:48	20.12.2001 00:27:48	c:\Diga
3	3	Batt0	TS000003	1 -	2	sonne	20.12.2001 00:33:21	20.12.2001 00:33:21	c:\Diga
4	4	Batt0	TS000004	1 -	2	sonne042	26.04.2005 13:22:19	26.04.2005 13:22:19	c:\Diga
5	5	Batt0	TS000005	1 -	2	sonne043	26.04.2005 12:13:57	26.04.2005 12:13:57	c:\Diga
6	6	Batt0	TS000006	1 -	2	sonne044	26.04.2005 13:22:27	26.04.2005 13:22:27	c:\Diga
7	7	Batt0	TS000007	1 -	2	sonne	20.12.2001 00:46:57	20.12.2001 00:46:57	c:\Diga
8	8	Batt00002	TS000001	1 -	2	sonne042	26.04.2005 13:35:55	26.04.2005 13:35:55	c:\Diga
9	9	Batt03	TS000001	1 -	2	sonne043	26.04.2005 12:27:33	26.04.2005 12:27:33	c:\Diga
10	10	Batt04	TS000001	1 -	2	sonne044	26.04.2005 13:36:03	26.04.2005 13:36:03	c:\Diga
11	11	Batt0	TS000008	1 -	2	Sonne	20.12.2001 00:55:20	20.12.2001 00:55:20	c:\Diga
12	12	Batt0	TS000009	1 -	2	Sonne	20.12.2001 01:10:27	20.12.2001 01:10:27	c:\Diga
13	13	Batt00002	TS000002	1 -	2	sonne042	26.04.2005 13:59:25	26.04.2005 13:59:25	c:\Diga
14	14	Batt0	TS000010	1 -	2	Sonne	26.04.2005 14:26:45	26.04.2005 14:26:45	c:\Diga
15	15	Batt0	TS000011	1 -	3	Sonne	26.04.2005 14:28:37	26.04.2005 14:28:37	c:\Diga
16	16	Batt0	TS000012	1 -	4	Sonne	26.04.2005 14:30:57	26.04.2005 14:30:57	c:\Diga
17	17	Batt0	TS000013	1 -	4	Sonne	26.04.2005 14:43:10	26.04.2005 14:43:10	c:\Diga
18	18	Batt0	TS000014	1 -	4	Sonne	26.04.2005 14:57:12	26.04.2005 14:58:12	c:\Diga
19	19	Batt0	TS000015	1 -	4	Sonne	26.04.2005 15:36:35	26.04.2005 15:37:35	c:\Diga
20	20	Batt03	TS000002	1 -	4	sonne043	27.04.2005 08:05:32	27.04.2005 08:06:32	c:\Diga
21	21	Batt04	TS000002	1 -	4	sonne044	27.04.2005 08:05:43	27.04.2005 08:10:01	c:\Diga
22	22	Batt0	TS000016	1 -	4	Sonne	27.04.2005 08:05:55	27.04.2005 08:10:07	c:\Diga
23	23	Batt00002	TS000003	1 -	4	sonne042	27.04.2005 08:06:05	27.04.2005 08:10:23	c:\Diga
24	24	Batt03	TS000003	1 -	4	sonne043	27.04.2005 08:08:43	27.04.2005 08:10:28	c:\Diga
25	25	Batt03	TS000004	1 -	4	sonne043	27.04.2005 08:12:52	27.04.2005 08:13:52	c:\Diga
26	26	Batt03	TS000005	1 -	4	sonne043	27.04.2005 08:14:44	27.04.2005 08:15:44	c:\Diga

5.2 Measuring Data Parameters

Each row of the spreadsheet represents one execution of a program and is considered a session representing data collected for that particular process. The displayed values may vary depending on the system (e.g. test or formation). For **BTS-600** the most important parameters are:

ID – ongoing number

Battery – type of battery used

Test section – name of the belonging test section

Program name – name of the program used

Program version – versions number of the program used

Circuit name – name of the circuit used

Start Time – time at start of the program

End Time – time at the end of the program. If the program is still running, no time is displayed.

File – location of the test section

5.3 Deleting Data

Rows from the measuring data table may be deleted by clicking on the **Delete** button. A dialog box appears for confirmation. By clicking on **Yes** you will delete the sections you have selected and all associated data. Data from programs still in progress cannot be deleted. Multiple rows can be selected to delete several data sets at once.

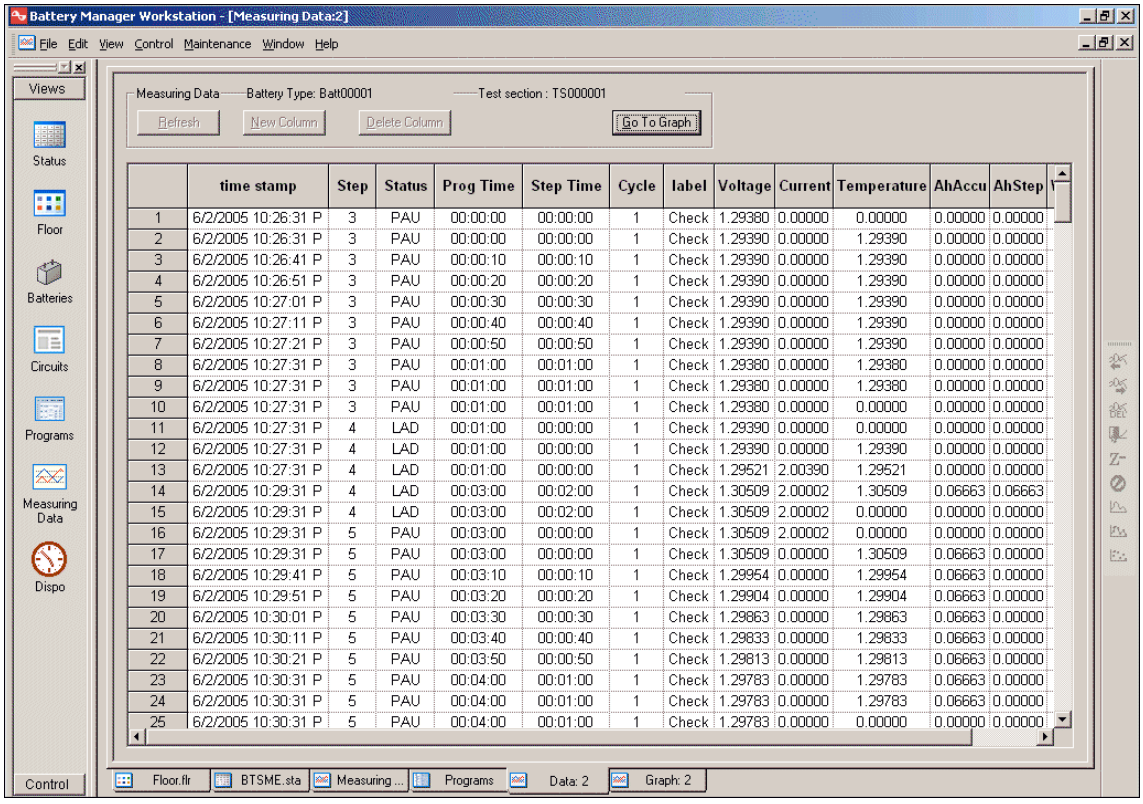
5.4 Exporting Data

Data may be exported for use with other programs such as Excel by a CSV file. You may select one or more rows for data export. The system creates one CSV export file for each entry selected. The CSV file contains all of the measuring data for that entry. If you select an entry and click on **Export**, a box appears for you to specify the output file name and directory. If you select multiple rows and click on **Export**, a box appears for you to specify a directory only. The system will automatically generate filenames for you.

5.5 Measuring Data Display

You may view data by selecting single or multiple data sets. To select one row, click the left mouse button on the very left column of the entry you wish to view. To select multiple rows, press the **Ctrl** key and hold it down while clicking the left mouse button on the numbers of the rows you want to select. During the execution of a program, the collection of data is optional.

You may select one or more data sets to be displayed or analyzed. To do this click on **Display**, and another screen that shows the numeric data stored during the program run appears. You may alternatively double-click an entry in the very left column or the **Battery** column. A table with data for this battery appears.



	time stamp	Step	Status	Prog Time	Step Time	Cycle	label	Voltage	Current	Temperature	AhAccu	AhStep
1	6/2/2005 10:26:31 P	3	PAU	00:00:00	00:00:00	1	Check	1.29380	0.00000	0.00000	0.00000	0.00000
2	6/2/2005 10:26:31 P	3	PAU	00:00:00	00:00:00	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
3	6/2/2005 10:26:41 P	3	PAU	00:00:10	00:00:10	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
4	6/2/2005 10:26:51 P	3	PAU	00:00:20	00:00:20	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
5	6/2/2005 10:27:01 P	3	PAU	00:00:30	00:00:30	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
6	6/2/2005 10:27:11 P	3	PAU	00:00:40	00:00:40	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
7	6/2/2005 10:27:21 P	3	PAU	00:00:50	00:00:50	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
8	6/2/2005 10:27:31 P	3	PAU	00:01:00	00:01:00	1	Check	1.29380	0.00000	1.29380	0.00000	0.00000
9	6/2/2005 10:27:31 P	3	PAU	00:01:00	00:01:00	1	Check	1.29380	0.00000	1.29380	0.00000	0.00000
10	6/2/2005 10:27:31 P	3	PAU	00:01:00	00:01:00	1	Check	1.29380	0.00000	0.00000	0.00000	0.00000
11	6/2/2005 10:27:31 P	4	LAD	00:01:00	00:00:00	1	Check	1.29390	0.00000	0.00000	0.00000	0.00000
12	6/2/2005 10:27:31 P	4	LAD	00:01:00	00:00:00	1	Check	1.29390	0.00000	1.29390	0.00000	0.00000
13	6/2/2005 10:27:31 P	4	LAD	00:01:00	00:00:00	1	Check	1.29521	2.00390	1.29521	0.00000	0.00000
14	6/2/2005 10:29:31 P	4	LAD	00:03:00	00:02:00	1	Check	1.30509	2.00002	1.30509	0.06663	0.06663
15	6/2/2005 10:29:31 P	4	LAD	00:03:00	00:02:00	1	Check	1.30509	2.00002	0.00000	0.00000	0.00000
16	6/2/2005 10:29:31 P	5	PAU	00:03:00	00:00:00	1	Check	1.30509	2.00002	0.00000	0.00000	0.00000
17	6/2/2005 10:29:31 P	5	PAU	00:03:00	00:00:00	1	Check	1.30509	0.00000	1.30509	0.06663	0.00000
18	6/2/2005 10:29:41 P	5	PAU	00:03:10	00:00:10	1	Check	1.29954	0.00000	1.29954	0.06663	0.00000
19	6/2/2005 10:29:51 P	5	PAU	00:03:20	00:00:20	1	Check	1.29904	0.00000	1.29904	0.06663	0.00000
20	6/2/2005 10:30:01 P	5	PAU	00:03:30	00:00:30	1	Check	1.29863	0.00000	1.29863	0.06663	0.00000
21	6/2/2005 10:30:11 P	5	PAU	00:03:40	00:00:40	1	Check	1.29833	0.00000	1.29833	0.06663	0.00000
22	6/2/2005 10:30:21 P	5	PAU	00:03:50	00:00:50	1	Check	1.29813	0.00000	1.29813	0.06663	0.00000
23	6/2/2005 10:30:31 P	5	PAU	00:04:00	00:01:00	1	Check	1.29783	0.00000	1.29783	0.06663	0.00000
24	6/2/2005 10:30:31 P	5	PAU	00:04:00	00:01:00	1	Check	1.29783	0.00000	1.29783	0.06663	0.00000
25	6/2/2005 10:30:31 P	5	PAU	00:04:00	00:01:00	1	Check	1.29783	0.00000	0.00000	0.00000	0.00000

This is a snapshot of the data available in the measuring data database at the time the worksheet is created. If the program is still running, the table will not update automatically. To get an update, click **Refresh**. The spreadsheet initially displays all available columns of data. The registration data is displayed in two groups of columns.

The group that appears on the left consists of the **Fixed Measuring Data**. These columns are:

- **Step** – The program step that was executing when the registration entry was collected.
- **Status** – The operation of the step that was executing when the registration entry was collected.
- **Step Time** – The time since the start of the step at which the registration entry was collected.

- **Prog Time** – The time since the start of the program at which the registration entry was collected.

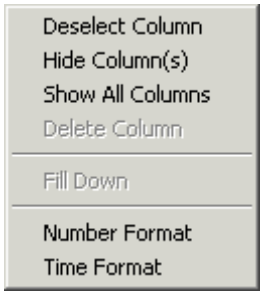
- **Cycle** – In programs that contain cycle (looping) instructions, the cycle number is a record of which iteration of the loop the registration entry was collected.

The fixed data columns can be moved right or left but are kept separate from the other data columns. The fixed data columns are kept on the display when the grid is scrolled to the left or right.

The fixed data columns are initially displayed in this order. The software allows the columns to be reordered or hidden, but they are kept separate from the fixed registration columns.

Certain devices supply data from extra channels called extension channels. Extension data columns are initially to the right of the fixed channels. The initial display order of the column is defined by the database. Column headings for the extension data are displayed according to entries made in the database.

The appearance of the table can be influenced from a menu that you call by clicking the right mouse button:

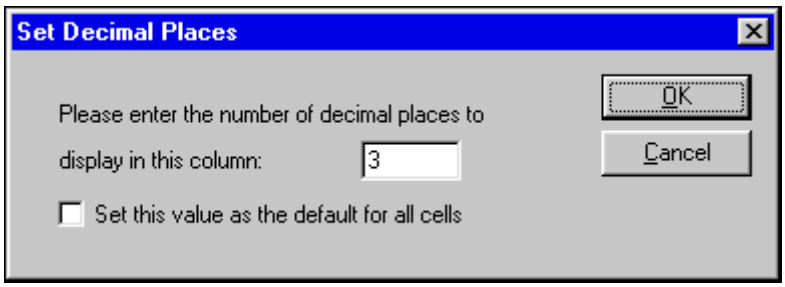


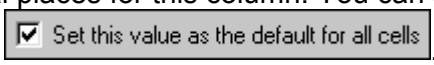
Choosing **Select Column** highlights the column you just clicked. You can select several columns by repeating this procedure. Calling up the menu again and then clicking **Hide Column(s)** makes the highlighted columns invisible. If you haven't marked a column and then pick **Hide Column(s)**, only the column you right-clicked is hidden.

All data can be made visible again with **Show All Columns**.

To undo the highlighting of a column, right-click it and pick **Deselect Column** from the appearing menu.

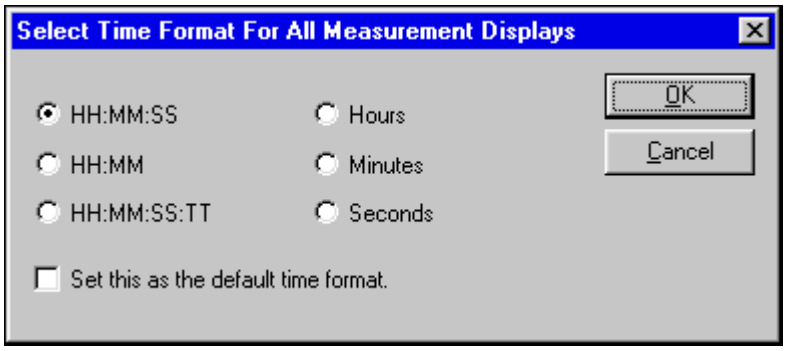
Use the context menu entries **Number Format** and **Time Format** to change the arrangement of data. After clicking **Number Format**, the dialog **Set Decimal Places** appears:



Enter the number of decimal places for this column. You can set this value as a standard for all cells of the table by clicking 

. Different then with the **Time Format** the settings made here are not stored when terminating the program.

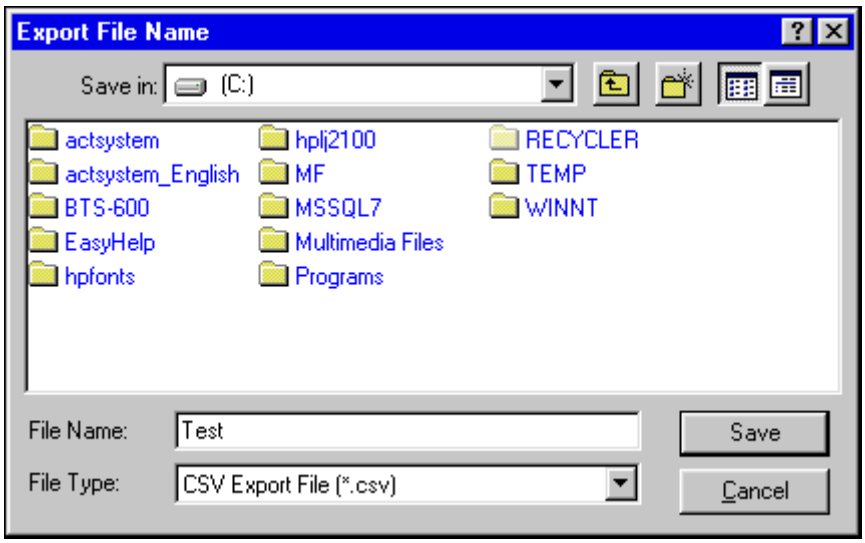
When clicking **Time Format** from the context menu you can choose between six display layouts:



The format **HH:MM:SS** (Hours:Minutes:Seconds) is the preset standard format. Instead you can choose for a display without seconds (**HH:MM**) or with 1/100 of a second (**HH:MM:SS:TT**). The formats **Hours**, **Minutes** and **Seconds** convert all times into decimal values, e.g. 2:45 turns into 2.75. You can set this format as a standard for all cells of the table by clicking 

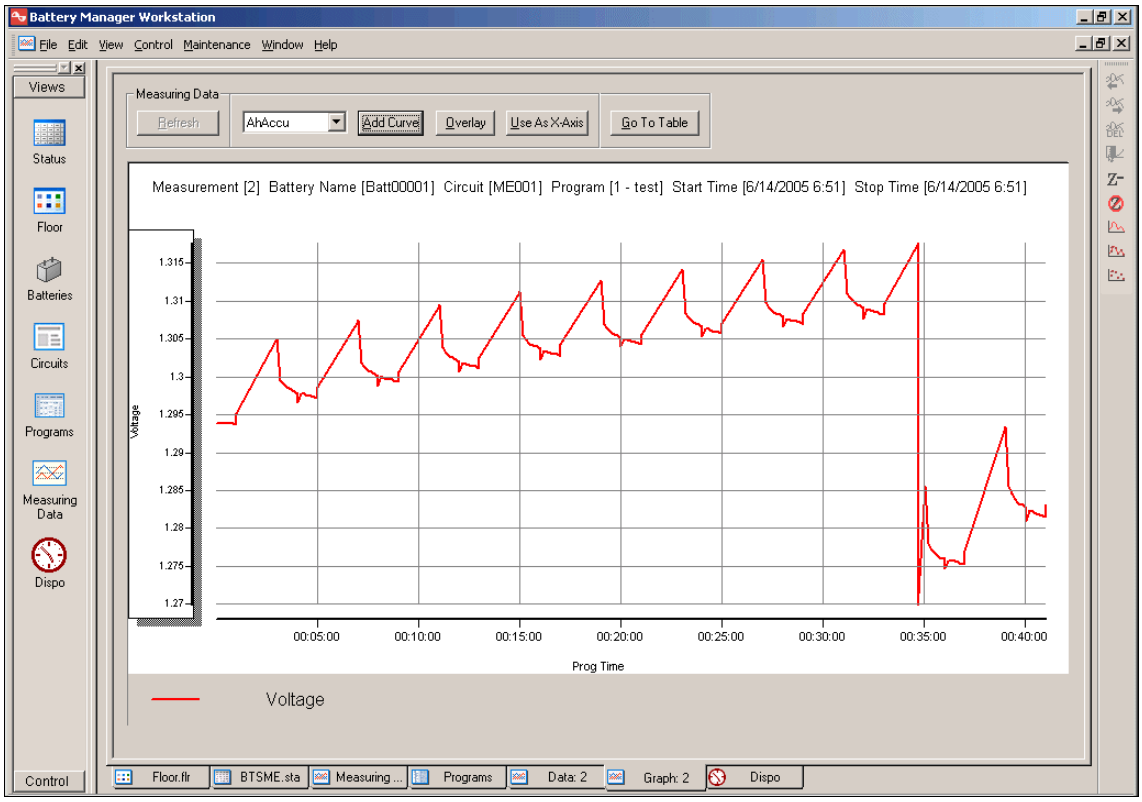
.

The **Export** button allows to convert the data from a selected measurement into a format that can be processed by other programs, e.g. Excel. Determine name and destination folder of the export file:



5.6 Graphical Display

To see the measuring data as a diagram, click the **Go To Graph** button. A new sheet is then created for the graphical display:

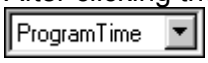


To show more than one curve in a diagram, pick a new value from the selection box 

and click the **Add Curve** button. A new Y-axis appears:

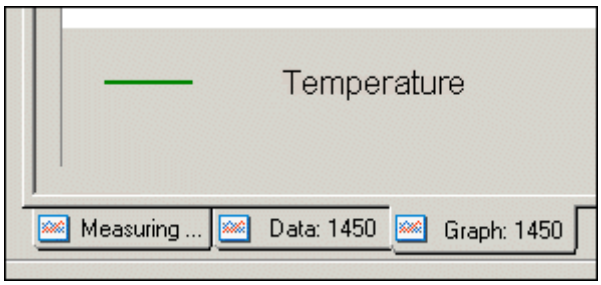


If you click **Overlay** instead of **Add Curve**, the new curve uses the Y-axis together with the previous curve.

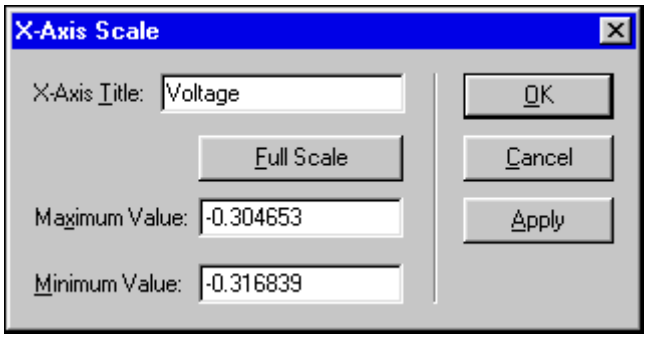
After clicking the **Use As X-Axis** button, the measuring unit shown in the selection box 

is used as X-axis. This means that this measuring unit is now represented by an even, while all other units are shown in dependence of this reference unit.

With the **Go To Table** button you return to the numerical display. Thereby the graphical display is not closed. On the bottom of the screen you can switch between the different sheets at any time:



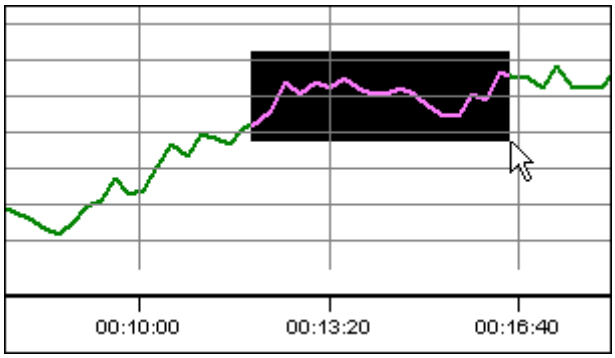
You can change the scale of X and Y-axis by double-clicking the axis. A new dialog appears:



Here you can enter a name for the axis and set maximum and minimum value for the graph. Note that when an axis represents time, maximum and minimum values are given in seconds. Click **Apply** to immediately see the impact on the diagram. Use the **Full Scale** button to return to the standard scale. Clicking **OK** closes the window.

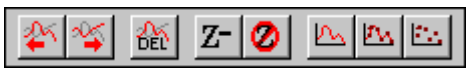
To make a snapshot of the current graph, click the **Copy** entry from the Menu **Edit**. Via Windows' clipboard you can insert the picture into another program.

To magnify a part of the diagram, position the mouse pointer on the upper-left corner and drag it to the lower-right corner of the section you want to view in detail. This section will then be marked as a rectangle:



After releasing the mouse button, the selection will be displayed greater. You can do this repeatedly. To undo a magnification, use  or  from the **graphical toolbar**.

You can use the **graphical toolbar** to modify the graphical presentation of data:



Please note that some functions are not always available.

 If more than one curve is displayed in a diagram, this button can be used to jump to the **previous** one. This curve then becomes the new active curve that can be recognized by the white background of its Y-axis.
When the first curve is active, clicking this button again selects the last curve. Thereby a repetitive run through all curves from right to left is possible.

 This button jumps to the **next** curve e.g. it allows you a run through all curves from left to right.

 This button erases the active curve from the diagram. The last remaining curve cannot be deleted.

 Use this button to zoom out of a diagram. Thereby you can take back every magnification step by step. This function is only available if you zoomed-in a part of the diagram before.

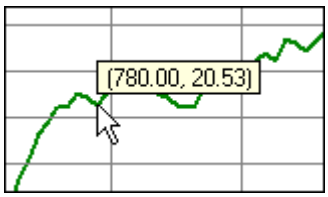
 This button immediately shows the graph in full scale e.g. all zooming steps are taken back. This function is only available if you zoomed-in a part of the diagram before.

 This button shows the active curve as a continuous line. This is the standard setting for all diagrams.

 This button shows a continuous curve with additional highlighting of the measuring points.

 Click this button to show only the measuring points of a curve.

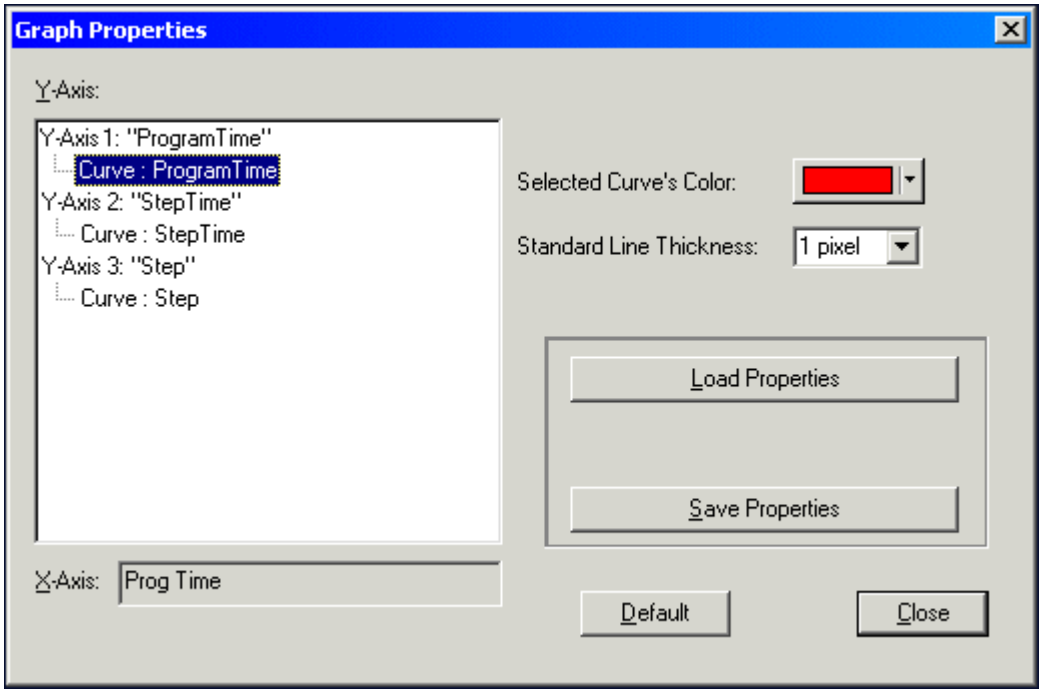
If you get the mouse pointer to a spot of a curve where a measuring point lies, the exact X- and Y-values will be displayed after a few seconds:



You can right-click a diagram to change more graph properties. Click with the right mouse button anywhere in the diagram, point at **Properties** and click **Graph**:



Now the **Graph Properties** window appears:



In the left field you select the curve whose appearance you want to modify. Then you can change the curve's color and line thickness. To save the changes to the referring program number, click **Save Properties**. Clicking **Load Properties** resets all changes you made to the graphical display since the last storage. Use the **Default** button to reset a diagram to its default settings.

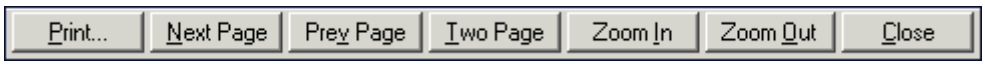
5.7 Printing

You have the possibility to print the currently displayed data or a graphical diagram. Therefore you can find several entries in the **File** menu:



With **Print** you get directly to Windows' printing dialog as known from many programs. **Print Preview** shows the arrangement of the current page the way it will appear on paper. The entry **Print Setup** enables you to change the printer's properties.

At the top of the Print Preview window you find several buttons:



Use the **Print** button to go to the Windows printing dialog. The **Next Page** and **Prev Page** buttons jump to the next or previous page (if available). Click the **Two Pages** button to show two pages at once. You can then use the same button to go back to the single page view.

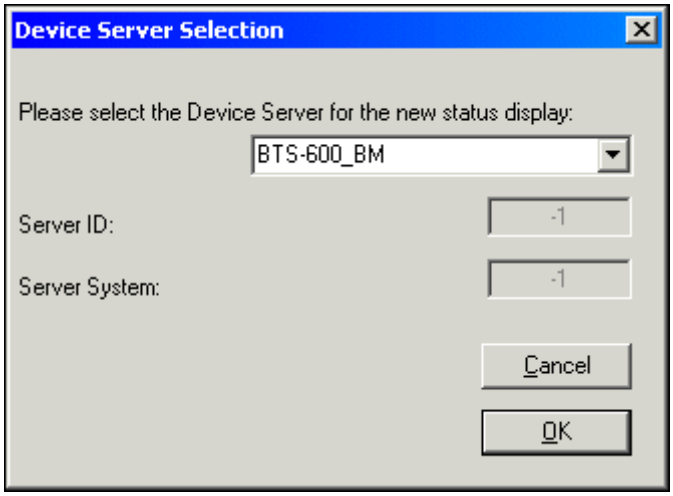
The **Zoom In** button enlarges the illustration by a step. You can do the same by clicking anywhere on the document with the mouse pointer that turns into a looking glass  . If there is no transition into a looking glass you have reached the greatest magnification step. The **Zoom Out** button works just the opposite way.

Use the **Close** button to exit the Print Preview. In comparison, clicking the  button at the top will terminate the whole evaluation program.

6 Status View

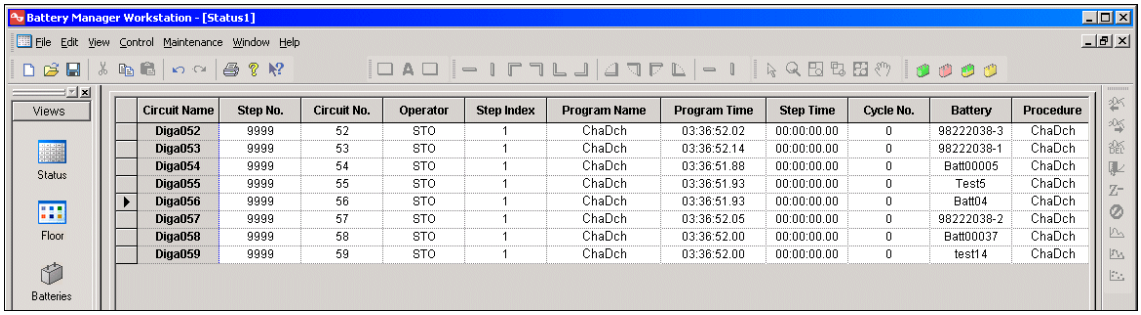
Status View presents measuring and registration data information in a tabular format. For every device server there is a separate Status View. To see Status Views from different device servers, select the entry **Status** from the **View** menu and pick the needed device server from the appearing dialog.

Example for a **BTS-600** device server:



Status View has many features to customize the view to suit a wide variety of user needs. Features include:

- Control
- Column Selection and Ordering
- Document Storage



Status View provides a view of the data organized by circuits and when a program is active, a view of the data associated with the circuit. By default, data within the Status View is refreshed permanently.

6.1 Invoking Status View

The user may invoke Status View in one of the following ways:

1. Selecting **Status** under the **View** menu. The View menu is one of the main menu items. The View menu is also available by right clicking in the main frame.

2. Pressing the Status icon in the Views Shortcut Bar. Hereby it is not possible to call a new Status View if another Status View is already displayed.



3. Opening a Status Grid file (***.sta**). Select **Open** from the **File** menu and pick the appropriate file. Also see Document Storage for more details.

6.2 Controlling Circuits in the Status View

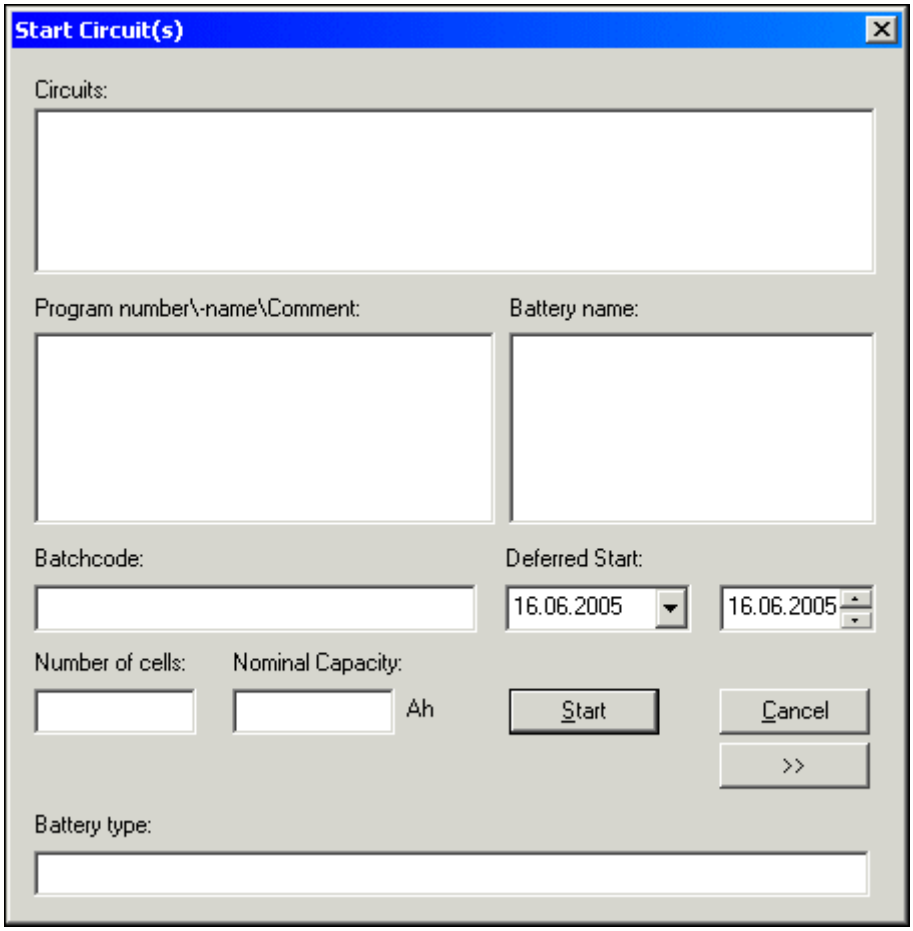
As with **Floor View**, the User may start, stop, interrupt, or continue a circuit (or a set of circuits) using several different methods. Common to all methods, the user may highlight whole rows of circuits to select multiple circuits or simply click anywhere in a row to select a single circuit. When several cells of a view are highlighted, but not full rows, then the row with the selection caret ► is the circuit selected. Likewise, when a single cell is selected, then the circuit of the row containing that cell is selected.

As with standard Windows, pressing the control key while selecting with the mouse allows the user to select multiple individual circuits. Holding the shift key causes all circuits between two individual selections to be selected. In addition, the user may simply start with one circuit and move up or down while depressing the left mouse button to select multiple circuits. The user must use the left-most column to make multiple selections.

As with Floor View, the user can exercise control (start, stop, etc.) through the icons, the **Control** menu option, and by right-clicking in either of the first left columns.

Depending on the installation there appears a dialog for the selection of various start parameters. When starting a **BTS-600** circuit after clicking **Start** the Dispo list appears instantly.

With the installation of a formation system the following dialog appears:



Perform the following steps here:

Click a program from the list. When starting a circuit, a deferred start is also possible. Thereby you can determine date and time of a circuit that is about to be started. If you want to run a circuit immediately, leave date and time fields unchanged. If you enter a Batchcode, it will be displayed in the measuring data overview later.

Click the button **Start** to perform the process you have chosen or click **Cancel** to abort. Afterwards you can recognize the new state of the circuit by its new Color in the **Circuit Overview**.

6.3 Moving Columns

The order of columns in the Status View can be changed. Select a column by clicking in the header of the desired column. Then depress the left mouse button in the column header, hold and drag to a new position. A red line appears on the left side of the column where the selected columns will be inserted when the mouse button is released. The user may move consecutive selected columns at the same time using the same technique.

6.4 Moving Rows

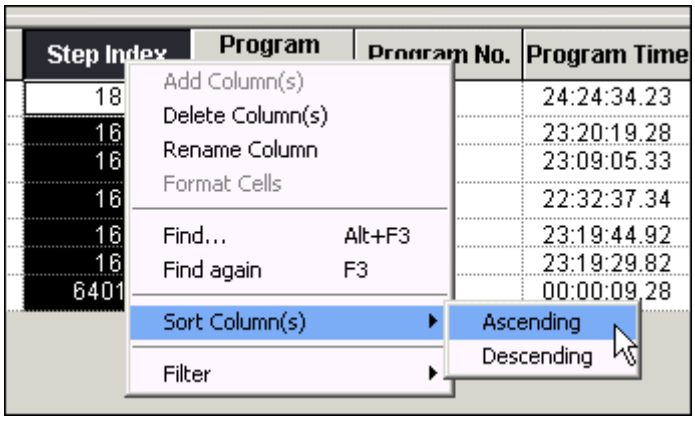
Select the row by clicking in the left-most column. Depress, hold, and drag to move. The selected row will be inserted just above the red line that appears while dragging. Select multiple consecutive rows for moving by first moving the cursor to the left-most column. Then depress the left mouse button and increase the selection by moving either up or down. Individual cells and groups of cells cannot be moved.

6.5 Resizing Rows and Columns

Select multiple columns or rows for resizing or simply resize an individual row or column. Place the cursor over the column or row border to the right or bottom until resize symbol appears.  Depress the left mouse button, hold and resize until the desired width is achieved, then release. Resize multiple rows or columns to achieve the same width or height for each.

6.6 Sorting Data

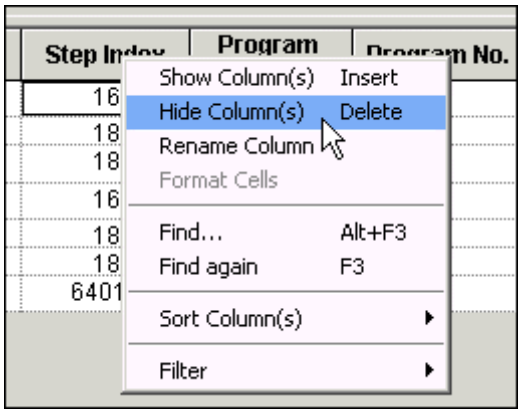
In Status View, there are two methods of sorting. The first is to double-click on the header of the column that contains the data to be sorted. The second is to right-click in the column header and choose **Ascending** or **Descending** from the menu that appears.



When the user double-clicks, if no sort is effect, then Status View first sorts the column in ascending order. Otherwise, Status View toggles the sort order. For example, to sort the Circuit column in descending order after first starting Status View, double-click on the Circuit column header to sort in ascending order, and then double-click again to sort in descending order.

The user can sort based on data in multiple columns using the menu sort commands. First, select the columns to sort. Status View evaluates selected columns from left to right. Move primary sort columns to the left of secondary sort columns before selection to change sort order.

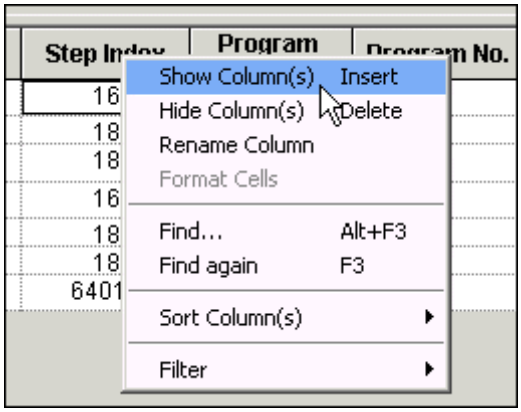
6.7 Hiding Columns



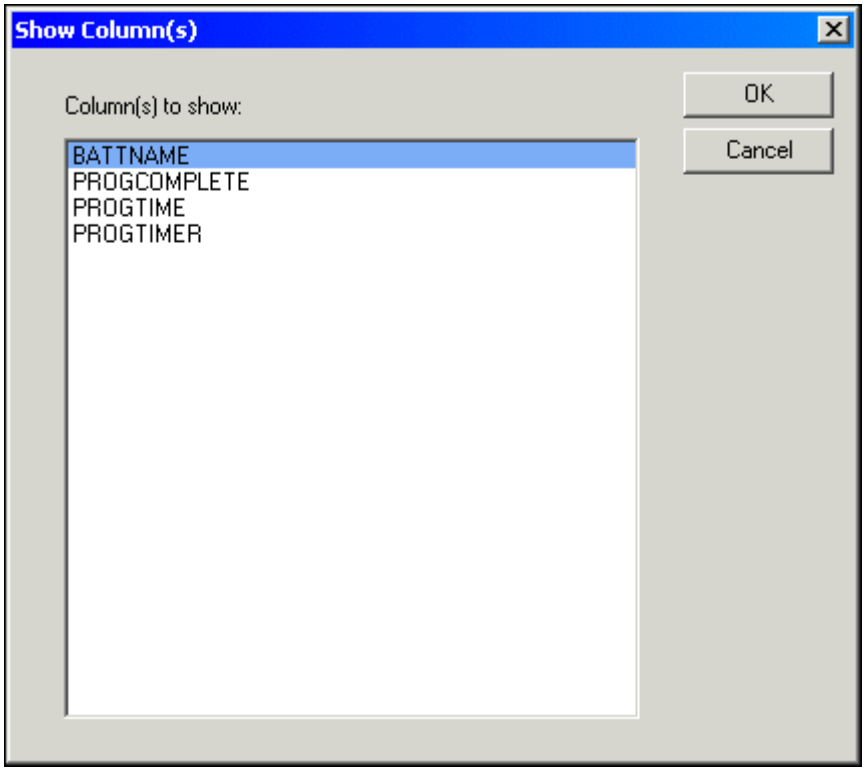
First select an unneeded column by clicking the column head. Then right-click the selected column and either pick **Hide Column(s)** from the appearing menu or press the **Delete** key. A warning message appears. Click **OK** or press **ENTER** to accept changes.

You may select several columns together by pressing down the **Ctrl** key. Always leave at least one column, so that columns may be added.

6.8 Showing Columns



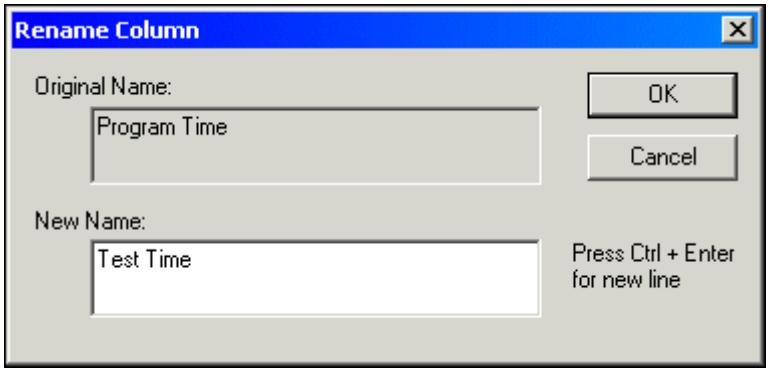
First select a column by clicking the column head. Then right-click the selected column and either pick **Show Column(s)** from the appearing menu or press the **Insert** key. A dialog appears listing columns currently not displayed.



Select columns to add. Use the **Ctrl** or **Shift** keys for multiple selections within the dialog. Click **OK** to insert the selected columns to the left of the column currently selected in the grid.

6.9 Renaming Columns

Column headers in Status View correspond to field names within the Battery Manager database. Right-click on the column header and select **Rename Column** to engage following dialog:



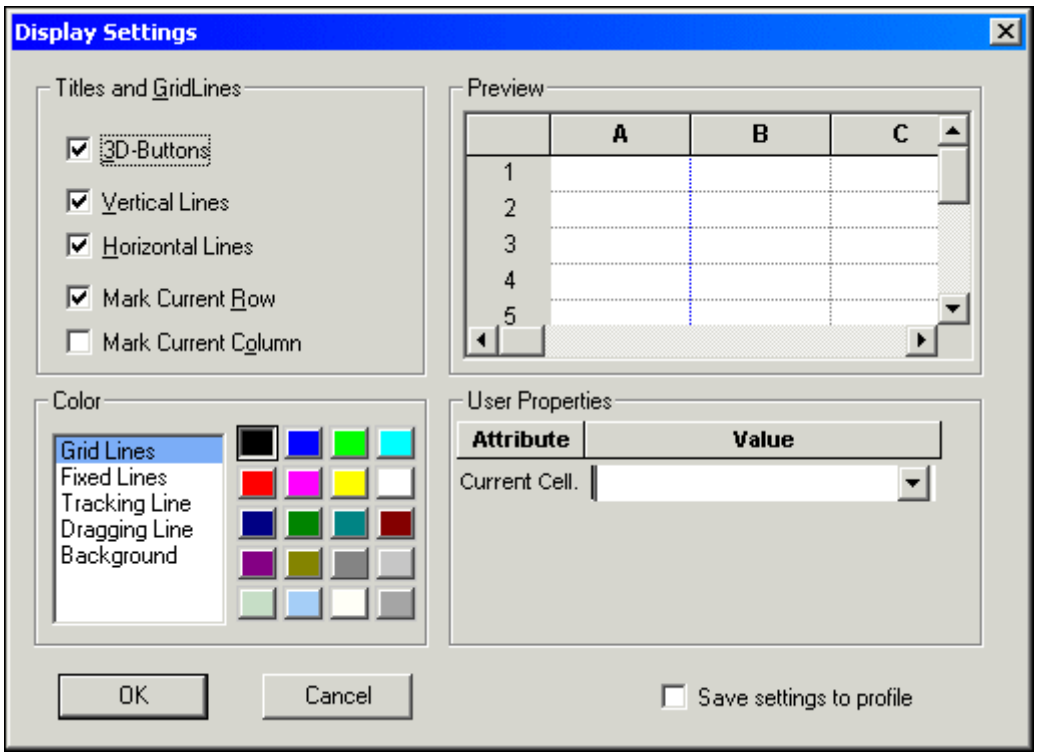
The **Original Name** is a name originally assigned by Battery Manager. Enter the new name for the column in the **New Name** edit box. Hold the **Ctrl** key and press **ENTER** to enter a carriage return; although, column names automatically wrap depending on the width of the column.

6.10 Cell Formatting

Status View allows formatting for cells containing floating-point data and data/time information. Right click in the column header of the cells to be formatted. **Format Cells** is enabled only for cells that can be formatted.

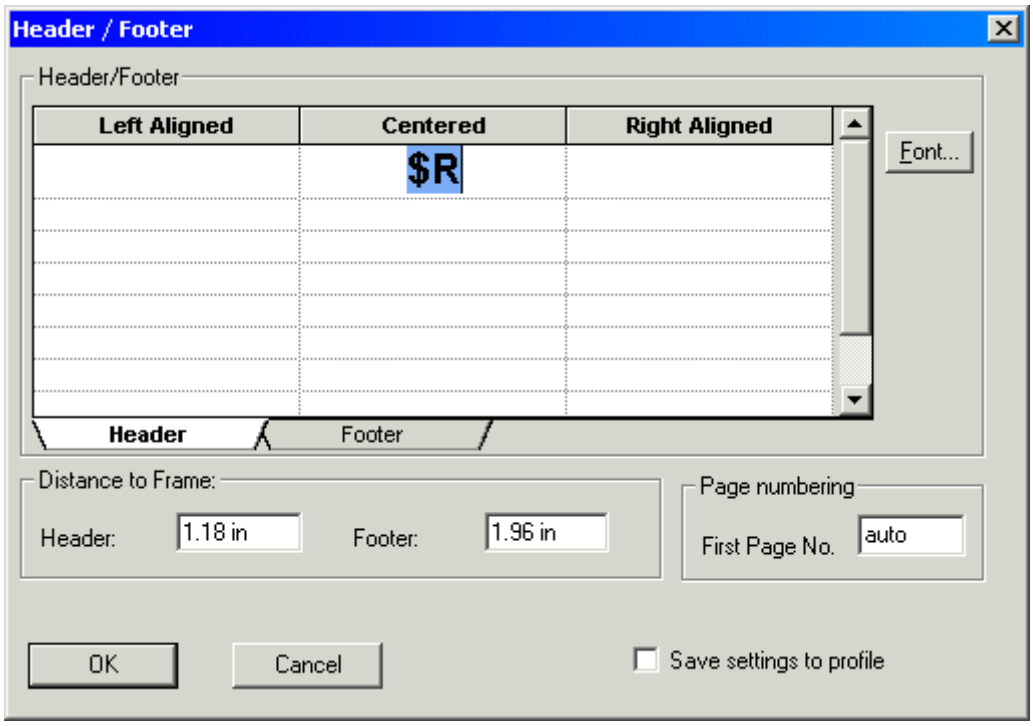
For floating-point data, the user can set the number of decimal places displayed. With time information, the user can display date, time, or both in different ways.

From the Status View you can use **Properties - Grid** from the **View** menu to adjust the display of the table view:



The option boxes in the left part alter the look of the representation immediately. The field **Colors** lets you change arrangement in terms of color. Most settings can be checked in the **Preview** window before confirmation with **OK**.

6.11 Header and Footer



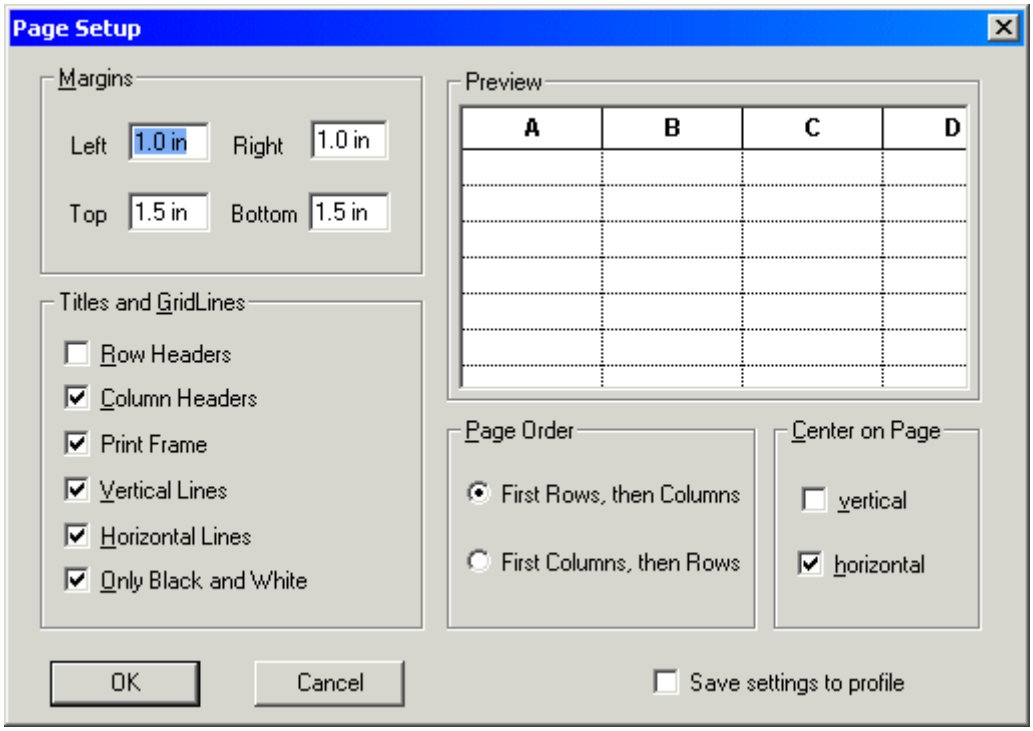
From the **File** menu you can click the entry **Header/Footer** to determine the text for printing the current Status View. Header and footer are thereby used for additional information on top and bottom of a page.

The effect of the settings made here can be verified by selecting Print Preview from the **File** menu.

Use the button **Font** to change font size and font type for header and footer.

6.12 Page Setup

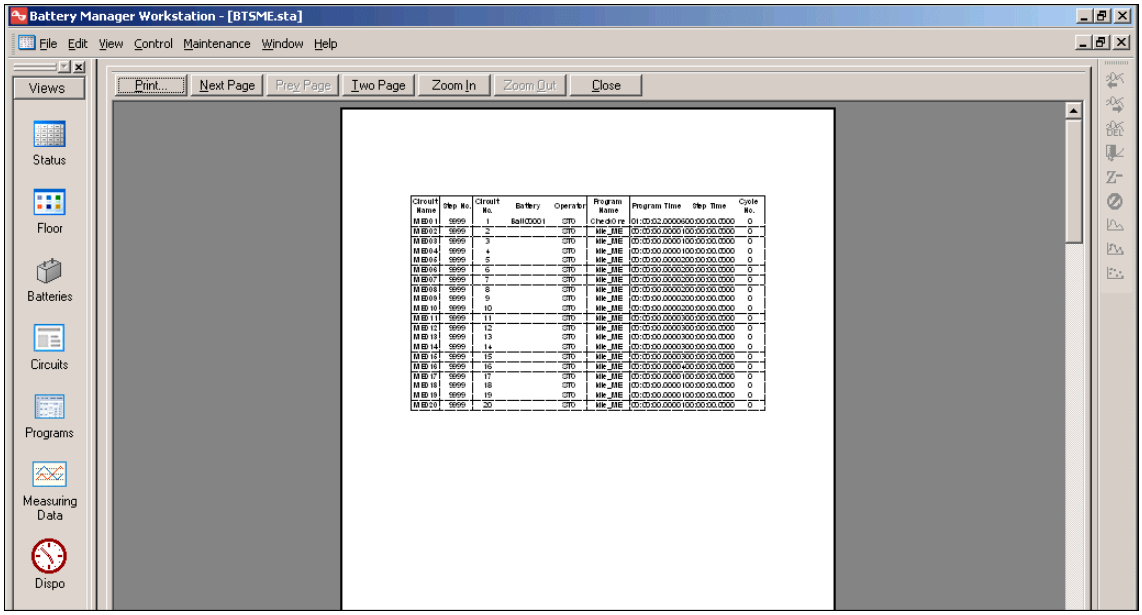
Click the **Page Setup** entry from the **File** menu to determine several parameters for your printout.



Use the field **Margin** to set page borders. With the field **Tiles and GridLines** you can define which table elements are actually printed. Finally you can determine the order of rows and columns when printing a page and the alignment inside the cells (vertical or horizontal).

6.13 Print Preview

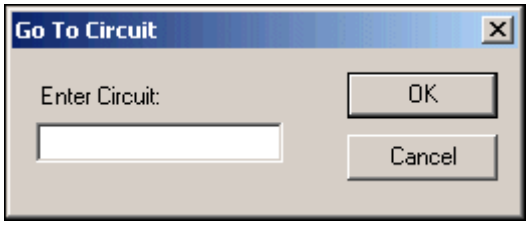
Click the **Print Preview** entry from the **File** to see the current page the way it will appear as a printout.



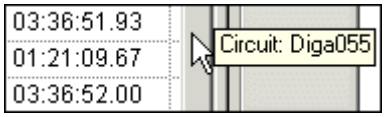
Above the "paper" the mouse pointer turns into a magnifying glass. Clicking with this magnifying glass makes the page appear larger. Terminate the Print Preview by clicking the button **Close** or start the printout with the Print button. All other buttons are self-explaining.

6.14 Find Circuit and Moving About

To find a circuit rapidly, double-click in the upper left corner of the grid and enter the circuit. Be sure to match case.



Alternatively, the user may grab the vertical scroll bar by depressing left mouse button, holding, and dragging. A hint window appears showing the circuit that the system will display in the top row when the mouse button is released.

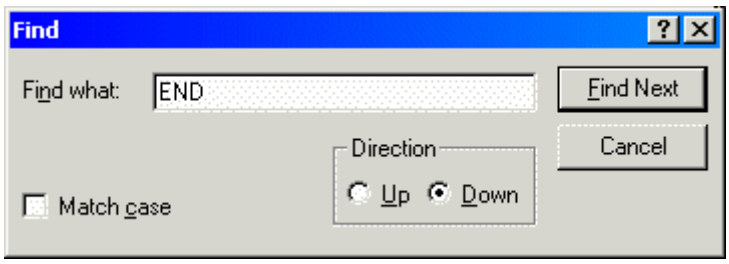


Likewise, when moving the horizontal scroll bar, the system displays a hint window showing the left-most column when releasing the mouse button.



Press **Ctrl-Home** or click once in the upper left corner of the grid to move to the first cell in the top row of the grid. Press **Ctrl-End** to move to the last cell in the bottom row.

Press **Alt-F3** or right-click on the column header for the **Find** dialog. The **Find** dialog looks for text in the same column as the currently selected cell, either up or down. Either click **Find Next** or press **F3** to find the next occurrence.



The user may leave the **Find** dialog displayed for continued use. Select a different column in the grid and press **F3** to search for the same text in another column.

6.15 Document Storage

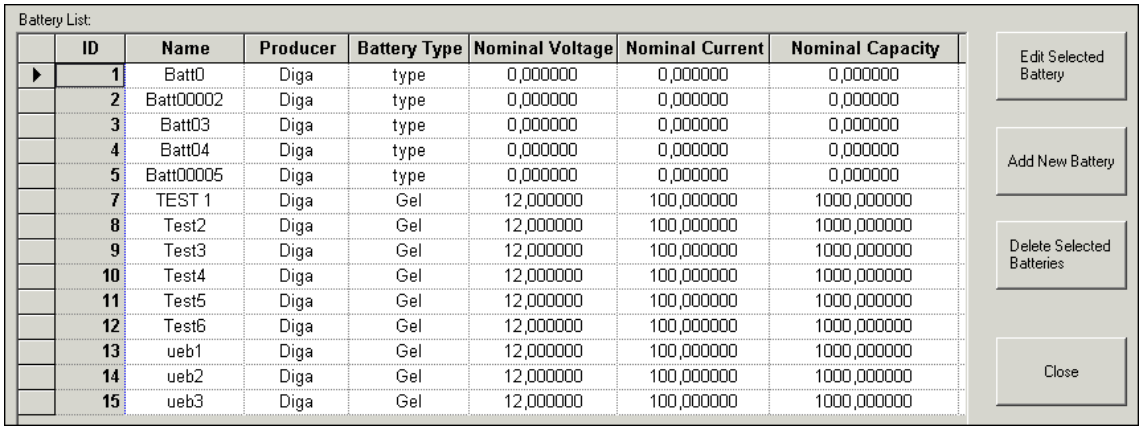
Whenever row height, column names, or column width are changed or a new sort is applied, etc., **Status View** prompts the user to save the view upon exit. If the user responds with “Yes”, the system responds with the **Save As** dialog. The user may choose a new file name or accept the default and then press **Save** to save the customized view.

At any point, the user may click the **Save** button  or select **Save** or **Save As** under the **File** menu to save the file.

If the saved Status View is active when closing Battery Manager, then Battery Manager reactivates the saved view when restarting. Pressing the **Status** icon or opening Status View through the **View** menu invokes the Default Status View.

Pressing the file open icon  or selecting **Open** under **File** on the main menu brings up a file selection dialog. Select a previously saved Battery Manager Status Grid file within the file selection dialog to open the view.

7 Battery Maintenance



Battery Manager provides the user with a utility called Battery Maintenance to maintain data associated with different types of batteries.

Invoke Battery Maintenance

The user may invoke Battery Maintenance in the following way:

1. Select **Batteries** under the **Maintenance** menu of the main screen.

or

2. Pressing the **Battery** icon in the Views Shortcut Bar. 

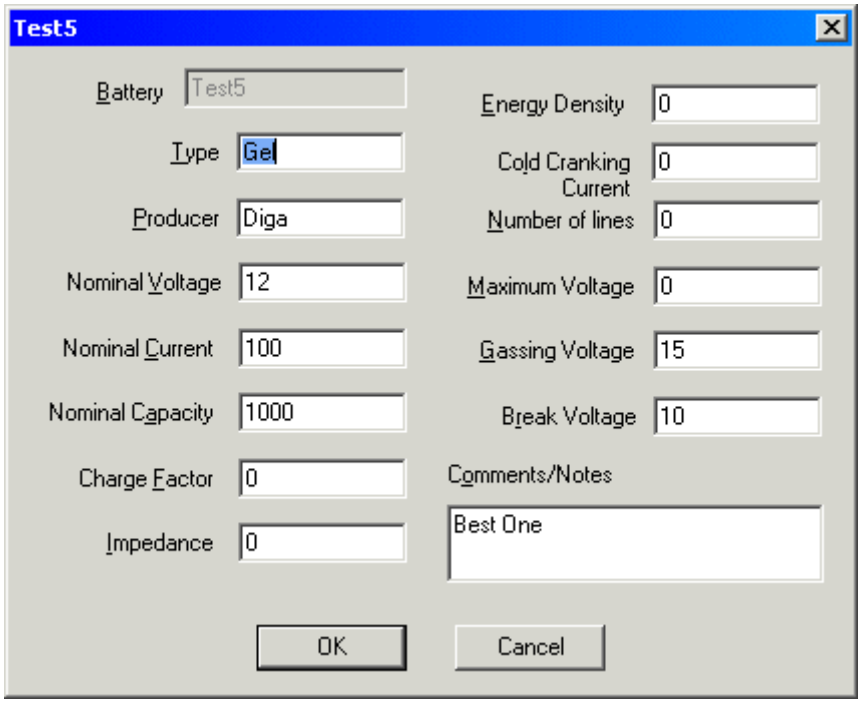
The user has access to Battery Maintenance if the Battery Maintenance privilege has been set in Password Maintenance.

Battery List

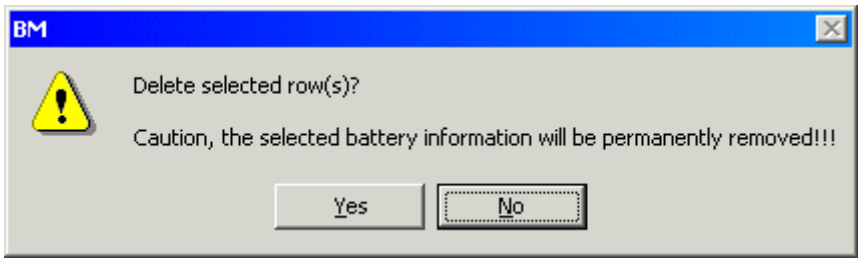
The Battery List is a grid reflecting the battery table within the Battery Manager databases. Each column in the grid corresponds to a field in the **Edit Battery** dialog (see below). The Battery List is read-only. Use the control buttons on the right side of the grid for additions, changes, and deletions. Additionally, most columns from the Battery List can be edited directly after a double-click.

1. **Edit Selected Battery** – Press this button to invoke the **Edit Battery** dialog with the currently selected battery.

2. **Add New Battery** – Press this button to invoke the **New Battery** dialog.



3. **Delete Selected Batteries** – Press this button to delete currently selected batteries. The system displays a warning dialog:



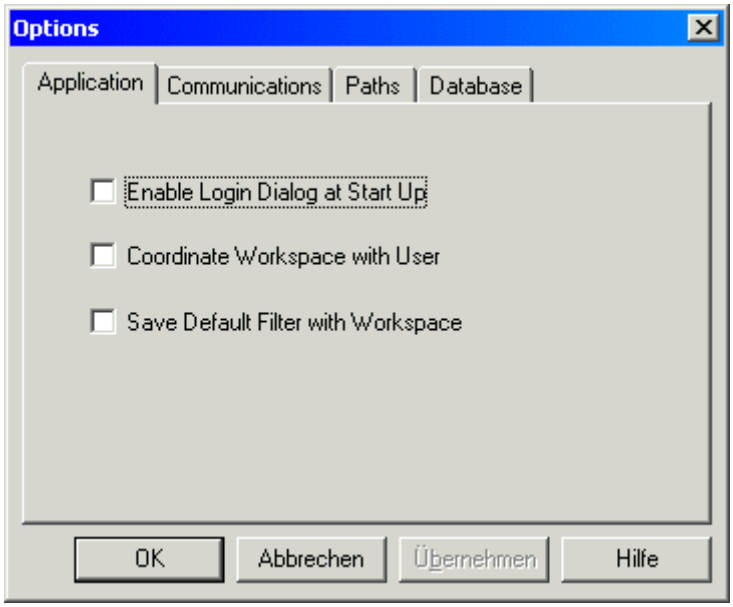
Click **Yes** to delete the battery and associated information or **No** to not delete. Although the system deletes the program assignment information, the associated programs are not deleted.

8 Maintenance Options

Maintenance Options are customer-configurable options available from within Battery Manager. Typically, the system stores these options in the registry, making them computer specific.

Typically, Maintenance Options are unavailable unless logged in as a system administrator. To invoke the Maintenance dialog, first log in as administrator (Supervisor). Then choose **Options** under the **Maintenance** item on the main menu. If the **Options** item is grayed and unavailable, contact Digatron for log in information.

The following dialog appears:



The only available options are under the **Application** tab:

1. **Enable Login Dialog at Start Up** – Check this option to force the Login dialog at application startup and user log off before allowing access to Battery Manager Workstation. This option works in conjunction with the **Enable Rights at Startup** option under the **Department Properties** dialog and the **Inactivity Period** setting under the **User Properties** dialog. For more information see Enhanced Capabilities.

With **Enable Login Dialog at Start Up** checked and **Enable Rights at Startup** checked Battery Manager Workstation is governed by the rights of the department (whose **Enable Rights at Startup** is checked) as normal. The only difference is that the Login Dialog appears at application start and whenever a user is logged off either by the system (because of the Inactivity Period setting) or because the user actually logged off.

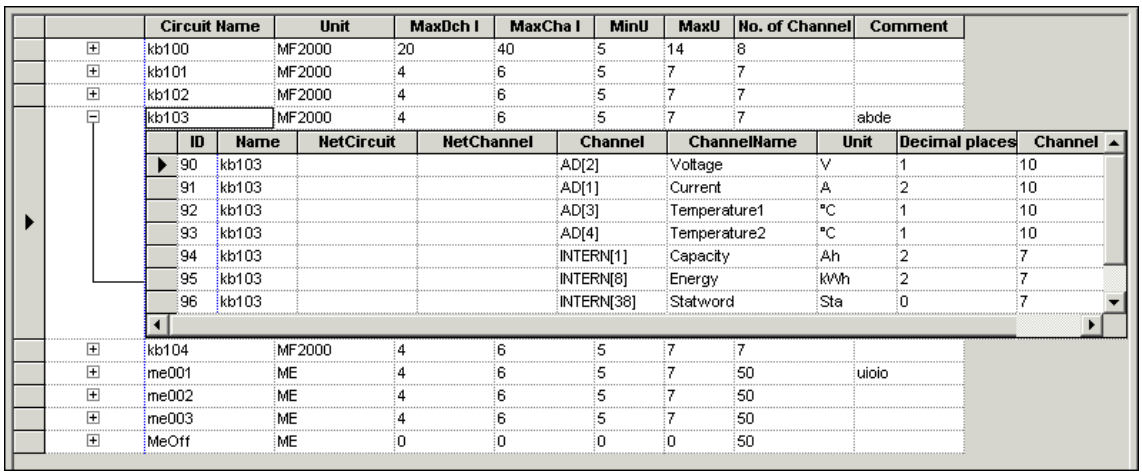
However, if the **Enable Rights at Startup** option is not checked for any department, then the Battery Manager Workstation application automatically exits if Login fails or if a user is logged off.

2. **Coordinate Workspace with User** – Check this option to cause the system to coordinate the workspace environment file with user Login. This allows each user to customize his or her view and return to the same view later when he or she logs in again. Logging out or not logging in displays the default workspace.

3. **Save Default Filter with Workspace** – this option is not yet implemented.

9 Circuit View

The **Circuit View** dialog is the central interface for the measuring data acquisition. Here you define which measuring channels are assigned to which circuits. To invoke Circuit View, select **Circuits** from the **Circuit** menu.



Circuit Name	Unit	MaxDch I	MaxCha I	MinU	MaxU	No. of Channel	Comment
kb100	MF2000	20	40	5	14	8	
kb101	MF2000	4	6	5	7	7	
kb102	MF2000	4	6	5	7	7	
kb103	MF2000	4	6	5	7	7	abde

ID	Name	NetCircuit	NetChannel	Channel	ChannelName	Unit	Decimal places	Channel
90	kb103			AD[2]	Voltage	V	1	10
91	kb103			AD[1]	Current	A	2	10
92	kb103			AD[3]	Temperature1	°C	1	10
93	kb103			AD[4]	Temperature2	°C	1	10
94	kb103			INTERN[1]	Capacity	Ah	2	7
95	kb103			INTERN[8]	Energy	kWh	2	7
96	kb103			INTERN[38]	Statword	Sta	0	7

kb104	MF2000	4	6	5	7	7	
me001	ME	4	6	5	7	50	uioio
me002	ME	4	6	5	7	50	
me003	ME	4	6	5	7	50	
MeOff	ME	0	0	0	0	50	

Each line of the list contains the circuit information for one test circuit. The main view will show you from left to right the circuit name, the unequivocal ID of the circuit in the system, the referenced device type, the maximum charge and discharge current, the maximum and minimum voltage and the number of channels of the circuit. The last column can be used for a comment.

Description of the Circuit View for BTS-600 systems:

In the column **Device** you can change various settings for this device type by clicking an entry followed by clicking .

Clicking a cross ⊞ left from a circuit name displays additional information on the channels of each circuit. These parameters are:

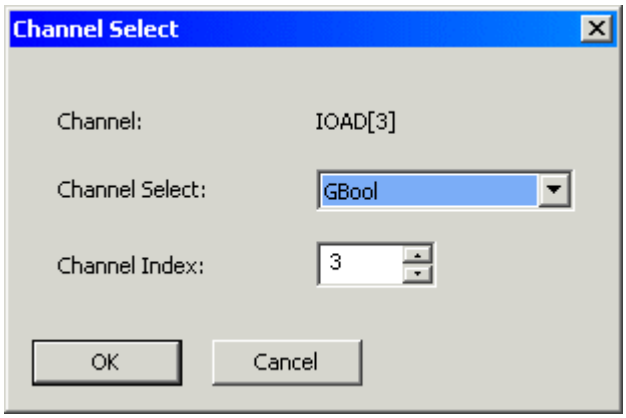
The **ID** is an ongoing number for all channels.

The **Netcircuit** column identifies other circuits by their name entered as they are registered in the circuit definition. Enter information in this column if you want to use measuring channels of a different circuit.

Column **Netchannel** contains information on which channel of another circuit should be linked to the present one. Enter here the physical name of the respective channel as it is defined in the circuit definition.

The **Channel** column contains the physical name for each channel which is exclusively used in the **Attribution circuit/channel** function. The names are automatically generated by BTS-600. They identify the channel by a descriptive term for the channel type plus an address, e.g. AD[1] for A/D converter channel No. 1, INTERN[3] for internal channel 3. Internal channels supply values that are not measured, but are calculated from the measuring values such as amp-hours, watt-hours, etc. It is not possible to edit the names contained in the **Channel** column.

When a line is empty you can create a new channel by clicking the **Channel** column followed by clicking . Select the channel and the belonging index from the appearing dialog:



The columns **Netcircuit** and **Netchannel** offer lists for selection, too.

The **Channel Name** represents the logic name of a channel, as it is used in any dialog. All titles for measuring data columns result of the entries made in this column. The names entered in this column can be freely selected. Press **ENTER** to confirm your entries.

The **Unit** column contains the physical or logical unit of the measuring channel. Physical units are used for instance for the measuring units (Current = A; Voltage = V) or for the units calculated there from (Capacity = Ah; Energy = Wh). If several measuring channels have the same physical unit (e.g. Ah and AhDCH), select logical units which clearly identify these channels. Certain channels, as for instance switching outputs, do not have a physical unit. In this case, the respective channel can be defined by 0 for "off“ and 1 for "on“. Use clear descriptions for logical units such as Relay_1, Valve_A, etc. The units defined in this column will be used in your test programs. The commands set in program editor columns **Nominal Value**, **Limit**, and **Registration** operate the individual measuring channels via their units.

The precision with which the respective measuring data are recorded differs subject to the preset measuring unit. We would therefore recommend you to adapt the **Decimal places** to the measuring range. Reducing the decimals displayed on the screen will make your display considerably clearer. For example, to reduce the decimals of the temperature values to one, open the corresponding field and enter 1.

The recorded measuring values may be treated with certain mathematical functions which will act as **Filters** to get the best possible display of measuring values.

The remaining columns show values that are used by the system internally, e.g. the version number.

10 Dispo List

The dispo list is only available for BTS-600 systems. The dispo list displays an overview of all circuits that have a dispo entry with all related parameters. the dispo list can be launched by clicking the  icon from the Shortcut Bar.

In addition, the dispo list is automatically displayed whenever you start a circuit from a BTS-600 system.

Circuit	Battery	Program	Test Section	Batch	ActiveTime	DeferredTime	StartTime	StopTime	Error
ME001	Batt00001	test	TS000001		6/2/2005 4:24:08 PM	6 / 2 /2005 4 :24:06 PM	6/14/2005 6:51:01 P	6/14/2005 7:16:20 PM	0
ME021	Batt00010				1/1/1970 1:00:00 AM	6 / 2 /2005 1 :40:19 PM	1/1/1970 1:00:00 AM	1/1/1970 1:00:00 AM	0
ME002	Batt00006	test	TS000001		6/2/2005 5:04:15 PM	6 / 2 /2005 5 :00:50 PM	6/2/2005 11:06:38 P	6/2/2005 11:33:21 PM	0
ME022					1/1/1970 1:00:00 AM	1 / 1 /1970 1 :00:00 AM	1/1/1970 1:00:00 AM	1/1/1970 1:00:00 AM	0

The parameters have the following meaning:

The column **Circuit** shows the individual circuits. The following three columns contain battery name, program name and the name of the associated test section.

In column **DeferredTime** you may enter a date at which you want the test to be started. The date should be entered in format DD.MM.YY (day.month.year). The time should be entered in the 24-hour format HH:MM:SS (hours:minutes:seconds).

The three times for **Active**, **Start** and **Stop** will be entered automatically by BTS-600 when the respective values are reached.

The lines in the dispo list are displayed in three different colors to represent the three different statuses of a test - namely **inactive**, **activated**, and **started**.

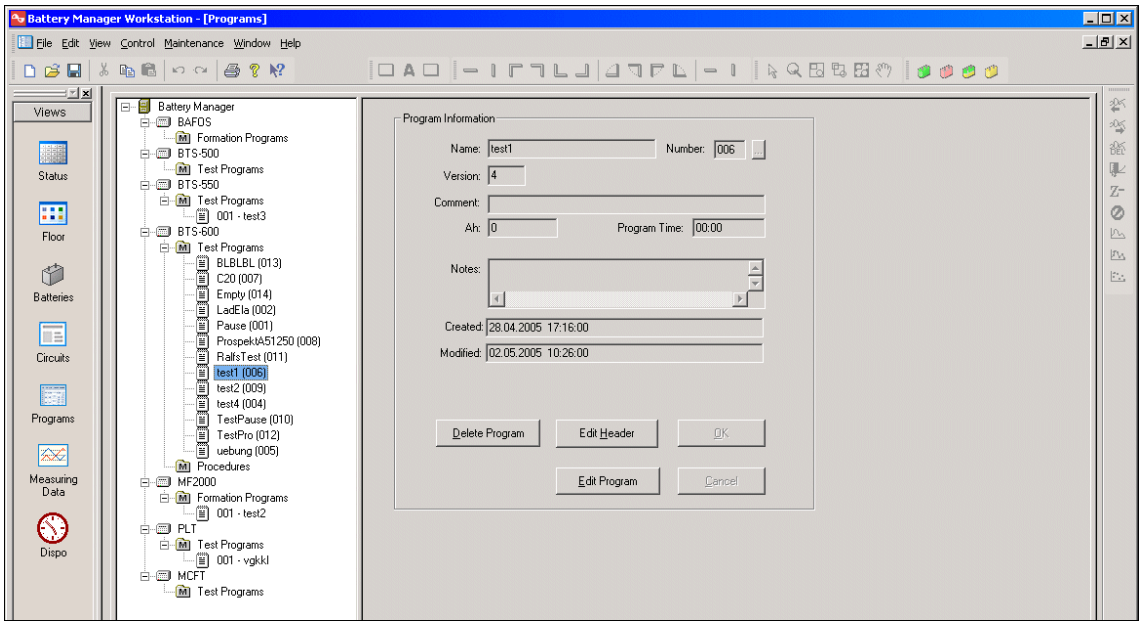
When you have entered a new test, it will first be inactive. The line will be displayed in **black and white**. Changes are still possible. The columns **Battery** and **Program** offer selection boxes.
When you activate a test, the color of the line display will change to **yellow**.
When a test is started, the respective line will be displayed in **red** color.
When a test is completed, the display will change from **red** to **green**.
This will help you to recognize immediately which tests are currently processed on the test circuits.

After selecting a line by clicking the very left column, a right-click inside the dispo list offers the possibility to add or remove rows as well as e.g. deactivate, interrupt or continue entries.

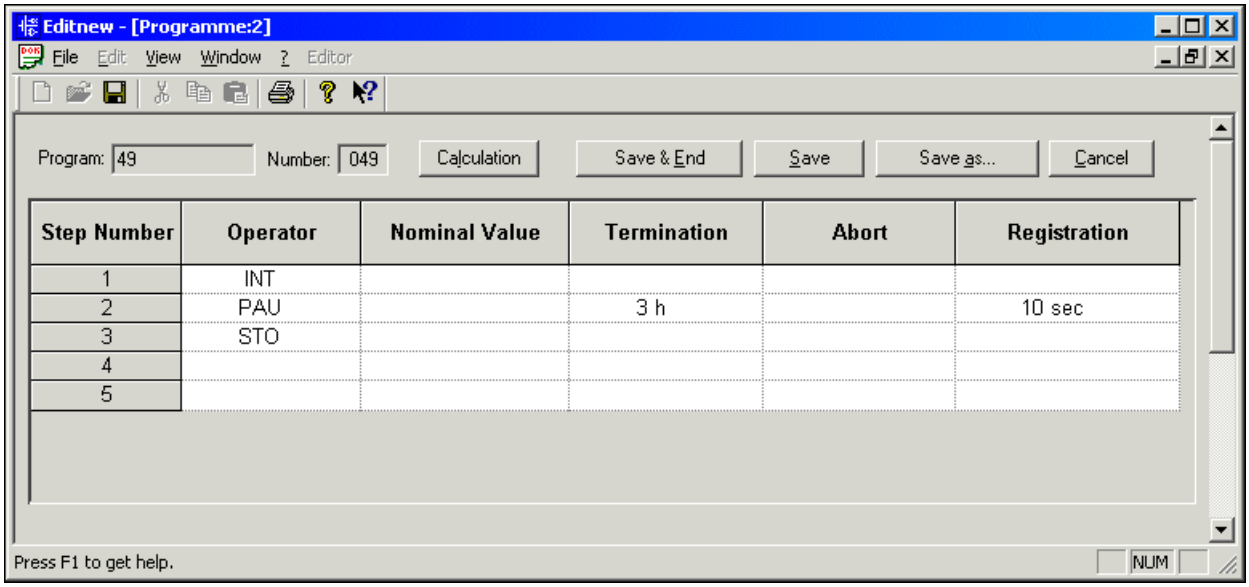
11 Programs

11.1 General Operating Instructions

To see a list of all programs, click the **Programs** entry from the menu **Maintenance** or click the **Programs** icon  from the Shortcut Bar. You then see a listing of programs from all your systems. Click a program from the list once to see more detailed information.

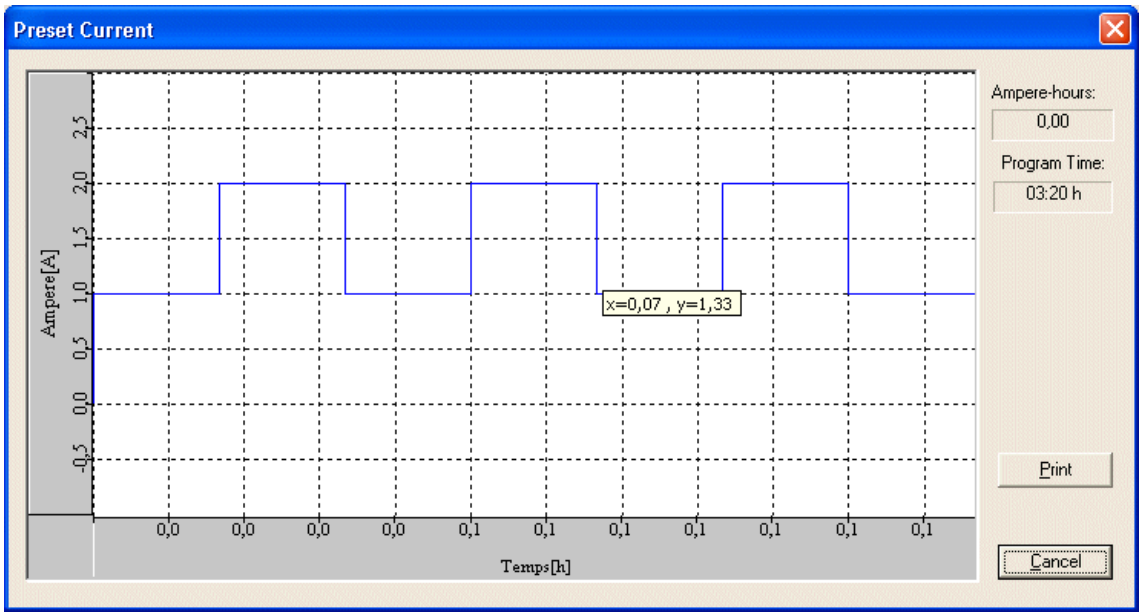


To start **Program Editor**, double-click a program name on the left or highlight the program name and click the **Edit Program** button from the **Program Information** window. There appears a new window where you can view and edit the program:



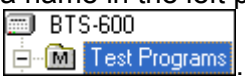
To return to the listing of programs, you can switch between windows in the **Window** menu.

For some systems you can click the **Calculation** button to see a graphical representation of the program's function. The following picture displays the current on the Y-axis while time is represented by the X-axis:

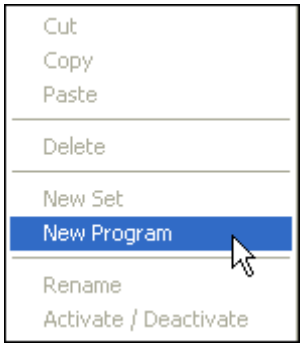


By right-clicking the diagram and then clicking **Properties** from the appearing menu the graph can be adapted to your requirements.

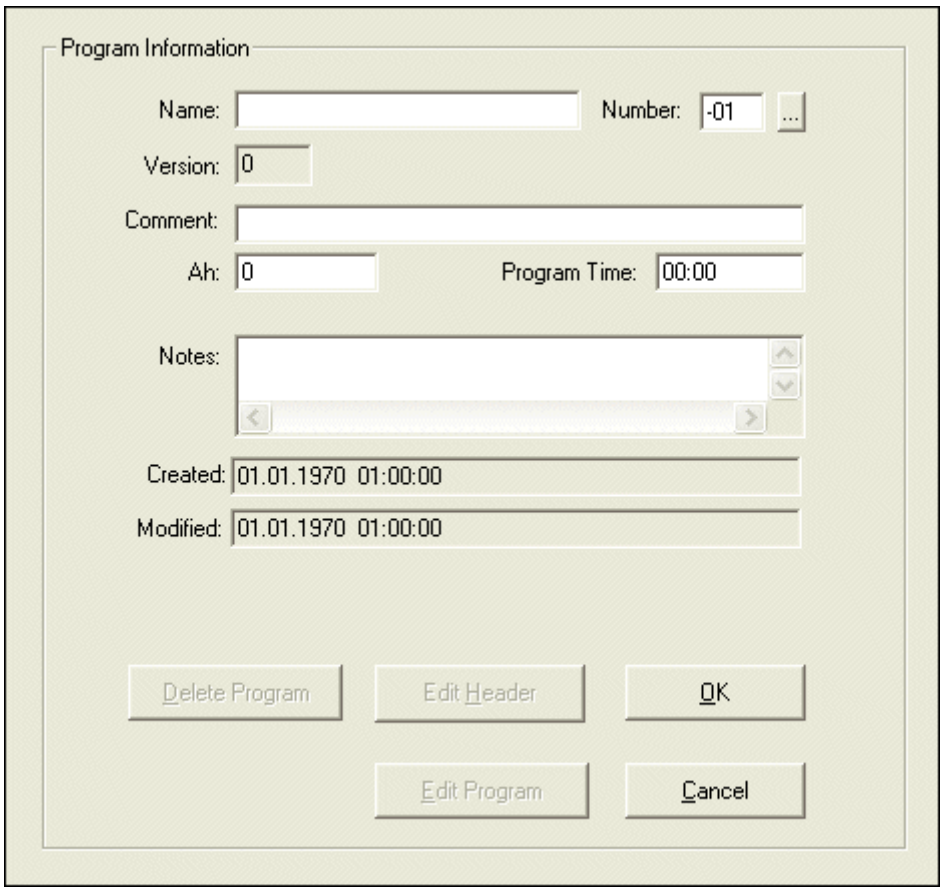
Create a new program

To create a new program, right-click a name in the left part of the window next to the symbol  to which the program shall belong, e.g. 

. In the appearing menu, click **New Program**:



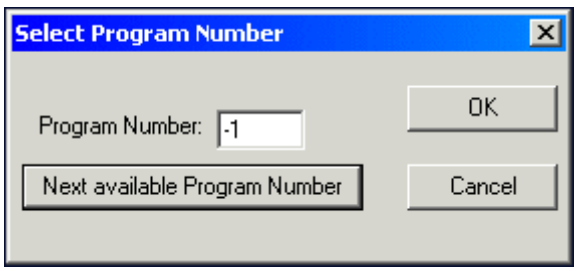
Now you have to enter a name for the program in the window **Program Information**.



Note

When creating test programs, make sure that the program names are unequivocal! If you are working with different battery types, **consider that the program name must clearly show for which battery type this program was written!** Confounding test programs may have fatal consequences on your batteries.

Select a free program number by clicking  .



When clicking the button **Next available Program Number** a free number is chosen by the program itself.

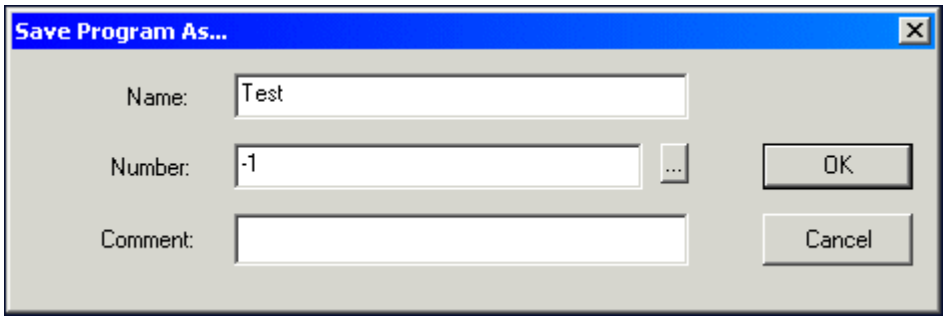
The program is listed in the left part of the window depending on its program number.

Save the program by clicking **OK**.

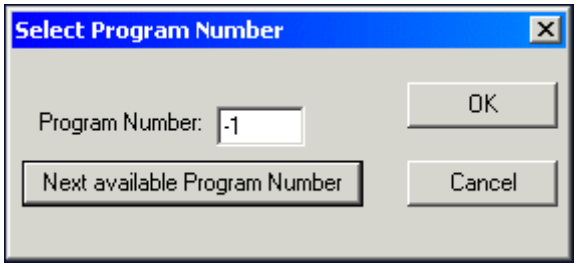
You may edit the program with the Program Editor.

Storing Programs

To store changes, click one of the **Save** buttons. In the **Save Program As** dialog you must enter a name and a free program number for the storage of the program.



After clicking  you have the possibility to enter a new program number:



When clicking **Next available Program Number** a free number is chosen by the program itself.

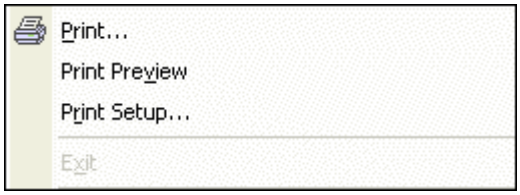
The **Cancel** button in the main window terminates **Program Editor**. In case you haven't saved all previous changes, you have to confirm whether or not you want to save the program now.

Deleting Programs

You may delete a program from the overview of programs by right-clicking the program name in the left part of the window and selecting **Delete** from the appearing menu.

Printing programs

You have the possibility to print the currently displayed program. Therefore you can find several entries in the **File** menu:



With **Print** you get directly to Windows' printing dialog as known from many programs. The entry **Print Setup** enables you to change the printer's properties. **Print Preview** shows the arrangement of the current page the way it will appear on paper.

At the top of the Print Preview window you find several buttons:



Use the **Print** button to go to the Windows printing dialog. The **Next Page** and **Prev Page** buttons jump to the next or previous page (if available). Click the **Two Pages** button to show two pages at once. You can then use the same button to go back to the single page view.

The **Zoom In** button enlarges the illustration by a step. You can do the same by clicking anywhere on the document with the mouse pointer that turns into a looking glass . If there is no transition into a looking glass you have reached the greatest magnification step. The **Zoom Out** button works just the opposite way.

Use the **Close** button to exit the Print Preview. In comparison, clicking the button at the top will terminate the whole evaluation program.

11.2 Programs for different systems

Battery Manager enables you to create programs for various application cases and systems. Depending on the system (e.g. for test, formation or laboratory) the procedure for writing programs might differ.

When you double-click a program it is shown in the program editor.

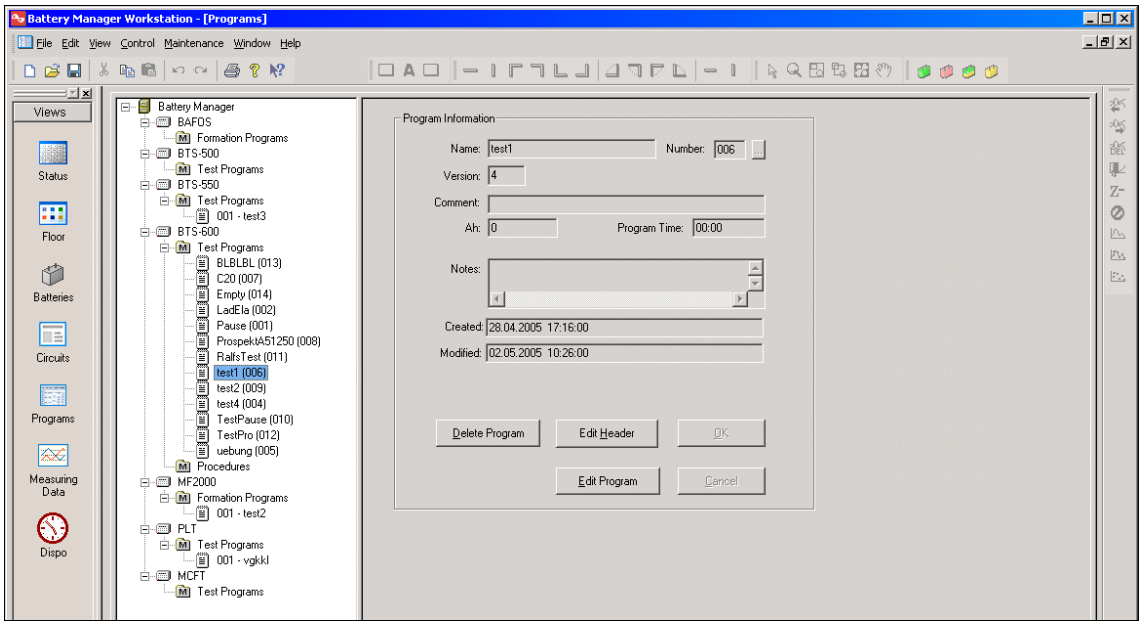
12 BTS-600 Programs

This chapters contains a detailed description of the BTS-600 Program Editor.

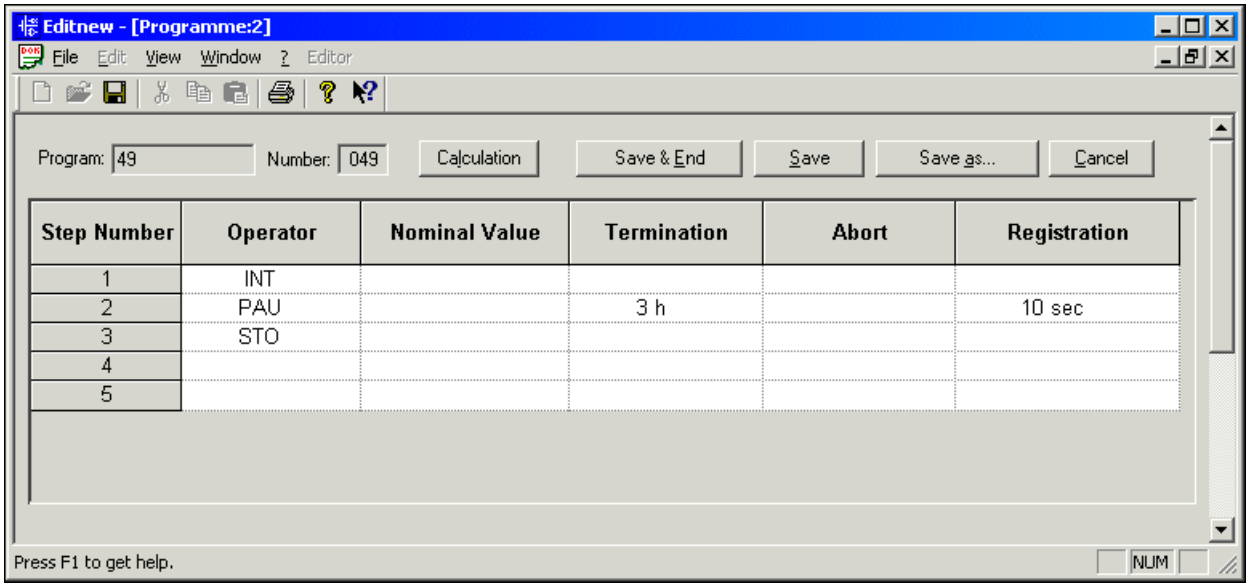
The **B**attery **T**est **S**ystem **BTS-600** provides extensive possibilities for testing, analyzing and evaluating batteries in a laboratory.

12.1 General Operating Instructions

To see a list of all programs, click the **Programs** entry from the menu **Maintenance** or click the **Programs** icon  from the Shortcut Bar. You then see a listing of programs from all your systems. Click a program from the list once to get more detailed information in the right part of the window.



To start **Program Editor**, double-click a program name on the left or highlight the program name and click the **Edit Program** button from the **Program Information** window. There appears a new window where you can view and edit the program:



You may as well create a new program (see chapter 12.1) and open it with a double-click.

12.1.1 Structure of the Editor

Columns

Each program consists of lines which, in turn, are organized in several columns. The number of columns depends on the currently utilized system.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		CHA	20.0 A	1.0 h	GOTO End	1.0 V

Step

Step
1

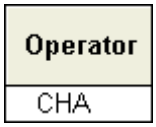
The first column shows the program step number. The consecutive number is generated automatically.

Label

Label

The second column will usually remain blank. GOTO commands are entered here in the form of a "label“.

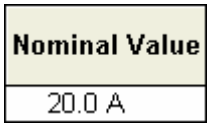
Operator



The third column contains the test operators. They indicate which type of test step should be performed (charge, discharge, pause, etc.). If you combine several operators to one procedure, enter the procedure name in this column.

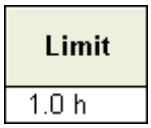
Find out more about operators in the chapter 12.2.2.

Nominal Value



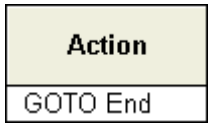
Most operators require nominal values - also called functions - such as charge (operator) with 10 A (nominal value). Enter your control settings here. A nominal value is always defined by numerical value and unit.

Limit



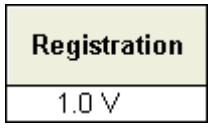
The fifth column shows **when** the test step will be terminated, namely if the limit entered here is reached.

Action



Enter here **how** BTS-600 should terminate the test step. This action will be performed as soon as the limit entered in the same line is reached. You can control the program process by entering different actions (next step, stop or interruption, GOTO command, load procedure). Leave this box blank if the program should continue processing the next test step.

Registration



The last column determines when, in which intervals, and under which conditions the measuring data are recorded and stored in the measuring data file.

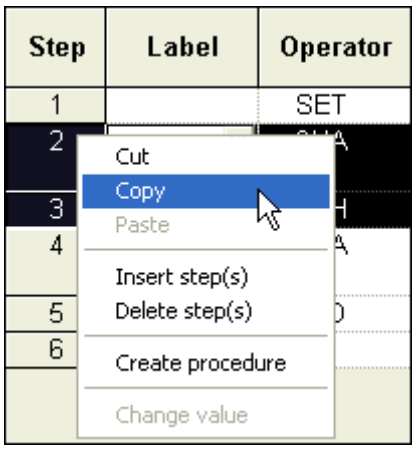
Find out more about **Registrations** in the chapter 12.2.1.

12.1.2 Editing Functions

Knowing how to use the editor most effectively will save you a lot of work and time. You can, for instance, avoid entering existing program parts by using the copy functions.

Copy

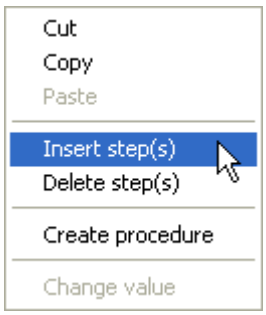
You may copy program steps and insert them elsewhere. To do this, Press the left mouse button while moving the mouse pointer along the **Step** column and thereby highlight the lines you want to copy. Only successive program steps can be copied together. When the steps are highlighted, right-click the area within the **Step** column and pick **Copy** from the appearing menu.



You can enter copied steps either elsewhere in the same program or in another program of the same system. To do this, go to the program and highlight the step where you want to insert the copied steps. Then right-click the program code and select **Paste** from the menu. The steps from the clipboard will be inserted above the highlighted step.

Insert steps

You can add new steps to a program by right-clicking the line after which you want to add the steps. Select **Insert step(s)** from the context menu:



There appears a dialog where you can define the number of steps to add.

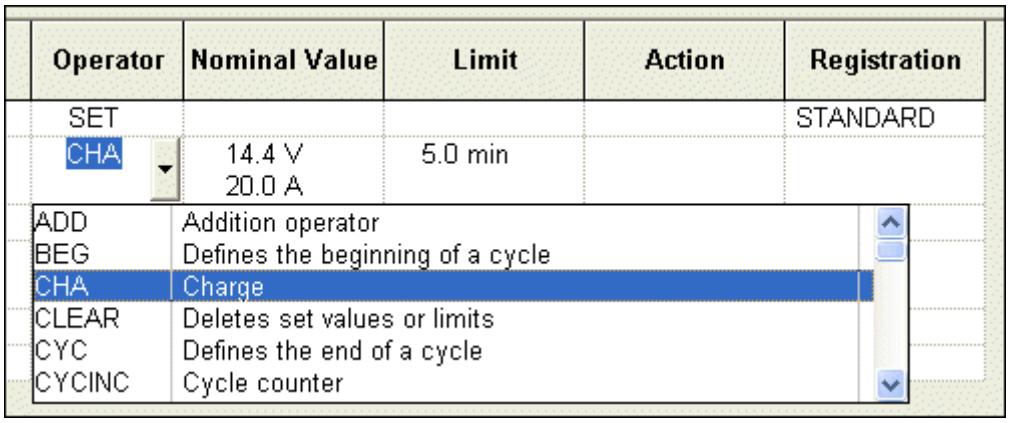


If you now want to insert steps from the Windows clipboard, right-click the step above which you want to add the steps and select **Paste** from the appearing menu.

Edit

To change a field's value, click into this field. A button or then appears. Clicking one of these buttons will call an input assistance.

For the **Operator** column there appears a list of all available operators. Please note that some operators might not be supported by your hardware.



Select the operator by clicking an entry. Find out more about operators in the chapter **Operators**.

The other columns might show a list of possible values depending on the kind of program, too when clicking into the field and then clicking the button. Select a value from the list or enter a value yourself.

Delete

You can either delete parts of a field, a complete field or one or several lines from the program editor.

To edit an individual field, first open it by double-click. Then delete its content by actuating the BACKSPACE or the DEL key.

If you want to delete a field completely, move the cursor to this field (without opening it) and actuate the SPACEBAR or DEL key. The field will be opened and its entire content will be deleted at the same time. You can now make new entries. If you do, however, decide that you had rather left this field unchanged, simply press the ESC key to restore the previous status. You can also click the UNDO button to take back changes step by step.

You can also delete one or more program steps by highlighting the steps, right-clicking them and selecting **Delete step(s)** from the appearing menu.

12.1.3 Comment Lines

BTS-600 permits you to add comment lines directly into the test program. A few comments already help to make comprehensive or complicated test programs much more clear and easier to understand.

To insert a comment line, move the cursor to the column **Label**. Now enter an exclamation mark **!** to insert a comment line. Confirm the input of an exclamation mark not by pressing ENTER but by using the arrow keys → or ↓.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1	!	Comment				...
2		SET				STANDARD
3		CHA	14.4 V 20.0 A	5.0 min		
4		DCH	21.0 A	< 9.0 V		0.01 V
5		CHA	14.4 V 20.0 A	16.0 h		
6		STO				

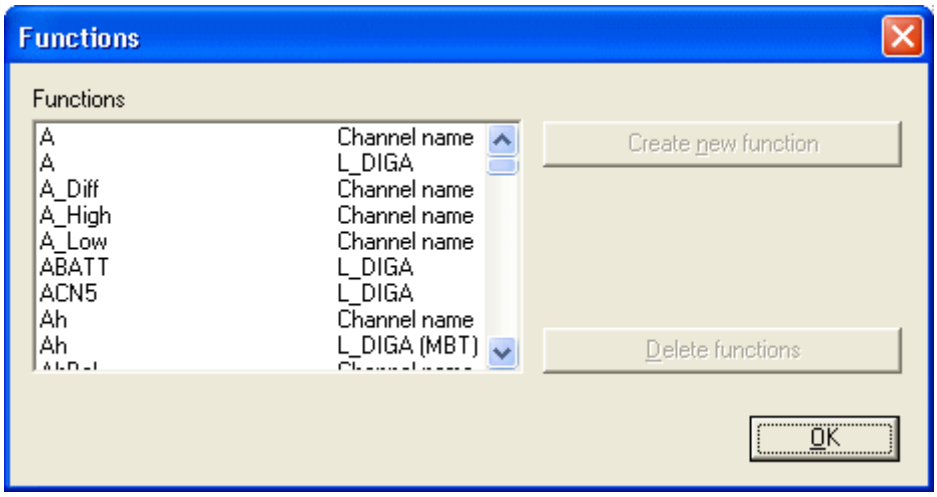
To enter text into the comment field, use the ... button or click into the field directly. You may add comment lines anywhere throughout the program by right-clicking the column **Step**, selecting **Insert step(s)** from the appearing menu and entering an exclamation mark.

If you want to enter more than one line of comment, you have to insert several commentary lines and edit one by one.

When finished writing a comment line, click the button ... to store the text.

12.1.4 Help Dialogs

To facilitate the creation of test programs, the program editor provides help dialogs. When you click the button or in the columns **Operator**, **Nominal Value**, or **Registration**, a dialog will appear that shows you all valid selections in alphabetic sequence.



You will merely have to select the corresponding list entry by clicking it. Confirm your selection by pressing ENTER or by clicking the **OK** button. The system will than replace the original entry by the right one.

Note
You can, of course, use this help dialog for programming purposes by displaying all valid selections. You will frequently not know which elements exist already or which have already been defined. The help dialogs will give you a quick survey and help you to take your decisions!

12.2 How to Create a Test Program

Important Note:

1. **Entries in screen masks:**
   All entries in the screen masks of this manual are intended as examples only. It is, therefore, useful and necessary to make your entries in compliance with your specific applications.

2. **Security criteria in programs:**
   The limit criteria - e.g. limits, actions, etc. - in a program should not be the only security measure in a test.

3. **Security measures to be taken by the user:**
   Damages to batteries, test units or peripherals due to a failure must be prevented by taking appropriate security measures. This is particularly important in the case of unattended test processes!

4. **Responsibility of the user:**
   It is in the user's sole responsibility to ensure a safe application of the program functions available in the BTS-600 system.

Now try to create your first test program. The following explanations will help you, step by step, to make correct and clear entries. This chapter is written in the sequence of the necessary operating steps.

After clicking the **New** button you will be in the program library. You will already have entered the program information (name, title, and comment).

12.2.1 Registration Formats

General

The first step to create a test program is to determine a registration format. You can determine here which of the existing measuring channels will be registered. The Registration format defines in which sequence BTS-600 registers the measuring data of which physical or logical channels. The *registration rate* can be set in column **Registration**, which is described in the following. The registration format setting is essential for the test evaluation. If you register more channels than necessary, the space required on your hard disk will increase considerably. If only few channels are disposable for evaluation, you will lack important information. BTS-600 therefore permits you to adapt and optimize the registration format to your individual requirements.

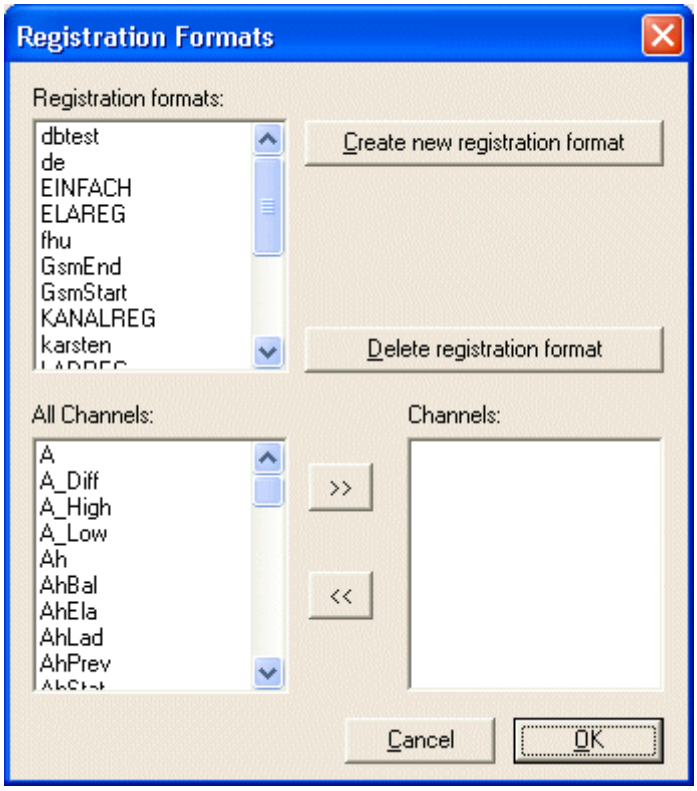
There is **no** registration format used per standard setting. i.e. only measuring values for current and voltage are recorded in the test section. In most cases this information will be a little poor, therefore it's a good idea to start a test program with the definition of a registration format.

Existing Registration Formats

Enter the **SET** operator in the third column **Operator**. The **SET** operator defines not only counters but also registration formats. The format used has to be entered in the **Registration** column. Click the field where you want to enter the registration and then click the appearing button ...



A help dialog will open.



Here you find all registration formats included in BTS-600. Make your selection from the list **Registration formats** and click **OK**.

The following table shows preset formats, which channels are registered with these formats as well as the memory they require per registration.

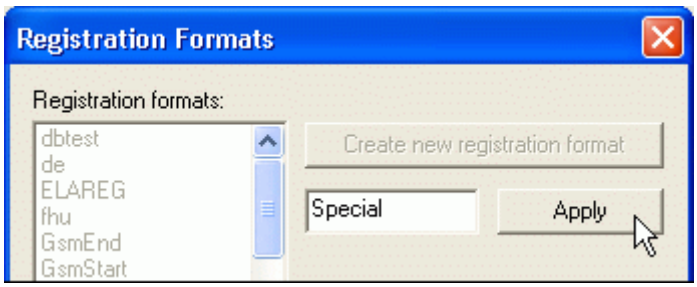
Name	Channels	Memory Required per Registration
SIMPLE	A, V	12 byte
STANDARD	A, V, C, Ah, AhStep, Wh, WhStep	42 byte
CHAREG	A, V, C, Ah, AhCha, AhStep, Wh, WhCha, WhStep	54 byte
DCHREG	A, V, C, Ah, AhDch, AhStep, Wh, WhDch, WhStep	54 byte
CHANREG	A, V, C, Ah, AhStep, Wh, WhStepV, V1 (= all channels of loggers V1),	42 byte + Number of Channels *6 byte
CYCLE	A, V, C, Ah, AhBal, AhStat, AhCha, AhDch AhStep, AhPrev, Wh, WhCha, WhDch, WhStep, WhPrev	90 byte
CYCLEC	A, V, C, Ah, AhBal, AhStat, AhCha, AhDch, AhStep, AhPrev, Wh, WhCha, WhDch, WhStep, WhPrev, V1	90 byte + Number of Channels *6 byte

Apart from physical channels (measuring data) A (current), V (voltage), and C (temperature) the system also records various logical channels. View the following table to find out which counter is linked with which channel.

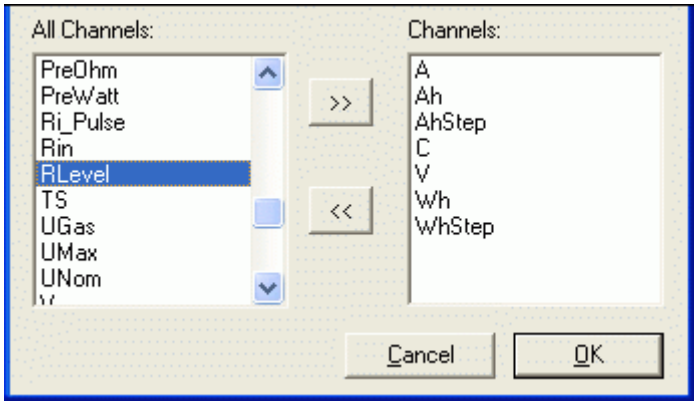
Channel	Description
Ah, Wh	up/down difference – may be < 0
AhStat	up/down difference of the amp hour balance, considering the charge factor
AhBal	difference (total Ah charged) – (total Ah discharged)
AhStep, WhStep	Counter for the actual test step
AhPrev, WhPrev	Counter for the preceding test step
AhCha, WhCha	Counter for all charge steps
AhDch, WhDch	Counter for all discharge steps

New Registration Formats

To create a new registration format, click the button **Create new registration format** from the dialog above, enter a name for the format and click **Apply**.



You may select the channels that are recorded with the new registration format by highlighting them in the list **All channels** and moving them to the right using the >> button:



Modify Registration Formats

To optimize registration you can also modify existing registration formats. For this purpose, select a registration format from the dialog **Registration Formats** and add channels with the >> button or take channels away with the << button.

Check whether the modifications are correct and store them by clicking the **Ok** button.
If you decide that you would rather leave the format in its original status, choose the **Cancel** button to return to the editor.

Note

Please take into consideration that registration formats are used for different programs. **Modifications will, of course, affect all test programs using this format**. In case of doubt, rather create a new format!

Change Registration Format

The SET operator at the beginning of a test program defines a registration format which is valid until you select another format by setting a new SET operator. To permit flexible reaction to existing testing conditions, you can change the registration format at any place in the program.

Caution!

If you use sub-programs - so-called procedures - in your test programs, you can, of course, also change the registration format there. These modifications are, however, only valid on the level of this sub-program. When the program returns to the main menu level, the registration formats defined for this level will apply!

To change the recorded channels for only one test step, change the registration format without defining a new SET operator. Enter the name of the requested registration format under column Registration in the corresponding line of the test step.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		DCH	10.0 A	10.0 sec		STANDARD

Please bear in mind that your modification will only apply to this test step! The next test step will be processed with the format defined by the SET operator.

How to Activate and Deactivate the Registration Function

Activate and deactivate a registration function by using the **RLevel** variable. The **RLevel** variable may be used to select the type of registrations that are recorded. There are two basic types of registration records that may be written to the hard disk, i.e. user defined data registrations and step change registrations. The data registrations include all the data the user has defined as the registration format. The step change registrations are primarily for indicating the beginning and end of a step. Step change registrations do not include all the user-defined data, e.g. cell voltages, temperature, pressure, etc.

The **RLevel** variable may be modified at any time. The following settings are possible:

RLevel = 3.0 no step change registrations (typically, a data registration is still recorded at the end of a step)

RLevel = 2.0 all registrations; i.e. data and step change registrations.

RLevel = 1.0 step change registrations only, no data registrations.

RLevel = 0.0 no registrations

To **deactivate** the registration, enter the **SET** operator in column **Procedure** and **RLevel = 0** in the **Nominal Value** column.
To **activate** the registration, use function **RLevel = 2**. Afterwards the original registration format is used again.

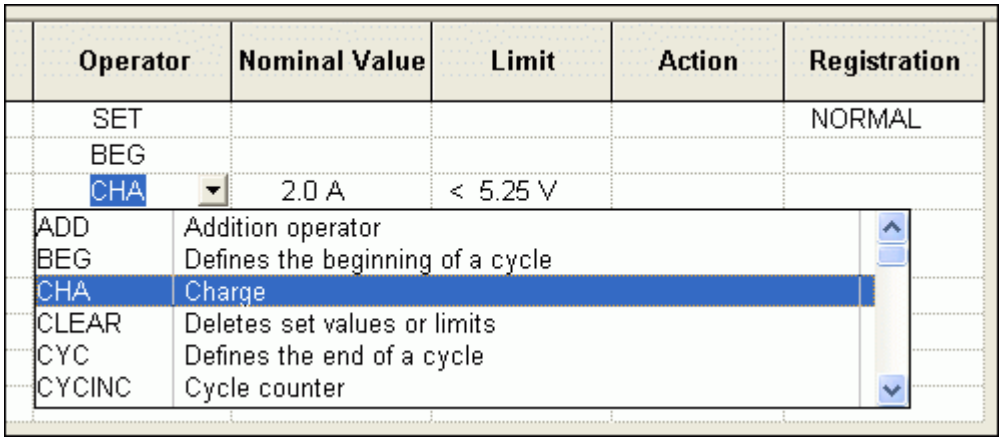
Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2	!	Registration will be switched off				
3		SET	RLevel = 0.0			
4	!	Registration will be switched on				
5		SET	RLevel = 2.0			

12.2.2 Operators

After you have determined the registration format, continue by programming the test steps. Move the cursor to the third column **Operator**. Enter here an Operator describing the function of the test step selected. For better clearness you may also enter the name of a procedure comprising several operators. For detailed information on procedures please see Procedures in chapter 12.3.2.

If you do not yet know which operators exist in the program, proceed as described for column **Registration**.

When clicking the button you activate the help dialog that lists all available operators and procedures. Select an entry by clicking on it.



Note

For safety reasons, we would strongly recommend you to enter a pause step of several seconds at the beginning of a test program. This will give you the opportunity of checking all test elements and cancel faulty tests in due time.

The following table describes all standard operators available with BTS-600.

You can have the system extended by further operators or procedures, e.g. for relay switching, polling additional inputs, controlling external outputs, etc. These customized options must be clarified with Digatron. They will not be considered in the following explanation.

Operator	To be used with column:	Description
SET	Nominal Value Registration	SET Operator: Sets registration formats, counters, etc. (also see chapter 12.2.8.1)
PAU	Limit	Pause Operator: Power contactors dropped until limit is reached.
CHA	Nominal Value Limit	Charge Operator: Charges with nominal value until limit is reached. - Charge relay is set.
RCH	Nominal Value Limit	Recharge Operator: same as charge operator In addition, the recharge relay is set. The charge LED on the front panel is blinking.

DCH	Nominal Value Limit	Discharge Operator: Discharges with nominal value until limit is reached. Discharge relay is set.
INT		Interrupt Operator: Interrupts the test program. To continue, use start button (front panel) or host computer (dispo list or tool buttons).
STO		Stop Operator: Terminates the test program; Can also be used in sub-programs. Therefore sometimes repeated several times in one test program.
BEG	(Nominal Value)	Cycle Beginning Operator: Defines the beginning of a cycle. Define cycle name in **Nominal Value** column.
CYC	Nominal Value	Cycle End Operator: Defines the cycle end. Nominal value indicates the number of cycles to be performed.
GOTO	Operator Action	Jump Operator: The jump destination, defined as procedure or action, must be entered under column **Label.**
REG	Nominal Value Registration	Registration Operator: Records the registration format once. (also see chapter 12.2.8.3)
SAVE		Save Operator: Saves all data of the actual registration format on the host computer hard disk.
REST		Restore Operator: Restores the data saved on the host computer.
TASK		Activating Operator: Activates parallel programs to control external processes: relays, temperatures, water levels, etc.
RET	Action	Return Operator: To return from a procedure into the main program or within the program to the next higher program level
SETMUX	Operator	Switch Operator: Activates the display of additional multiplexer channels in the case of serial data acquisition.
CYCSET	Nominal Value	Cycle Set Operator: Operator for presetting the "battery-referred" cycle counter with an initial value. Nominal value "0": no presetting. (See chapter 12.2.7, example 12)
CYCINC		Cycle Counter Operator: Adds one count to the host computer's battery-referred cycle counter in each step where this operator is entered. (See chapter 12.2.7, example 12)
FILE	Nominal Value	File name operator: Determines in the program the file name for the following registrations. (See chapter 12.2.8.12)
ADD	Nominal Value	Addition operator: adds the figure values of the channels entered in the nominal value column
BATT	Nominal Value	Battery Operator: (Battery Simulator) Preset of constant voltage and upper and lower current limit (see below).
ONERROR	Nominal Value	Label: See chapter 12.3.4
ONEXIT	Nominal Value	Label: See chapter 12.3.4

PROT		Protocol operator: Operator for starting a special program in the execution of BTS-600. The special program is entered in the file BTS600.INI in section [BTServer]. The entry reads SpecLogger.Exe = C:\Windows\notepad.exe.
RegOff		Switch operator: Operator to switch off the present registration. Valid for all following steps until RegOn. RLevel = 0.0 (see chapter 12.2.1).
RelOn		Switch operator: The switch output **IOOUT[13]** is set to the value "1.0" (⇔ ON). With this option it is possible to switch on an external relay. (optional)
RelOff		Switch operator: Operator to switch off the relay activated with the RelOn operator.
RegOn		Switch operator: Operator to switch off the present registration. RLevel = 2.0 (see chapter 12.2.1)
SUB	Nominal Value	Subtraction operator: Subtracts the figure value of the second channel entered in the nominal value column from the figure value of the first one.

How to use the operators **ADD** and **SUB**:

Step	Label	Operator	Nominal Value
1		SET	Sum = 50.0
2		ADD	Sum 5.0

Here the value 5 is added to **Sum**. The subtraction with SUB works exactly the same. **Sum** could be assigned with a unit name in the circuit/channel attribution list. **Sum** will change from 50 to 55 or 45 depending on ADD or SUB. If you want to move the result to another variable use SET like this:

Step	Label	Operator	Nominal Value
1		SET	Sum = 50.0
2		ADD	Sum 5.0
3		SET	Ah = Sum

After this step the variable **Ah** will contain the result of ADD or SUB.

How to use the operator **BATT** (Battery Simulator):

10		BATT	100.0 V 150.0 A -150.0 AG	1.0 h		1 min

Preset values for the BATT operator are the voltage and upper and lower current limits. The units of these preset values are as follows:
Voltage: V, upper current limit: A, lower current limit: AG.

In the above example a constant voltage of 100 V is defined. The current is regulated between -150 A and +150 A. Perhaps the use of a special TASK is required to adjust to the respective hardware.

Special Operators:

Operator	Description
FILTER	Permits to set digital filters for all channels in the **Attribution circuit/channel**.
TABLE	Loads a nominal value list from the hard-disk of the PC.
ERR	Initiates a message in line **System extern** of the main menu with interruption of the test program.
MSG	Initiates a message in line **System extern** of the main menu without interrupting the test program.
CLEAR	Deletes set nominal values or limits, if any.

Detailed explanations on above operators can be found here:

Operator "FILTER": chapter 12.2.8.11
Operator "TABLE": chapter 12.2.8.9
Operator "ERR": chapter 12.2.8.5
Operator "MSG": chapter 12.2.8.6
Operator "CLEAR": chapter 12.2.8.10

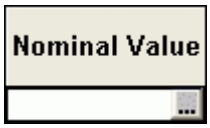
As already mentioned, you should insert a pause step after the SET operator. More details on how to insert a pause step will follow later in this manual.

After the pause step, enter a charge operator CHA. Your editor should now read as follows:

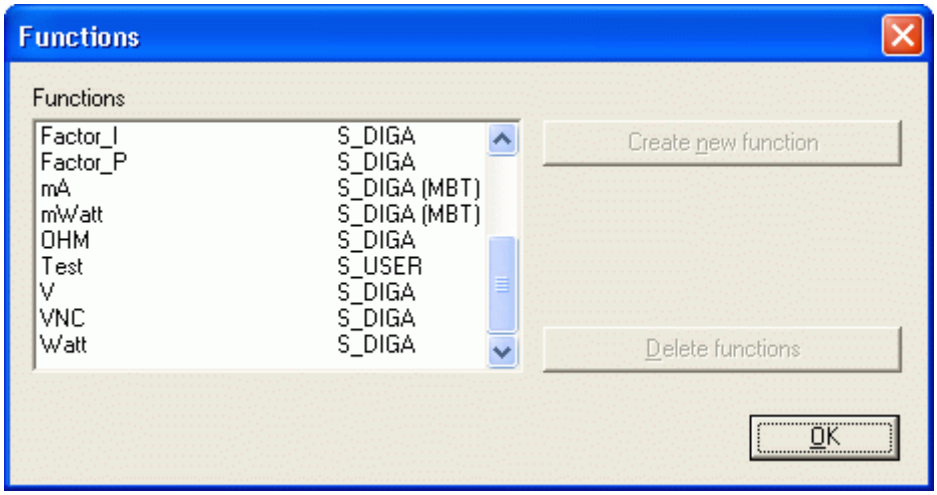
Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU				
3		CHA				

12.2.3 Nominal Values/Functions

When entering a charge or a discharge step, you will, of course, have to specify the conditions for charge or discharge. For this purpose, you will need functions consisting of numerical values and of a descriptive name or a physical unit. Move the cursor to the **Nominal Value** column. Activate the help dialog like in the **Operator** column by clicking the button ...:



The help dialog will appear showing all available functions for this column.



The selection list displayed contains two types of functions: standard functions (DIGATRON) and supplementary functions, which are marked with S_USER. Select a function by clicking it and copy it to the editor by clicking **Ok**. The buttons **Create new functions** and **Delete functions** are usually for use by Digatron personnel only because modifying the functions will have fundamental effects on the test process.

Please do not hesitate to contact Digatron for supplements to or extensions of these functions to meet your individual requirements.

The functions supplied with the BTS-600 standard software version are listed in the following tables:

Standard Functions

used in program steps for CHA, RCH, and DCH operators alike:

Function	Nominal Value - Example -	Description
Current	1.0 A	Current control at constant 1 amp
Voltage	14.4 V	Voltage control at constant 14.4 volts
Power	1000 Watts	Current and voltage control to a power of 1000 watts
Resistance	1 Ohm	Current and voltage control to a resistance of 1 ohm

Important:

In all program steps with CHA, RCH or DCH operators you always have to enter a nominal current, power or resistance! Never enter a voltage value only.

Supplementary Functions

The functions are based on the battery data provided by columns **Nom. Capacity** and **Cells** of the battery definition. These data are required for creating programs with battery parameters. The supplementary functions are also used for charge and discharge steps alike. They can, however, not be used for ramps.

Function	Nominal Value Example -	Description
5-hour discharge current, also named "C5“	10 ACn5	With a nominal capacity of e.g. 100 Ah - constant current control to 10* (100 Ah / 5 h) = 200 A.
8 ACN1		Current eight times the nominal capacity/ 1 as regulating quantity
8 ACN2		Current eight times the nominal capacity/ 2 as regulating quantity
8 ACN4		Current eight times the nominal capacity/ 4 as regulating quantity
8 ACN10		Current eight times the nominal capacity/ 10 as regulating quantity
8 ACN20		Current eight times the nominal capacity/ 20 as regulating quantity
Cell Voltage	2.35 VnC	With a cell number of e.g. 60 the voltage is regulated to 2.35 V/cell * 60 cells = 141 V.

Ramp Functions

Functions cannot only regulate to constant values, but also process ramp settings. You can define in a test step the initial and the end value of a ramp as well as the time in which the ramp should be executed. You can program either current, voltage, power, and resistance ramps or a combination of nominal ramp values, for instance a common ramp for current and voltage ramp.

Function	**Nominal Value - Example -**	**Description**
Ramp	1000- 5000 Watt 300 sec	Current and voltage control in a way that the power will increase from 1000 watts to 5000 watts within 300 seconds.

Parallel Function

You can combine several functions in one test step to set two nominal values. Depending on whether a charge or a discharge operator is set, the respective quantity will not exceed or fall below the nominal values.

Function		**Nominal Value - Example -**	**Description**
Constant Voltage/ Constant Current Charging	CHA	100 V 15 A	Controls voltage and current. Neither the nominal voltage nor the nominal current value may be exceeded. The charge current is regulated to constant 15 A until the voltage reaches 100 V. Then the voltage is maintained at 100 V, i.e. the charge current is reduced.
Constant Voltage Discharging	DCH	1000 Watt 80 V	Controls power and voltage. The power may not exceed the nominal power value and the voltage may not fall below the nominal voltage value. When, with a discharge power of 1000 watts, a voltage of 80 V is reached, the voltage remains constant.

Special Nominal Values:

Function	**To be used with column:**	**Description**
Factor_I	Nominal Value	Each value of a nominal current value table will be multiplied by the entered factor. See chapter 12.2.8.9 B.
Factor_P	Nominal Value	Each value of a nominal power value table will be multiplied by the entered factor. See 12.2.8.9 B.
OUW	Procedure	Activates the calculation of resistance and power with the entry "1.0 OUW" after the "TASK" operator.
CGRE	Procedure (Task)	Option: Operator to activate a temperature dependent relay through the output **IOOUT[16]** (⇔ Rel. 8).

CLESS	Procedure (Task)	Option: Operator to deactivate a temperature dependent relay through the output **IOOUT[16]** (⇔ Rel. 8).
TCONTR		Option: The switch output **IOOUT [16]** (⇔ Rel. 8) is used to switch a relay in dependence of the temperature.

The nominal values listed in the following table are exclusively valid for the operators mentioned below. For further details please refer to paras 12.2.8.3, 12.2.8.9 and 12.2.8.12.

Nominal Value	Description
File Name	File name of nominal value list in connection with operators **TABLE**, **REG** or **FILE**.
Filter Name	Functional name of the filter which you want to store under the respective number.
TIMER 1 to TIMER 3	Timers set and started by the **SET** operator.
MaxCHAI	General current limit for charge operations.
MaxDCHI	General current limit for discharge operations.
MaxChaU	General voltage limit for charge operations.
MaxDchU	General voltage limit for discharge operations.
MaxChaW	General power limit for charge operations.
MaxDchW	General power limit for discharge operations.
Number	Message number for operators **MSG** or **ERR**.

This multitude of functions guarantees highest flexibility in executing tests.

Now back to our program example.
Select current function **A** in the help dialog and return to the editor. You can, of course, also enter the shortcuts for these functions (A, V, Watt, Ohm, Acn5 and VnC) direct via the keyboard.

Note
Please do not enter "W“ to choose the power function (which is frequently done by users). The system will only accept "WATT“.

Your program should now read as follows:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU				
3		CHA	1.0 A			

Example for the Nominal Value TIMER:
In this example the program goes into STOP after 5 minutes:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	Timer1 = 5.0 min	Timer1	STO	

12.2.4 Limits

The column **Limit,** defines **when** and **under what condition** a test step is ended. As you can choose between a variety of conditions, you will also have to define which actions should be executed under what condition. This is why the **Limit** and **Action** columns are closely linked. The action is performed as soon as the limit set **in the same program line** is reached. When you select several limits for one test step, you can assign a different action to each limit.

The simplest way to set a limit is to set a fixed time. To do this, move the cursor to the **Limit** column in the second test step PAU. Here enter a time of several seconds. In this case, type in **30 sec**.



To meet the battery testing requirements, it should also be possible to terminate a test step depending on actual battery values. By making the respective entries in the **Limit** column, you can have all registration channels scanned. The limit is reached, when the respective value of the channel exceeds or falls below the value set in the **Limit** column.

The following table explains the possible entries in the **Limit** column.

Limit	**Example**	**Description**
Time	10 sec	The following action is executed when the set time is elapsed.
Lower Limit	< 10.0 K	The following action is executed when channel K supplies a value **less than** 10.0. You may select both physical and logical channels such as A, V, Ah, Wh, WhDch, etc.
Upper Limit	> 10.0 K	The following action is executed when channel K supplies a value **greater than** 10.0. You may select either physical or logical channels such as A, V, Ah, Wh, WhDch, etc.

When entering a time, please only use integer numbers with decimals or hh:mm:ss.

Example:
Seconds: 2.5 sec
Minutes: 1.9 min
Hours: 3.8 h

Please Note:
The time value refers to the step and not to a day time!

Special Limits:

Switching Value	Description
10.0 X	The following action will be executed when the difference between the actually measured or calculated value X and the value measured at the beginning of the program step is greater than 10.
< 10.0 X &	AND function- The AND sign links two or more limits. The action following the last limit set will be performed only after all conditions have been met. This function is valid for **A, V, C Ah, AhStep, WhPrev,** etc. (also see chapter 12.2.8)
> 10° Tm	The following action will be carried out when the temperature increases by more than 10 degrees per minute. The appropriate channel is **C**.
> 10° T	The following action will be carried out when the temperature increases by more than 10 degrees per hour. The appropriate channel is **C**.
Timer 1 to Timer 3	Terminates a step when **Timer1**, **Timer2**, or **Timer3** is elapsed. The timers are set and started by the **SET** operator.
> 1.0 AhDef	The present Ah value will be defined as 100% for the capacity. The step will be terminated immediately after its start.
> 80 PercAh	The step will be terminated when the Ah counter reaches 80% of the value fixed by AhDef.
ABATT	Special customer's option for switch functions.
ACN5	Current referred to the nominal capacity at 5 hours discharge
MINDEF	Special customer's option for setting global time limits.
MINT	Special customer's option for setting global time limits.
OHM	Resistance value calculated from the actual voltage and current values
PERCCNOM	When the calculated capacity reaches the value indicated in percentage of the nominal capacity, the present program step will be terminated
VBATT	Special customer's option: Actual value of battery voltage
VNC	Voltage value referred to the nominal voltage of a cell
WATT	Power value calculated from actual voltage and current values.

Caution!

When the limit is a timer that can also accept negative values such as Ah or Wh, please observe the respective sign. An upper limit for the discharged capacity should be entered such as **< -100 Ah.**

Our program example now contains one pause step of 30 seconds and one charge step with 1 A charging current. To terminate the charge step, you will have to set another limit. For instance, enter an upper voltage limit of 14.1 V in the respective column. The correct input is simply: > 14.1 V. As soon as the system measures a voltage of more than 14.1 V, the test program will execute the action indicated in the same program line. Please also refer to chapter 12.2.5.

Your editor will now read as follows:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	1.0 A	> 14.1 V		

GradT, GradI, GradU

With these limits a step can be terminated if a given increase of temperature, current or voltage occurs within an hour or minute (defined by adding an "m" to the function, e.g. GradU**m**).

Example:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
3		CHA	1.0 A	2.0 GradTm		

The step is terminated on a rise of temperature of at least 2 degrees within one minute. Use a limit with a negative sign to respond to the decrease of a value.

The functions work like this:

Example: GradUm

At the beginning of the step, the present voltage Y is defined as reference point X (X=Y). Now for one minute with each new measuring value, i.e. every 100 ms, GradUm is calculated from the recently measured voltage (Y) and the reference point (X) as follows:

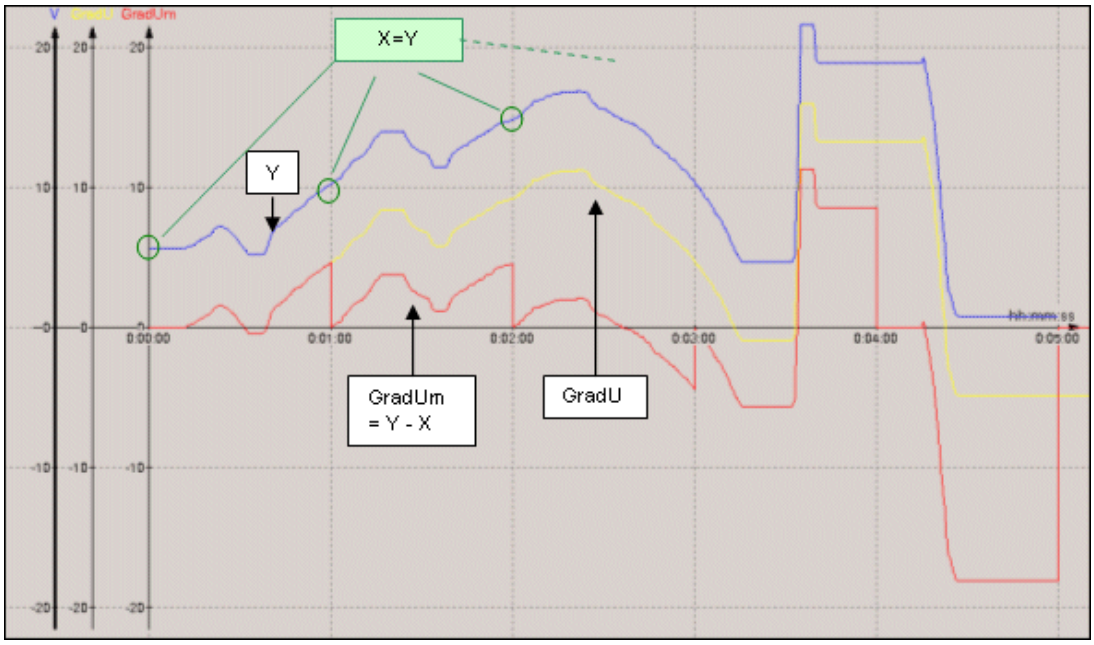
GradUm = Y – X

With GradU**m** a new reference point (X) is defined every minute, i.e. X=Y and thus GradUm is reset to 0.

With **GradU** a new reference point is defined only every hour.

The check of these limits takes place with the maximum measuring speed, i.e. every 100 ms.

Graphical representation GradU(m):



12.2.5 Action

The **Action** column indicates **how** a test step is terminated as soon as the appertaining limit is reached.

The simplest case is that, when the limit is reached, the program will continue with the next test step. In this case, the **Action** column will remain blank.

With certain battery conditions the test process may, however, require a carefully directed program flow. The following actions are disposable:

Action	**Description**
Blank	When the appertaining limit is reached, the next test step will be executed.
INT	When the appertaining limit is reached, the program will be interrupted.
STO	When the appertaining limit is reached, the program will be stopped.
GOTO - *Jump Destination*	When the limit is reached, the program will jump to the program step that contains the entry *Jump Destination* in column **Label**.
Procedure Name	When the limit is reached, the procedure indicated here will be loaded. Then the following program step will be activated. After having executed the procedure, the program will go to the next program step.

Special Actions:

Action	Description
ERR	Message in the measuring file with interruption of the test program. Please also see chapter 12.2.7, Example 11.
MSG	Message in the measuring file without interruption of the test program. Please also see chapter 12.2.7, example 11.

Actions have a fundamental effect on the test process. You should therefore read and observe the following information very carefully:

An **Interruption** of the test program (INT action) should be activated with unusual or dangerous battery conditions, e.g. over-temperature, imminent over-charge, exhaustive discharge, etc. The test process can be continued only after you have given the expressive command to do so. This can be done either by actuating the "Start“ button on the front panel of the test circuit or by entering commands in the host computer (tool buttons or dispo list). Prior to giving this command, give your battery a thorough check.

Stopping the program by entering the STO command in the **Action** column will merely stop the actual test, but not the remaining tests indicated in the dispo list. If, for instance, a charge test is followed by a discharge test and charging has been terminated for whatever reason, the discharge test will start nevertheless. This may cause an exhaustive discharge!

Caution!

The STO action does not terminate the test process, but only the actual test program! The system will execute further tests contained in the dispo list! If, due to premature interruption of the test, the battery could not be fully charged, the subsequent discharge test may lead to an exhaustive discharge!

GOTO Commands only apply to the same program level! It is, for instance, not possible to jump from a procedure to the main menu or to another procedure!

If you enter a procedure in the action column you can promptly react to certain events or statuses such as error handling, activating recharge procedures, etc.

How to use procedures and GOTO commands is described in paras 12.3.2 and 12.3.3.

Our program example now contains a charge step which shall be terminated when the voltage limit of 14.1 V is reached. To realize this setting, enter **STO↵** in the **Action** column in the same line as the limit entry > 14.1 V.

> 14.1 V	STO

In addition, set a temperature limit which will interrupt the charge step when a temperature of 50° C is exceeded. Move the cursor to the field below the last entry in the **Limit** column. Now enter the temperature limit **> 50 C↵**. Enter the respective action in the same line in the **Action** column. As, for safety reasons, no tests should be performed when overtemperatures occur, enter **INT↵** in this column.

> 14.1 V > 50.0 C	STO INT

The next step should be a pause step. Therefore, edit the fourth step in the same manner as the second step. Use the copy function to repeat the second test step below the third test step.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	1.0 A	> 14.1 V > 50.0 C	STO INT	
4		PAU		30.0 sec		

The test program in its present form is now ready for operation. It does, however, not yet produce sufficient measuring data. Though you did define, by entering the SET operator, which channels will be registered, you have not yet defined the registration rate. This setting is made in the last column. It will be described in the chapters 12.3.2 and 12.3.3.

12.2.6 Registration

The last column of the program editor serves to define the frequency in which the measuring data are recorded. The frequency can be set individually for each test step.

If this column is left blank for a certain step, the measuring data will, in any case, be registered at the beginning and at the end of the test step. Please observe that the quantity of registered data effects the time needed for subsequent test evaluations. Carefully consider which registration rate to set and adapt it to your actual requirements.

BTS-600 offers many variations for registration. The following table will give you a survey on all available registration formats. The quantities and counters listed in the table stand for all existing registration channels.

Registration Type	Example	Description
Delta	2.0 A 5 min	registers when the current fluctuates by 2 A registers every 5 minutes
Threshold <, >	< 0.5 A > 50.0 C	registers when the current falls below 0.5 A registers when the temperature exceeds 50 °C
Range << & >> &	>> 1 h & 2.0 V << -100.0 Ah & 1.0 C	registers each voltage fluctuation by 2 V one hour after beginning of step registers each temperature change by 1 °C after a 100 Ah discharge
Maximum *	15 * >> 10000 Watt 5* >> 30 sec 1*	registers exactly 15 times with the highest resolution of the A/D converter, i.e. in intervals of 0.1 seconds registers 5 times with the highest resolution when a power of 10 kW is exceeded registers once exactly 30 seconds after the begin of a step

Special registrations:

Registration	Description
Registration Format Name	only in connection with operators **SET, CHA, DCH, RCH, PAU, REG, SAVE**. Defines the registration format used in a program.

You can, of course, also combine several of the above registration types in one test step.

The first type, "Delta“, is the registration type used most frequently. If, for instance, you are interested in obtaining an evaluation of the current curve, you will record whenever the current changes by a certain value. It might also be important for you to know the exact time response. You will then need measuring data recorded in constant intervals.

You will have to determine for each test which measuring data are of importance and enter the registration setting correspondingly. We would, at any rate, recommend you to make economic use of the registration function.

Now back to our program example. Our example is a simple charge program with constant charge current. So it would not make sense to have registrations with current fluctuations. What might be of interest is the voltage curve during the charging process. Therefore, BTS-600 should record whenever the voltage changes by 0.5 V. To set this registration function, move the cursor to the last column of the third test step. Enter **0.5 V↵** in this field.

As the voltage may remain constant for some time, enter an additional time registration (e.g. every five minutes) to register the data measured in the meantime. Double-click the field with the registration just created, perform a line break and enter **5 min↵**.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	1.0 A	> 14.1 V > 50.0 C	STO UNT	0.5 V 5.0 min
4		PAU		30.0 sec		

If the registration column contains several entries for one test step, the sequence of the entries is of no importance.

12.2.7 Program Examples

In the following you will find several examples describing the applications used most frequently in battery testing.

The examples are applicable for operators DCH, CHA, and RCH. The battery will be charged or discharged with constant current, voltage, power or resistance.

Example No. 1: Discharge

The following is a discharge step with constant current, which will be terminated as soon as the voltage falls below a preset limit voltage. The registration will, above all, supply measuring points for the voltage curve.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		DCH	10.0 A	< 9.8 V		0.5 V
4		STO				

Example No. 2: Charge in two steps

The battery will be charged with constant 20 A until it reaches a voltage of 14.1 V. Then BTS-600 regulates this voltage value, i.e. the charge current will be reduced while the charge operation advances. The registration was specially defined for this current curve.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	20.0 A	> 14.1 V		0.5 V
4		CHA	14.1 V	< 1.0 V		1.0 A
5		STO				

Example No. 3: Discharge in two steps

The first step ends when 50 Ah have been discharged.

Caution: Discharge limits must be entered with negative sign!

The second step contains a limit of the admissible battery temperature. When the temperature exceeds that limit, the test will be interrupted. When the voltage falls below 10.5 V, the registration rate for this step will be reduced from 5 minutes to 2 minutes.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		DCH	0.5 OHM	< -50.0 Ah		5.0 min
4		DCH	600.0 Watt	< 9.8 V > 50.0 C	INT	5.0 min << 10.5 V & 2.0 min
5		STO				

Example No. 4: Charge and Discharge

It often makes sense to combine charge steps (example No. 2) and discharge steps (example No. 3) in one test program. In this case, you should, at any rate, reset the counters to quickly view the charge level of the battery.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	20.0 A	> 14.1 V		0.5 V
4		CHA	14.1 V	< 1.0 A		1.0 A
5	!	Counter reset after charge				
6		SET	Ah = 0.0 Wh = 0.0			
7		DCH	0.5 OHM	< -50.0 A		5.0 min
8		DCH	600.0 Watt	< 9.8 V > 50.0 C	INT	5.0 min << 10.5 V & 2.0 min
9		STO				

Example No. 5: Parallel Functions

Example No. 2 shows a charge program with two charge steps, which can also be combined in one step. For this purpose, two nominal values are assigned to the CHA operator. These values will be observed simultaneously, i.e. none of them may be exceeded. The battery will be charged with maximal 20 A. During this process, the voltage can increase to maximum 14.1 V. When this nominal value is reached, the voltage will be kept constant, i.e. the charge current decreased.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	14.1 V 20.0 A	< 1.0 A > 50.0 C	 INT	1.0 Ah 0.5 C
4		STO				

Example No. 6: Programs with battery parameters

Using battery parameters permits you to use your programs for multifold applications. The following example contains the same charge program as described in example No. 5, the nominal capacity being 100 Ah and the battery being constructed of 6 cells. The current of 1.0 Acn5, supplied for five hours, then equals 20 A, 0.05 Acn5 equals 1 A, a cell voltage of 2.35 V per cell equals 14.1 V battery voltage.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	2.35 VNC 1.0 ACN5	< 0.05 ACN5 > 50.0 C	 INT	1.0 Ah 0.5 C
4		STO				

Example No. 7: Ramp Function

Steps 3 and 4 contain discharges in ascending or descending ramp form. Apart from the initial and the end value of the ramp, you will also have to define the ramp duration in the **Nominal Value** column. The time values entered in the **Limit** column may *not* correspond with the ramp duration.

If the ramp is executed before the limit is elapsed (step 3), the battery will be discharged with the ramp end value for the remaining time. Our example shows a one-minute ramp discharge, followed by another minute discharge with 1000 watts.

If the limit elapses before the ramp is executed (step 4), the limit has priority. After 45 minutes, the next test step will be executed. At that moment, the discharge current will be approx. 35 A. Step 3 contains a voltage limit (<9.8 V) in the limit column, leading to a test stop (STO action). If this voltage limit is reached in step 4, the test will not be stopped as step 5 merely contains a pause.
This pause serves to monitor the voltage after the end of the discharge process.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		DCH	0.0 - 1000.0 Watt 1 min	2.0 min < 9.8 V > 50.0 C	 STO INT	100.0 Watt 0.5 V
4		DCH	80.0 - 20.0 A 1 h	45.0 min < 9.8 V > 50.0 C	 INT	6.0 A 0.5 V
5		PAU		3.0 h		0.5 V
6		STO				

Example No. 8: Jump

If the test is stopped prematurely in step 3 of the previous example (STO with < 9.8 V), no measuring data will be recorded after the end of discharge. This can be avoided by using a GOTO command. When the limit is reached, the program will be continued with the jump command, which is "Pause“ in our example. A jump command must be entered in the **Label** column. For further information on jumps please see chapter 12.3.3.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		DCH	0.0 - 1000.0 Watt 1 min	2.0 min < 9.8 V > 50.0 C	 GOTO Pause INT	100.0 Watt 0.5 V
4		DCH	80.0 - 20.0 A 1 h	45.0 min < 9.8 V > 50.0 C	 INT	6.0 A 0.5 V
5	Pause	PAU		3.0 h		0.5 V
6		STO				

Example 9: Interruption

The test program is interrupted. To continue the program, either actuate the "START" button on the test circuit or reactivate the program on the host computer (direct functions or tool buttons in the main menu).

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2		INT				
3		CHA	10.0 A	> 14.0 A	INT	10.0 min
4		DCH	50.0 A	< 10.5 V		10.0 min
5		STO				

The program starts with an interruption. After the program is started on the host computer, it will remain in hold state until you either actuate the "START" button on the test circuit or continue the program on the host computer. The following charge will be interrupted when 14.0 volts are reached. The program will have to be reactivated manually also with this interruption.

Example 10: Actions in Test Steps

Actions defined in test steps serve to react with different actions on different limits. If a limit is reached, the action set in the same line will be executed. No entry means: Go to next step.

The following example shows different limits with different actions.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2	Charge	CHA	10.0 A 14.4 V	< 0.5 A 10.0 h > 100.0 Ah > 40.0 C	 ERR 7 INT GOTO COOL	10.0 min 0.2 A
3		DCH	1.0 ACN5	< 10.5 V		0.1 V
4		STO				
5	COOL	PAU		1.0 h	GOTO Charge	

Explanations on the Individual Limits and Actions:

< 0.5 A	

If the current reaches a value smaller than 0.5 amps, the charge process will be terminated. The program will jump to the next step.

10.0 h	ERR 7

If the charge process takes more than 10 hours, it will be interrupted. An error message will be written to the measuring file.

> 100.0 Ah	INT

If the 100 Ah limit is reached prior to the 0.5 A limit, the program will be interrupted.

> 40.0 C	GOTO COOL

When the battery temperature rises to 40° C, a 1-hour pause will be initiated. After elapse of this pause, the program will jump back to the charge process.

Please Note:

A limit will always terminate the running step. There are only two exceptions to this rule, namely if **MSG** or **ERR** are set as actions.

MSG will merely send a message to the measuring file and to the host computer. It will not terminate the program step.

ERR will send a message and interrupt the running step. You can have the program continued at the same position of the same step by actuating the "Start" button on the test circuit.

Certain actions only function if used with certain operators. Combined with general limits and the **MSG** command the **SET** operator can, for instance, send a message to the measuring file. Combined with the **ERR** command, the **SET** operator will interrupt the current step. In both cases the step will **not** be terminated when the limit is reached!

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET		V > 15.0 V A > 100.0 A	ERR 4 STO	STANDARD

For further information please refer to paras 12.2.8.5 and 12.2.8.6.

Example 11: &-linked Limits

To link limits, enter an & sign after the limit. This links the limit in the next line with the preceding limit. You can thus define complex conditions as switching values.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET		V > 15.0 V A > 100.0 A	ERR 4 STO	STANDARD
2		PAU		1.0 h		
3	Charge	CHA	10.0 A 14.4 V	< 0.5 A & 10.0 h & > 100.0 Ah > 40.0 C	MSG 4 GOTO COOL	10.0 min 0.2 A
4		STO				

The charge step will be terminated when the current is smaller than 0.5 amps, the step runtime has reached 10 hours, and the Ah counter has exceeded 100.0. After 10 hours, however, a message will be sent to measuring file and host computer.

Example 12: Battery referred cycle counter

Control of the battery referred cycle counter is made with the two commands **CYCSET** and **CYCINC**. With this cycle counter, it is possible to determine a certain number of steps to be counted for the respective battery as a "cycle", independent from the cycles within a program.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	Ah = 0.0 Wh = 0.0			STANDARD
2		CYCSET	15.0			
3		BEG	Example			
4		CHA	10.0 A	> 14.0 V		Special
5		DCH	1.0 ACN1	< 10.5 V 5:30 h	GOTO End	10.0 min
6		PAU		30.0 sec		
7		CYCINC				
8		CYC	120 *			
9	End	STO				

In the above example, the counter is preset to the value "**15**" with the command "CYCSET" and the value entered in the nominal value column for step 2. After the start of the program, this value will appear in the column "Cycle from:" of the respective test section. If no value is entered with the command "CYCSET", the counting will begin with "1".

Each step with the command "CYCINC" will increment the counter by "1". The actual count will be indicated in the column "Cycle to:" of the respective test section. In the following test section counting of cycles will then continue with the next value.

In the list of test sections you can then see exactly which battery cycle has been executed in which test section. This will make it easier to retrieve particular measuring data of the respective battery.

12.2.8 How To Use Special Operators and Actions

This chapter contains supplementary explanations on the application of special program steps.

12.2.8.1 How to Set Variables

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	Thomas = 43.0 Harry = 12.0 Volt = 14.1 CutVolt = 11.0 ChargEnd = 14.0			STANDARD

With the **SET** command you may define and set your proper variables. To do this, simply equate the variable name to a certain value.

Please Note:

If you use a name already assigned to a channel in the **Attribution circuit/channel**, this channel will be set to the value this name is equated to. This may be advisable for presetting counters such as **Ah** and **Wh**.

Why Variables?

You may use these variables wherever the corresponding values would usually appear in a program. That is to say that, whenever you use values frequently in a program, you had better use variables. If you change this variable, the program will automatically adapt each program step containing this variable to the new value. You can also use these variables to preset cycle counters. Please also see para 12.3.1.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	Cycle = 543.0 CurValue = 20.0			STANDARD
2		GOTO	into			
3		PAU		1.0 sec		
4		BEG	Cycle			
5	into	CHA	CurValue A	1.0 sec		
6		DCH	CurValue A	2.0 sec		
7		CYC	545 *			

In above figure the cycle counter is preset to value 543 and variable CurValue is defined and set to value 20. Please note that "cycle" is a freely selected name, i.e. a variable. To assign this variable to a cycle, you will have to enter the same variable name after the cycle beginning **BEG**. Setting the **GOTO** command effects that the program jumps direct into the cycle. This avoids counter resets. In our case, this effects that three cycles will be executed successively. Both the charge and the discharge step will be executed with 20 amps, as the variable **CurValue** was set to 20.

12.2.8.2 How to Set Global Nominal Values or Limits

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	MaxLadI = 50.0 MaxElaI = 100.0 MaxLadU = 15.0 MaxElaU = 10.0	A > 110.0 V > 16.0 Watt > 1200.0	ERR 1 ERR 1 ERR 1	STANDARD

Global nominal values are limits that affect the entire program, that means each program step. The corresponding values do not exceed or fall below these limits. If, for instance, this limit should be reached with a certain watt value set in the program, the test circuit will automatically regulate the power to this limit.

The same applies to global limits. A limit does, however, perform the action preset thereafter in the same line. Possible actions are **ERR** and **MSG**.

Please also refer to the explanations given in chapter 12.2.8.10.

12.2.8.3 Registration as Program Step

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2		BEG				
3		CHA	10.0 A	> 14.0 A		10.0 min
4		DCH	50.0 A	< 10.5 V		10.0 min
5		REG				Special
6		CYC	500 *			

The **REG** command will register once in the registration format entered in this line.

How to store a registration in a separate data file:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2		BEG				
3		CHA	10.0 A	> 14.0 A		10.0 min
4		DCH	50.0 A	< 10.5 V		10.0 min
5		REG	Cycdat			Special
6		CYC	500 *			

If you enter a file name in the **Nominal Value** column, the registration initiated by the **REG** command will be recorded in this file. This permits to write evaluation files already while the test program is running. You can, for instance, determine that only the capacity values are recorded in a separate file at the end of each cycle. This permits you to have the capacity curve of a battery displayed as graph or in numerical figures, without having to evaluate large measuring files that contain all registrations of a test program.

12.2.8.4 How to Use the "TASK" Operator

This operator activates a program running in parallel to the BTS program. This program serves to control relays, temperatures or similar.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		TASK	40 TCONTR			

The **TASK** command permits you to integrate into the test program up to 12 parallel processes. In above example the temperature of a water bath is controlled to 40 degrees Celsius. Please consider that activation of this process is not possible without the appropriate hardware! BTS-600 allows to integrate a multitude of **TASK** enhancements. Please contact Digatron for further information.

12.2.8.5 How to Use the "ERR" Operator

Error status with interruption of test program.

ERR as independent test step:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2	Newstart	BEG				
3		CHA	10.0 A	> 14.0 V		10.0 min
4		DCH	50.0 A	< 10.5 V 5:30 h	GOTO OUT	10.0 min
5		REG	Cycdat			Special
6		CYC	500 *			
7		STO				
8	OUT	ERR	5			
9		GOTO	Newstart			

The test run will be interrupted when the program arrives at the 9th test step. The "Error" LED and the error relay output will be set on the test circuit. Beyond that, a message will be displayed on the screen in line **System extern**. This message will show the number entered under the **Nominal Value** column.

ERR as Action:

You can also set the **ERR** operator in column **Action**.

4		DCH	50.0 A	< 10.5 V 5:30 h	ERR 5	10.0 min

12.2.8.6 How to Use the "MSG" Operator

Error status without interrupting the test program.

MSG as Independent Test Step:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		TASK	40 TCONTR			
2		SET				STANDARD
3		BEG				
4		CHA	10.0 A	> 14.0 V		
5		DCH	50.0 A	< 10.5 V 5:30 h	ERR 5	10.0 min
6		REG	Cycdat			Special
7		MSG	5			
8		CYC	500 *			
9		STO				

The message belonging to the number behind **MSG** in the Nominal Value column will appear on the screen in future versions of the software. The test will continue without interruption.

MSG as Action:

You can also enter the **MSG** operator under the **Action** column.

5		DCH	50.0 A	< 10.5 V 5:30 h	MSG 5	10.0 min

12.2.8.7 How To Use the "SAVE" Operator

This operator saves all battery data on the host computer in the current registration format.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2		BEG				
3		CHA	10.0 A	> 14.0 V		10.0 min
4		DCH	50.0 A	< 10.5 V		10.0 min
5		CYC	50 *			
6		SAVE	Test			STANDARD
7		STO				

When a test is finished, the counts from channels that are assigned in the given registration format are buffered on the hard-disk of your host computer under this battery. You can now run any tests with this battery, without these counts getting lost for this battery.

The **SAVE** operator stores the data into a file whose name has to be entered in the **Nominal Value** column, with the file extension **.sav**. Several files for one battery are possible.

You can restore the counts for further tests with this battery by setting the **REST** operator. For further explanations please see chapter 12.2.8.8.

12.2.8.8 How to Use the "REST" Operator

This operator restores data previously saved on the host computer. This function permits you to continue a test at a later date with all counts (**Ah**, **Wh**, etc.).

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2		REST	Test			
3		BEG				
4		CHA	10.0 A	> 14.0 V		10.0 min
5		DCH	50.0 A	< 10.5 V		10.0 min
6		CYC	50 *			
7		STO				

When you start this test, the counts previously stored by the **SAVE** command will be restored. Therefore, with **REST** the name of the appropriate file has to be given in the **Nominal Value** column. The restoration only works for channels that have been saved with the same name as used in the registration format that has to be declared before **REST**. In above example, this registration format is **STANDARD**.

12.2.8.9 How to Use the "TABLE" Operator

A: Creating a Nominal Value Table with Notepad

To create a nominal value table you require a text editor e.g. Notepad (started by clicking Start-Programs-Tools-Accessories-Notepad). You can define a nominal value table line by line using the following pattern:

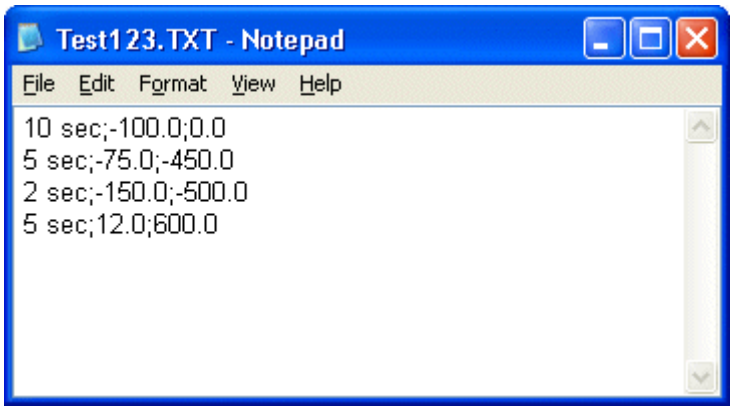
X sec;Y;Z;

with

X = duration in seconds
Y = current in A
Z = power in W

Current and wattage values for discharge steps must be entered with negative sign. **Make sure that the entered current and wattage values must be within the admissible range of the respective power unit.**

The table could for example look like this:



Save the nominal value table as a text file with the extension **.TXT** in the directory **C:\Battery Manager\Server\BTS-600\Table**. By calling the file name you can use the table for your programs. See the following paragraph for more details.

B: Loading a Nominal Value Table from the Hard-Disk of your PC

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				TABREG
2		CHA	400.0 A 14.1 V	< 14.0 V		1.0 min 0.5 A
3		SET	Ah = 0.0 Wh = 0.0 AhCha = 0.0 AhDch = 0.0 WhDch = 0.0 WhCha = 0.0			
4		TABLE	FTP2			
5		CHA	400.0 A 14.1 V	< 50.0 A		1.0 min 0.5 A
6		STO				

In step No. 4 of above test program a nominal value list named FTP2.TXT will be loaded and executed.

The nominal value list on the hard-disk must have the extension **.TXT**, as described under **A:** and must be stored in directory **C:\Battery Manager\Server\BTS-600\Table**.

C: Entry of Factors for the Current and Power Values in a Table

If you wish to use a nominal value table in several tests with different current and power values, you have to enter appropriate factors into the **Nominal Value** column:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				TABREG
2		CHA	400.0 A 14.1 V	< 14.0 V		1.0 min 0.5 A
3		SET	Ah = 0.0 Wh = 0.0 AhCha = 0.0 AhDch = 0.0 WhDch = 0.0 WhCha = 0.0			
4		TABLE	FTP2 0.5 Factor_I 0.5 Factor_P			

Step 4 uses the nominal value table with the name FTP2.TXT. Each value in the table for current and power is multiplied in the above example with the factor "**0.5**".

The factors must be entered in the Nominal Value column under the table name. The order of entry is at random.

D: FUDS Test

By using the TABLE operator it is now possible to run the FUDS test (**F**ederal **U**rban **D**riving **S**chedule) which is an automobile industry standard vehicle time-velocity profile for urban driving that has been used for a number of years for electric vehicle performance testing. Furthermore a table for the Dynamic Stress Test (DST) and a table for the simplified FUDS test are implemented. The tables for the FUDS and DST tests contain relative values of the peak power. The table for the SFUDS contains values rated to W/kg. The following description gives an example how to use the FUDS test.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1	!	Set registration format for every table step.				
2		SET				fudsreg
3	!	Implement the Task for calculating the actual power.				
4		TASK	1 OUW			
5	!	Activate FUDS Table and set the PeakPower to 100 Watt.				
6	!	Use the predefined table fudstbl for the FUDS test.				
7	!	This table includes the relative values for the PeakPower.				
8	FUDS	TABLE	fudstbl 100.0 Factor_P	< -50.0 Ah < 11.3 V	GOTO End GOTO End	
9		GOTO	FUDS			
10	End	STO				

The FUDS test is based on discharging the battery with a given power. The test program must include the TASK **OUW** for calculating the actual power which is taken from or given to the battery. The channel name and the units name is Watt. Thus the actual power can be used in a registration format by using Watt in a certain format.

The table used for the FUDS test is given in relative values. Thus a factor has to be added to each relative value to get the absolute value of the Peak Power used for each step of the FUDS test. This is done by using the **Factor_P** function. The value standing before this function is the used Peak Power. In the example program above it is 100 Watt.

The FUDS test is normally terminated by a certain amount of amperehours removed from the battery or if the battery reaches a certain voltage level. The example above terminates either if the battery voltage is below 11.3 V or 50 Ah has been removed from the battery. If none of these terminations are valid during a complete FUDS test that contains 1372 steps, the FUDS test starts again at the beginning of the table.

Using the DST or SFUDS is equal to FUDS test however the table of the SFUDS is based on W/kg. Thus the weight of active material has to be used as the factor for the SFUDS table and not the absolute value for the peak power.

The names of the tables that are available and stored in the directory **C:\Battery Manager\Server\BTS-600\Table** are as follows:

fudstbl for the FUDS test
sfudstabl for the SFUDS test
dsttbl for the DST

12.2.8.10 How to Use the "CLEAR" Operator

This operator deletes the global nominal values or limits previously set with the **SET** operator.

See chapter 12.2.8.2 for more details

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	MaxChaI = 50.0 MaxDchI = 100.0 MaxChaU = 15.0 MaxDchU = 10.0	A > 110.0 V > 16.0 Watt > 1200.0	ERR 1 ERR 1 ERR 1	STANDARD

The **CLEAR** command deactivates the global nominal values or limits previously set.

5		CLEAR				

12.2.8.11 How to Use the "FILTER" Operator

The definition of filter functions can, for the time being, only be made by Digatron personnel. Please contact us for further information, if required.

12.2.8.12 Application of the "FILE" Operator

For certain tests it might be useful to save the measuring data registered during a program run in files with different names. This can be done by the entry of the operator **FILE** and the desired file name into the **Nominal Value** column.

The following program example shows the alternatives:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				STANDARD
2	!	Writing from here into measuring file Test_A!				
3		FILE	Test_A			
4		CHA	10.0 A	10.0 sec		LADREG
5	!	Writing from here into automatically generated measuring file!				
6		FILE				
7		DCH	1.0 A	5.0 sec		ELAREG
8	!	Writing from here into measuring file Test_B!				
9		FILE	Test_B			
10		CHA	15.0 A	10.0 sec		LADREG
11		BEG	CyclFile			
12	!	Writing from here into measuring file TSZxxxxx (xxxxx = actual cycle number)				
13		FILE	CyclFile			
14		CHA	1.0 A	10.0 sec		
15		DCH	1.0 A	10.0 sec		
16		CYC	10 *			
17		STO				

The file with the respective name and the extension ***.dat** is stored into the folder **C:\Battery Manager\Battery\[Battery name]** as a test section.

12.3 Helpful Program Elements

The program example as well as the chapter 12.2.7 should give you a first impression on how to create a test program. BTS-600 contains a program language that can realize also complex tasks in a simple and easy-to-survey manner. The following elements should assist you in creating and structuring complex test programs.

12.3.1 Cycles

Cycles are essential program elements. You can use them whenever you want to repeat the same steps. A brief description of the necessary operators, namely **BEG** and **CYC** is given in chapter 12.2.2.

Programming a cycle is most simple:

Set the beginning of the cycle by entering the **BEG** operator in the **Operator** column. Enter a name for the cycle in column **Nominal Value**. This is necessary if you will have to preset the counter of this cycle with a certain value.

The cycle end is defined by the **CYC** operator. Define the number of cycles by entering a figure and an asterisk " * " (e.g. 10*) in the **Nominal Value** column.

It is, of course, also possible to interlace up to 16 different cycles.

The following example shows two interlaced cycles:

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		BEG	outer			
4		BEG	inner			
5		CHA	28.0 A	> 14.1 V		0.5 V
6		CHA	14.1 V	< 0.84 A		2.0 A
7		CHA	0.84 A	3.0 h		20.0 min
8		DCH	28.0 A	< 9.8 V		0.5 V
9		CYC	5 *			
10		PAU		12.0 h		0.5 C
11		CYC	10 *			
12		STO				

The *inner* cycle (steps 4 thru 9) contains a two-step charge process, followed by a discharge step. After the 5th repetition follows a 12-hour pause. Then the *outer* cycle (steps 3 thru 11) will start from the beginning. The inner cycle is executed 5*10 = 50 times.

12.3.2 Procedures

BTS-600 permits you to combine several test steps to one command - to create a procedure.

If you want the same sequence of test steps to be executed in different programs, it is useful to combine these steps to a procedure. Procedures can be used in any program. Furthermore, they make your test programs more easy to survey.

How to Create a Procedure

Procedures are created in the program editor. For example: a three-step charge process should be combined to a procedure. On the one hand, the reduced number of test steps will make your program more clear. On the other hand, you can use this procedure in any other test program used for the same battery type. Another positive aspect is that procedures reduce the error frequency in program creation.

The original charge process comprises test steps 3 thru 5:

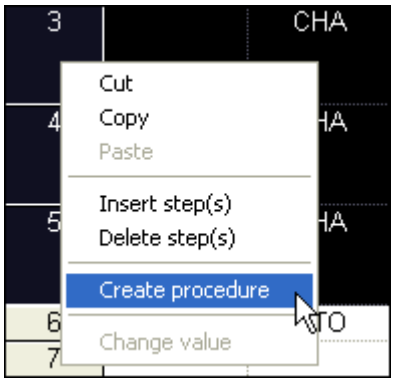
Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	28.0 A	> 14.1 V > 100.0 Ah > 50.0 C	 STO INT	0.5 V 15.0 min
4		CHA	14.1 V	< 0.84 A > 100.0 Ah > 50.0 C	 STO INT	15.0 min 2.0 A
5		CHA	0.84 A	3.0 h > 100.0 Ah > 50.0 C	 STO INT	20.0 min
6		STO				

First select all lines that should be contained in the procedure. Lines are selected by dragging the mouse along the **Step** column.

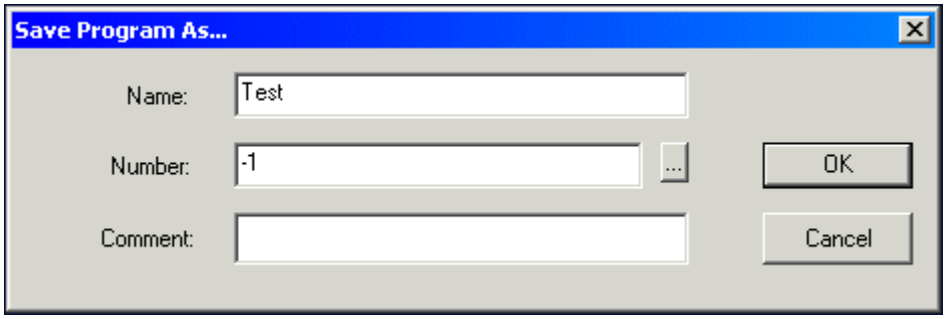
The selected program lines are now shown inverse.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		CHA	28.0 A	> 14.1 V > 100.0 Ah > 50.0 C	STO INT	0.5 V 15.0 min
4		CHA	14.1 V	< 0.84 A > 100.0 Ah > 50.0 C	STO INT	15.0 min 2.0 A
5		CHA	0.84 A	3.0 h > 100.0 Ah > 50.0 C	STO INT	20.0 min
6		STO				

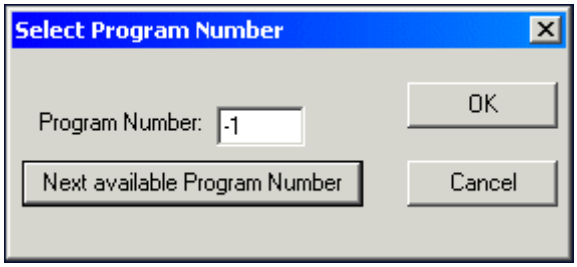
Then perform a right-click in the **Step** column of a highlighted column. Select **Create procedure** from the appearing menu.



You are then prompted to enter a name and a free program number for the storage of the procedure.

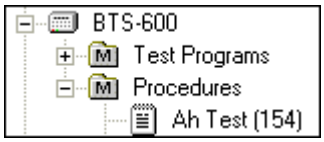


After clicking  you have the possibility to enter a program number:



When clicking **Next available Program Number** a free number is chosen by the program itself.

After saving the procedure it can be found in the overview of all programs under BTS-600 in the section **Procedures**.



You may also create a procedure by right-clicking the entry **Procedures** in the list of all BTS-600 programs as shown above and then click **New Procedure** from the appearing menu. After entering an new and clicking the **Edit Program** button you can create procedure steps in the program editor.

Once a procedure has been created, it can be used by entering its name in the **operator** column of a test program.

The **Cancel** button in the main window terminates **Program Editor**. In case you haven't saved all previous changes, you have to confirm whether or not you want to save the program now.

Note

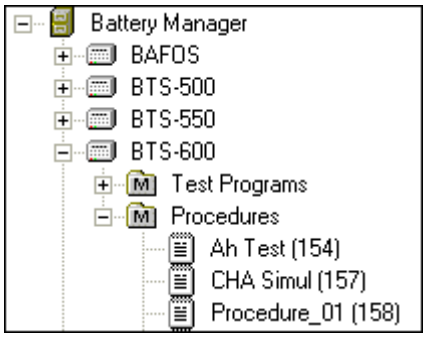
As with test programs, also with procedures the name and the title should clearly show for which type of battery the procedure was created, respectively for which type of battery it is suited. Using the wrong procedures may have fatal effects on your battery!

Instead of the original test steps, the corresponding columns in a program now merely contain the names of the selected procedure. The following test steps are renumbered automatically.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET				NORMAL
2		PAU		30.0 sec		
3		Ah_Test				
4		STO				

Display Procedures

Procedures can be found in the overview of all programs under **BTS-600** in the section **Procedures**.



By double-clicking a procedure name the procedure is displayed in the program editor. As you are in the editor window, you can edit the procedure like a normal program.

Caution!

Please note that, when you have modified a procedure, all other test programs using this procedure will also be effected. In this case, you 'd better decompose this procedure, then modify it and create a new procedure under a new name!

12.3.3 Counter Settings, GOTO Commands

Counter Setting

During testing operation you will often have to interrupt tests and continue them afterwards in a slightly modified form. When a test is stopped - i.e. terminated completely - the test program must be re-compiled before the test is continued. This will, of course, also reset all counters of the preceding tests, i.e. all cycle counters will start with 1, all other counters (Ah, Wh, etc.) will start with 0. Many test programs use counts as limits. Therefore, write down the counts prior to the program stop and redefine these counts in the program before continuing the test.

Enter a jump command to the point where the program was aborted to continue the test at the very point.

Use the SET operator to set the counters and use GOTO operator to enter jump commands.

Step	Label	Operator	Nominal Value	Limit	Action	Registration
1		SET	Cycle = 27.0 Ah = -53.21 Wh = -6521.76			
2		PAU		30.0 sec		
3	!	Caution: Continue at 27 cycles at label break!				
4		GOTO	Break			
5		BEG	Cycle			
6	Break	DCH	1200.0 Watt	30.0 sec < -80.0 Ah	GOTO End	
7		DCH	500.0 Watt	30.0 sec < -80.0 Ah	GOTO End	
8		CYC	1000 *			
9	End	PAU		1.0 h		

The first step is to define the cycle counter and the Ah and Wh counters with the values of the last registration before the program was aborted. To preset the cycle counter, first enter a cycle name. Enter any name in the **Nominal Value** column at the beginning of the cycle (BEG operator) - in our case "Cycle“. It goes without saying that you can also preset the counters with values for several or for interlaced cycles. **Please make sure that the Ah and Wh counters show negative values!** If you forget to enter a minus sign, the battery will be discharged from 53,21 Ah to - 80 Ah, i.e. by another 133,21 Ah! Please see the entries in steps 1, 6 or 7 of above example.

Jumps

After a 30-second pause, the program will execute the jump to the abort point, marked with "Break“ above. If the cycle counter was not programmed with a SET command, the count will start with 1. In our example the counter will start with 27.

The program will be processed as usual. Unless the cycle counter shows 1000, the two discharge steps will be repeated until the limit of -80 Ah is reached. The GOTO command effects that the program quits the cycle and goes to label "End“.

The following brief example shows the most important features of the GOTO operator.

You can enter a jump command either as **Operator** or as **Action**. When used as operator, the jump destination should be entered in the **Nominal Value** column, while with actions jump destinations should be entered in the **Action** column. In both cases, the jump destination must be clearly defined in the **Label** column. With a jump command, you can also enter a cycle or quit it.

The only restriction is that you cannot use a jump command to go to a sub-program (procedure) or to return from a sub-program to the main program. **The jump destination must always be on the same program level!**

To realize conditional branches, use the GOTO command in the **Action** column.

5		DCH	1200.0 Watt	30.0 sec < -80.0 Ah < 9.8 V > 50.0 C	 GOTO End GOTO Charge GOTO Error	

Depending on which limit is reached, the test will branch at four different program points: Either to the next program step after 30 seconds or, before that, to one of the three labels "End“, "Charge“ or "Error“.

This permits a dynamic organization of the test process, based on battery parameters.

12.3.4 Special Labels

Function	Description
ONERROR	If an error is reached, the program jumps to this label. The program steps entered here will be executed.
ONEXIT	If the program end is reached, the program jumps to this label. The program steps entered will be executed.

ONERROR:
The entry "ONERROR" is made in the column **Operator**. The name of the label is to be entered into the nominal value column. In the case of an error, the programs skips to the step with that label name.

The program part provided for the case of an error should be placed at the end of the program, as all steps to the operator "STO" will be executed.

Example:

6		ONERROR	Error			
7		DCH	10.0 A	10.0 min		5.0 sec
8		CHA	80.0 A 14.4 V	20.0 min		1.0 V
9	Error	DCH	20.0 A	10.0 min		

Due to the ONERROR function in step 6, the battery will be discharged in the case of an error for a time of 10 minutes with 20 A.

ONEXIT:
The entry "ONEXIT" is made in the column **Operator**. The name of the label is to be entered into the nominal value column. The steps under the label will be executed after the end of the program.

Example:

6		ONEXIT	Exit			
7		DCH	50.0 A 14.4 V	20.0 sec		0.5 V
8	Exit	MSG	5			

The ONEXIT function in step 6 in this example ensures that a message is transmitted to the host computer after the end of the program.

13 MF-2000 Programs

13.1 Program Parameters

The program language of **"MicroForm 2000"** is a language that has specifically been developed for writing battery formation programs.

The parameters of this language are listed hereunder:

Operators:	**CHA, RCH, DCH, LOM, PAU, INT, SET, BEG, CYC, STO, M1, M0**
Nominal values:	Current in amperes **"A"**, voltage in **"V"**, Resistance in **"Ohm"**, wattage in **"Watt"**.
Nominal value ranges:	Current: 0.00x - xxxxxxxxxA 1) Voltage: 0.00x - xxxxxxxxV 1) Resistance: 0.00x - xxxxxxxOhm 1) Wattage: 0.00x - xxxxxxWatt 1)
Switch value ranges: (Termination)	Current: 0.00x - xxxxxxxxxA 1) Voltage: 0.00x - xxxxxxxxV 1) Time: 0 - 999:59:59h 1) Capacity: 0 - ±xxxxxxxxAh 1) Energy: 0 - ±xxxxxxxkWh 1) Temperature: -30.0 C to + 100.0 C 2) (marked by **">"** or **"<"**), e. g.: **>100 A**, **<200V**
Abort values:	same as for switch values
Registration:	same as for switch values

1) Maximum 10 characters available for the entry, incl. sign, decimal point, blank and dimension. The number of available places after the decimal point and the unit to be entered depend on the technical data of the power unit.

2) Only for units with temperature measuring channel (option).

13.2 General Program Data

Number of programs:	max. 999 programs
Number of fields:	max. 100 (1 field = 1 entry)
Program length:	max. 50 lines
Number of cycles:	max. 32000 (display max. 9999)
Control data:	current, voltage, wattage, resistance
Switch data:	current, voltage, time, temperature, capacity, and energy, additional channels (option) *)
Functions:	charge, discharge, recharge, pause, ramp and cycles

The indicated program data are only valid for the MF 2000 standard software for host computer operation. If a special software version is implemented on your host computer, the program data may differ considerably.

*) **Number of additional channels per circuit:** max. 80
All other program data same as stated above.

13.3 Operators

The following table shows a list of the operators available for the function control (**CHA**, **DCH**, **RCH**, **PAU**) and the additional operators for the process control (**BEG**, **SET**, **STO**, **INT**, **CYC**). Within a program, every operator is marking the beginning of a new program step.

Operator	Function	Signification in Program, Comments
CHA	Charge	Charge with programmed nominal values
RCH	Recharge	Charge with programmed nominal current value
DCH **LOM**	Discharge **LO**ad **M**anagement	Discharge with programmed nominal current value Operator to limit the current during peak load times. 2 channels are available, see description of **LOM**
PAU	Pause	Rest position, no input of nominal values allowed!
INT	Interrupt	Inactive status of the program, recommended e.g. for performing measurings manually when certain formation steps of the battery have been processed. The program can be continued by pressing the "**Start"** button at the power circuit (= power unit)
SET	Setting	Setting of new start values for the **Ah** and **kWh counters** at any point in the program. The entry is made in the column **Registration**. For example **"22.5 Ah"** or **"100 kWh"**.
BEG **CYC**	Beginning Cycle	Start marking for the beginning of a repetition for all following program steps till the end marking operator **CYC** (so-called program loop) End marking for the repetition of preceding program steps. (All program steps between the operators **BEG** and **CYC** will be repeated automatically). The number of cycles is to be stated in the **Nominal Value** column, e. g. with **"3*"** for three repetitions of the respective program steps.
M1	Begin	Start operator for single cell measuring for the following program step(s).
M0	End	End operator for single cell measuring
STO	Stop	End of program. This operator is automatically set at the end of each program.

Operators for function and process control of the power circuits

13.4 Nominal Values

The nominal values determine the general conditions for the different program steps. The entry in the program column **Nominal Value** is made in the following form:
<figure> <unit>, e.g. 500 A or 144 V.

Each entry of nominal value must always be made with value **and** unit. A different form of entry is only allowed if a ramp function is programmed. In this case, the entry must be made as follows: **5 – 200 A** for an increasing current or **100 – 2 A** for an decreasing current. See following program example.

Example: Program lines for a ramp function

Operator	Nominal Value	Termination	Abort	Registration
CHA	**5- 200 A**	1 h		5 sec
PAU		30 min		0.1 V
DCH	**100 - 2 A**	20 min		0.1 V

If several nominal values for one program step are entered one below the other, the control in the power unit will ensure that none of the values is exceeded. The following example shows a typical IU charge.

Example: Program lines for an IU charge

Operator	Nominal Value	Termination	Abort	Registration
CHA	**200 A 284 V**	880 Ah		30 sec
STO				

13.5 Switch Values (Termination)

The switch values determine the moment for the transition to the next program step. In principle, the following entries are permitted in the program column **Termination** (also see chapter **13.1 Program Parameters**):

Voltage:	Values in **V** with one digit after the decimal point	**1)**
Current:	Values in **A** with one digit after the decimal point	**1)**
Temperature:	Values in **degrees Celsius** with one digit after the decimal point: e. g. 98.5 C or - 30.0 C	**2)**
Capacity:	Values in **Ah** with one digit after the decimal point	**1)**
Energy:	Values in **kWh** with one digit after the decimal point	**1)**
Time:	Values in **hours : minutes : seconds** up to maximum "999:59:59h", "32767:59h" or "32767h". The time may as well be entered in the form of "30 sec" or "1,5 min".	

1)	The number of possible digits after the decimal point depends on the technical data of the power unit.
2)	Only if equipped with the respective option!

It is also allowed to set the sign **">"** (= greater than) or **"<"** (= smaller than) as a prefix to the switch value. If no prefix is entered, **">"** (greater than) will always be assumed for program steps with the operators **CHA** and **RCH**, and **"<"** (smaller than) for program steps with the operator **DCH**. If **several switch values** for one program step are entered one below the other, the one reached **first** will trigger the transition to the next program step.

Please note that during the charge steps (CHA) and recharge steps (RCH) the capacity and energy values are added and during discharge steps (DCH) they are subtracted. Therefore, the sign must be observed for the entry of capacity and energy values.

If a value in **Ah** or **kWh** is entered as switch value, please be aware of the fact that during the entire program process these values will be added in "charge steps" and subtracted in "discharge steps". Please take account of this "balancing" when entering the termination values. The capacity and energy values defined with the **SET** operator will also be taken into account at the respective point of the program. Therefore, you have to make sure that any switch value entered in Ah or kWh will not yet be reached in the preceding program step, because otherwise the program step with that switch value would be skipped.

The automatic adding of the Ah and kWh counter values at the beginning of each program step can be suppressed. For a reset of the counters after **each program step** you have to change the DIL switch setting. For further details see "DIL switch setting: 64" in the calibration instructions.

13.5.1 Number of Cycles, Cycle Termination Values

A cycle can only be terminated by the entry of the number of cycles with "*" in the **Nominal Value** column or by means of a step skip.

13.5.2 Fixed Step Times and Program Operation Times

The programming of fixed step and program operation times described in this chapter does not apply for software versions with single cell measurement. These can be distinguished in the master menu by their version number V**C** 10.**XXX**.XXX.

By means of the switch sign **"/"** it is possible to program fixed step times or fixed program operation times.

To program a fixed step time enter the "numerical value/time unit" as termination value after the operator **PAU** (= pause), e.g. 3/h. As time units also min or sec are permitted.

Example 1:

Operator		Nominal Value	Termination	Abort	Registration
1	DCH	400 A	210 V		
2	PAU		3/h		
3	CHA	200 A 288 V	200 Ah		
4	PAU		5/h		

The entry **3/h** in the second line limits the time for the first two program steps to maximum 3 h.

With the entries in program lines 3 and 4 the battery is charged to a termination value of 200 Ah according to IU characteristics. When the capacity of 200 Ah is reached, there will be a pause until the total time is 5 hours.

Important:
Make sure that the fixed step time **"x/h"** is long enough to ensure that the termination condition of the previous program step is reached within this step time.

Further to fixed step times, it is also possible to enter fixed program operation times. This mode of operation can be programmed with the **SET** operator and the termination value **"0/h"**. See line 2 in the following example. The program time meter is set back to zero and the time of the following program steps will be counted until a termination value entered as **"x/h"** is reached that deactivates this mode of operation. See second last line in the following example.

Example 2:

Operator	Nominal Value	Termination	Abort	Registration
1 DCH	200 A	2 h		
2 SET		0/h		
3 CHA	100 A	148 V		
4 CHA	148 V 100 A	< 5 A		
5 CHA	100 A	5/h		
6 DCH	200 A	105 V		

The above example shows:

If an operation time of 5 hours is not exceeded in the program steps 3 and 4, the second last step will be continued until the termination value of **"5/h"** is reached.

The 3 program steps with the **CHA** operator are in this case limited to a total time of 5 hours.

13.5.3 Charge Factor

The programming of steps with charge factor described in this chapter does not apply for software versions with single cell measurement. These can be distinguished in the master menu by their version number V**C** 10.**XXX**.XXX.

If certain current quantities are discharged from fully charged batteries, the current quantities required for a subsequent full charge will be higher. The relation of the current quantity required for the full charge to the current quantity discharged before (= charge factor) can be taken into account in program steps with the **CHA** operator.

Example 1:

Operator	Nominal Value	Termination	Abort	Registration
DCH	100 A	1 h		
PAU		30 min		
CHA	100 A 144 V	110 %Ah		

At the start of each program step (except pause steps) the counts of the Ah counter and of the kWh counter are automatically set back to the start values **"0 Ah"** and **"0 kWh"**, if the DIL switch "64" is set accordingly. (See calibration instructions MicroForm 9000, page 4-3.)

In the above example, the Ah counter has counted -10 Ah before the start of the second pause step. If a charge factor of **"1.1"** shall apply for the subsequent charge step, this must be programmed by entering the termination value **"110 %Ah"**. In this case the charged current quantity refers to the current quantity discharged in the last discharge step before.

A programmed charge factor may also apply to several charge steps if the **SET** operator with the termination value **"0 %Ah"** has been entered before. This entry will set the Ah counter to 0 Ah and the current quantities will be added till the next termination value entered as **" %Ah"**.

Example 2:

Operator		Nominal Value	Termination	Abort	Registration
1	DCH	200 A	2 h		
2	SET		0 % Ah		
3	CHA	100 A	148 V		
4	CHA	148 V	< 5 A		
		100 A			
5	CHA	100 A	110 %Ah		

In the above example, the Ah counter shows a value of -400 Ah after the discharge step. Due to the termination value of 110 %Ah (⇔ charge factor 1.1) the battery will be charged with a total of 440 Ah.

Example 3:

Operator		Nominal Value	Termination	Abort	Registration
2	DCH	200 A	105 V		
3	CHA	100 A	110 % Ah		
4	CYC	10 *			

Example 3 shows a programming of cycles with the two program steps to be repeated ten times. If the charge factor is matched for the battery to be tested, the latter will have the same charge level at the begin of each cycle. Therefore, the cycle times will be constant, although no discharge time has been entered.

Example 4:

Operator		Nominal Value	Termination	Abort	Registration
2	DCH	200 A	2 h		
3	SET		0/h 0%Ah		
4	CHA	100 A	148 V		
5	CHA	148 V 150 A	< 5 A 110%Ah		
6	CHA	120 A	5/h		
7	CYC		300 h		

Example 4 shows a cyclic repetition of one discharge step followed by charge steps. After the discharge the Ah counter shows a value of -400 Ah. The termination values **0/h** and **0%Ah** after the **SET** operator determine the start conditions for the following charge steps: time counter and Ah counter are reset to zero. The time of the first charge step until the termination value of 148 Volts is unknown. This also goes for the second charge step. The latter will only be terminated when the charge current is less that 5 A or when the charged current quantity in both charge steps has reached 440 Ah (= charge factor 1.1). The time of the third charge step is automatically limited to the effect that the total charge time is 5 hours and the time of a cycle 7 hours.

13.6 Abort Values

If one of the values entered into the program column **Abort** is reached, the program operation will be interrupted immediately. It is distinguished between exceeding (marked by **">"**) and falling below (marked by **"<"**).

The program operation may be continued after an interruption. This can be done by a new start command at the host computer or by pressing the start button at the formation circuit. If an abort value is reached, this will be indicated by a message on the screen stating the respective circuit number.

Abort values should always be entered if undesired battery conditions are likely to be expected. In principle, you may enter into the program column **Abort** the same values as those allowed as switch values mentioned in chapter 13.5.

13.7 Registration

Values entered into the program column **Registration** determine the conditions under which or the time when the measured values will be registered. If **no** entry is made in this column, the measured values will only be registered **at the beginning and at the end of each program step**. This will be the same for program steps within cycles. In principle, the same entries as for the switch values are allowed as registration conditions.

The most common entry is normally a time value. A time entered as registration condition means a time interval after which one measuring data line each will be created.

If time intervals are entered for the registration you should observe the length of the respective program step. If you choose your registration conditions appropriately you will get clear and informative data files, avoiding the printout of endless lists of measured data.

An entered time value may also be defined as an absolute value if it is preceded by an equal sign "=". In this case, an additional registration will be made when the entered time has elapsed since the start of the program step. If several time values are entered for the same program step **with or without** the equal sign "=", the entered time interval will be fixed again when the absolute value has been reached. See "**Example for registration with absolute value**" on the next page.

By means of the **SET** operator, you may enter random start values for the following program step for the amphour counter and the watthour counter. It is as well possible to take over the counting from the preceding program step, just by entering the **SET** operator before the new program step.

Example for a program line with "SET" registration condition

	Operator	Nominal Value	Termination	Abort	Registration
1	SET				440 Ah
2	CHA	100 A	148 V		

In the above example the counter is set to a value of 440 Ah by means of the **SET** operator, thus fixing the start value of the internal amphour counter for the following program step.

Example for a program line with "SET" operator

	Operator	Nominal Value	Termination	Abort	Registration
1	DCH	200 A	2 h		
2	SET				
3	CHA	100 A	148 V		

In this example the **SET** operator is used to take over the counter values from the preceding step as start values for the next following charge step.

Example for registration with absolute value

Operator	Nominal Value	Termination	Abort	Registration
1 CHA	**200 A**	**2 h**		**10 V**
2 DCH	**100 A**	**55 V**		**2 min = 180 sec**

Registration in the second program step:

1. Registration at the start of the step
2. Registration after 2 minutes
3. Registration after 180 seconds
4. Registration after 5 minutes (= 180 sec + 2 min)
5. Registration after 7 minutes

etc.

13.7.1 Extended Registration Conditions

General Description:

The software for the formation system MicroForm 2000 has been extended. As from the software version VC10.001.010 (delivery from March 1996) it is possible to program three different degrees of registration:

- Degree 0: no registration Entry: 0 Reg
- Degree 1: registration at the end of a program step or a cycle Entry: 1 Reg
- Degree 2: standard registration Entry: 2 Reg

The degrees of registration may be entered for the whole program, for certain program steps or for a single program step.

1. Degrees of registration for the whole program

The entry is made with the **SET** operator in the program column **Registration**.

Operator	Nominal Value	Termination	Abort	Registration
1 SET				0 Reg
2 SET				1 Reg
3 SET				2 Reg

The respective entry is valid until a new degree of registration is entered with the **SET** operator.

2. Degree of registration for a single program step

The entry is made in the program column **Registration** of the respective program step.

Operator	Nominal Value	Termination	Abort	Registration
4 CHA	50 A 14.4 V	105 Ah		0 Reg
5 DCH	20 A	10.5 V		1 Reg
6 CHA	10 A 14.4 V	15 h 18		2 Reg

The respective entry is only valid for the respective step and it has priority over the entry as per para. 1. It is thus possible to program random program steps with their own degrees of registration.

3. Examples for Registration Conditions

11-MAR-1996 10:49:09 MF		Program		Modify
Program No. : 3 - Program with 100 lines and GOTO Capacity : 100 Ah			Time : 0.30 h	
Operator	Nominal Value	Termination	Abort	Registration
1 CHA	50 A	180 min 1 h		0 Reg 1 sec

No registration is made in this step, due to the fact that the entry "0 Reg" invalidates the registration condition "1 sec".

Operator	Nominal Value	Termination	Abort	Registration
1 SET				0 Reg
2 CHA	15 A 14.4 V	105 Ah		2 Reg 1 sec
3 DCH	40 A	<10.5 V		1 sec

No registration is made in the first step.
In the second step a registration is made at the beginning and at the end of the program step and during the step in intervals of one second.

No registration is made in the third step due to the fact that the entry "0 Reg" in the first step is also valid for the third step.

Operator	Nominal Value	Termination	Abort	Registration
4 BEG				1 Reg

5 DCH	1000 Watt	< 11 V 1 h		

After the **BEG** operator only the entry "1 Reg" has a function. This is also valid for the **CYC** operator.

9 CYC	150 *			1 Reg

With these two entries it is possible to have only the measuring values at the beginning and at the end of a cycle registered, under the condition that "0 Reg" has been entered with the **SET** operator before the beginning of the cycle.

13.8 Program Examples

13.8.1 Input of Formation Programs

When a formation program is created, the system will automatically add a program step number to each line that begins with an operator. This program step number is automatically corrected if lines are added or deleted. The program step numbers will also appear on the printouts.

In the case of formation programs with a length of several hundred steps this number will give you direct information on the present position within the formation program.

13.8.1.1 Examples of Formation Programs

11-MAR-1996 9:09:22 MF	Program.	Modify

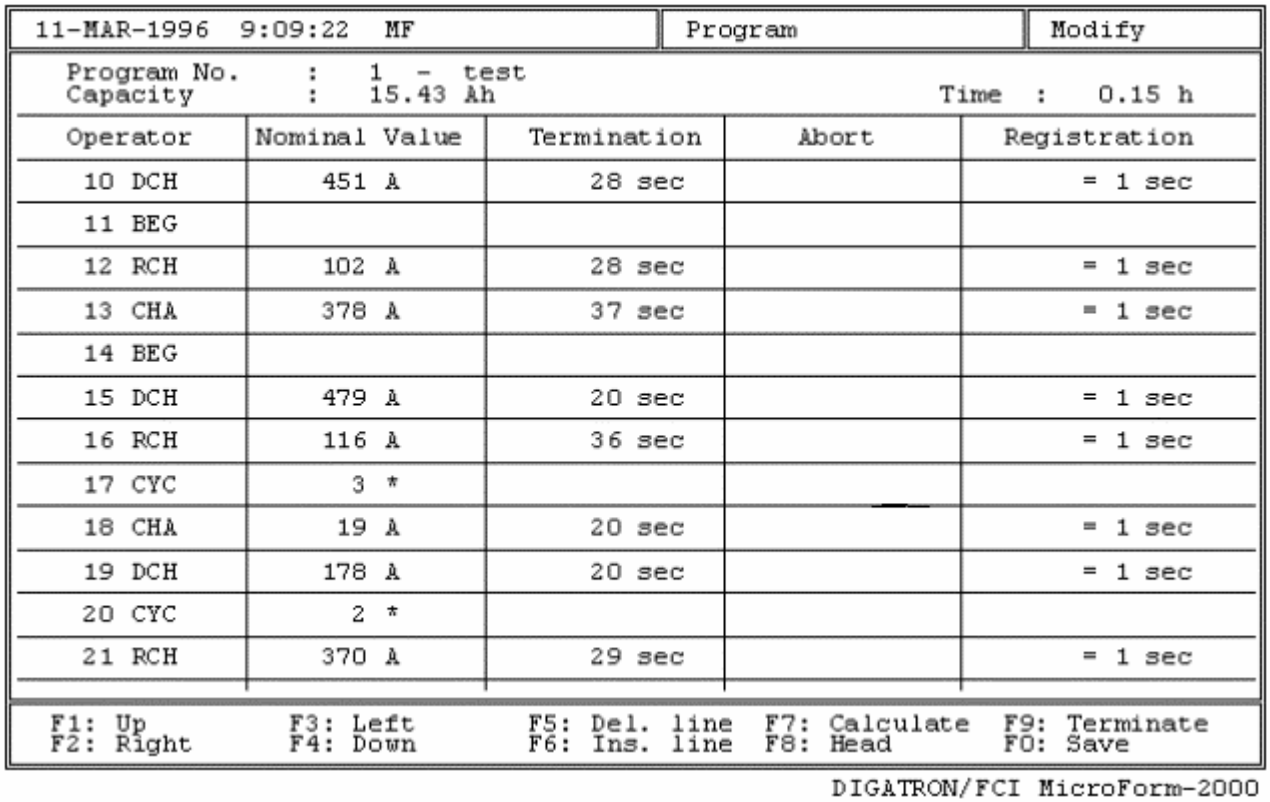
Program No. : 1 - test
Capacity : 15.43 Ah Time : 0.15 h

Operator	Nominal Value	Termination	Abort	Registration
1 DCH	10 A	20 sec		= 30 sec
2 DCH	430 A	20 sec		= 30 sec
3 RCH	136 A	26 sec		= 30 sec
4 CHA	159 A	20 sec		= 30 sec
5 CHA	186 A	20 sec		= 30 sec
6 DCH	41 A	20 sec		= 30 sec

F1: Up F3: Left F5: Del. line F7: Calculate F9: Terminate
F2: Right F4: Down F6: Ins. line F8: Head F0: Save

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Program step numbers in a formation program



11-MAR-1996 9:09:22 MF | Program | Modify

Program No. : 1 - test
Capacity : 15.43 Ah
Time : 0.15 h

Operator	Nominal Value	Termination	Abort	Registration
10 DCH	451 A	28 sec		= 1 sec
11 BEG				
12 RCH	102 A	28 sec		= 1 sec
13 CHA	378 A	37 sec		= 1 sec
14 BEG				
15 DCH	479 A	20 sec		= 1 sec
16 RCH	116 A	36 sec		= 1 sec
17 CYC	3 *			
18 CHA	19 A	20 sec		= 1 sec
19 DCH	178 A	20 sec		= 1 sec
20 CYC	2 *			
21 RCH	370 A	29 sec		= 1 sec

F1: Up F2: Right F3: Left F4: Down F5: Del. line F6: Ins. line F7: Calculate F8: Head F9: Terminate F0: Save

DIGATRON/FCI MicroForm-2000

Interlaced cycles

Explanation:

The first cycle begins with program step number 11, ends with step number 20 and shall be executed two times. The second cycle begins with formation step number 14, ends with formation step number 17 and shall be executed three times. The second cycle will be executed six times in total, as it is part of the first cycle.

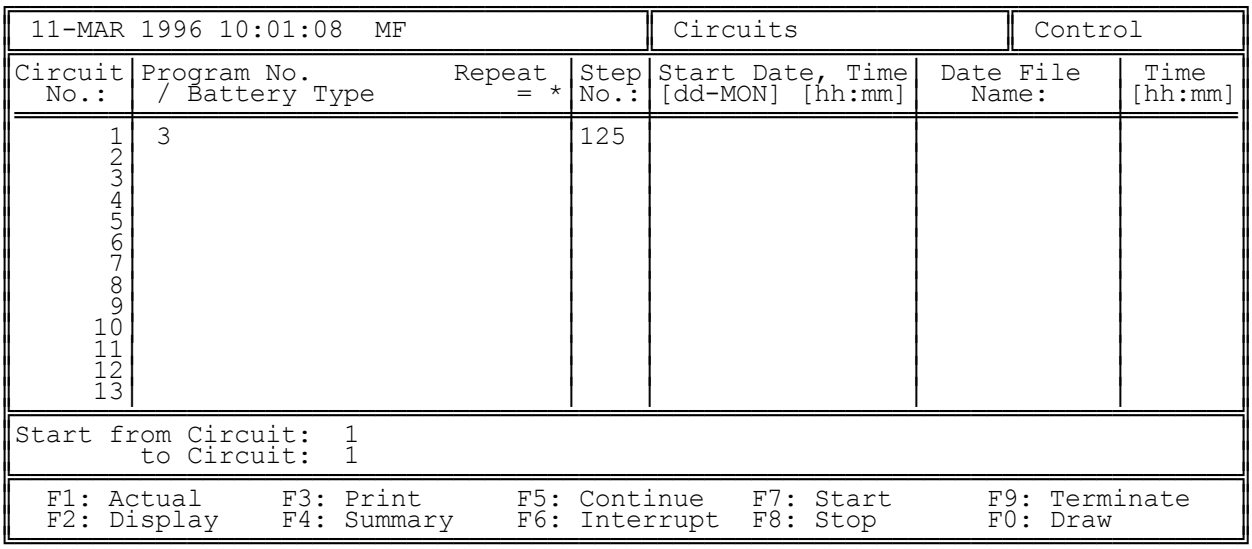
13.8.2 Start of a Formation Program

If a program is started directly with program step 1, the operation is the same as described in chapter 6.2.1 of the MF-2000 manual. If, however, the formation program shall start with a certain step, you have to enter the respective step number into the column Step No.:.

Please note:

Depending on the length of the formation program it may happen that the program run at the formation circuit starts only after a few minutes (approx. 5 minutes at maximum program length).

13.8.2.1 Example for the Start of a Formation Program



Start with Step Number 125

13.8.3 Step Skipping in a Formation Program

The different program steps are, as explained under 1.1, numbered and thus defined. Any general termination value may be used to trigger the skipping to another step. Enter the target step number directly after the entry of the respective termination value or nominal value, leaving a blank between the value and the step number. This is also valid for the entry of the number of cycles into the nominal value column.

Please note:

The entry of the target step number must always be separated by a blank, as otherwise it will not be possible for the program compiler to identify it.

13.8.3.1 Admissible Step Skips

In general, skipping is allowed to any step in the program that has been assigned with a step number, i.e. also to those steps that do not involve a direct action, such as **BEG** or **SET**. Skipping out of a cycle or into a cycle is also possible, however, under respect of certain conditions. See chapter 3.5 of the MF-2000 manual.

A target step number may be added to the following termination values or conditions:

- Current values in amperes (A)
- Voltage values in volts (V)
- Temperature values in degrees Celsius (C)
- Capacity values in amperehours (Ah)
- Energy values in watthours (Wh)
- Capacity values with charge factor (%Ah)
- Fixed program run times in hours (/h)
- Number of cycles ("*")
- Switch values for additional channels, e.g. 1.7V1

13.8.3.2 Syntax

The entry of termination values to limit a step is explained in the manual. To include a step skip in the program, add the desired target step number to the termination value, leaving a blank between the termination value and the step number. If a step shall be terminated for example as soon as the voltage is higher than 14.4 Volts, you have to enter **14.4 V** into the termination value column. If, however, you wish to skip to another but the following step, e.g. step number 15, when this voltage is reached, you have to enter **14.4 V 15**. The next chapter shows some examples for the input.

13.8.3.3 Input Examples

Operator	Nominal Value	Termination	Abort	Registration
		10 A 5		
		10 min 20		
		>14 V 6		
		<50 Ah 3		
		>10 Wh 1		
		1.7 Vz 15		
		etc.		

13.8.3.4 Program Example for the Skip Function (GO TO)

11-MAR-1996 10:49:09 MF		Program		Modify
Program No. : 3 - Program with 100 lines and GOTO Capacity : 100 Ah			Time : 0.30 h	
Operator	Nominal Value	Termination	Abort	Registration
9 BEG				
10 DCH	1000 Watt	< 11 V 14 1 h		1 min

If the voltage falls below 11 Volts before the elapse of one hour, the program will be continued with step number 14, otherwise it will be continued with the next following step.

11 DCH	100 A	10 min		1 min
12 CHA	15 A 14.4 V	105 Ah		1 Ah
13 PAU		1 h 15		

After elapse of one hour, the program will be continued with step number 15.

14 CHA	20 A 14.4 V	105 Ah		1 Ah
15 DCH	20 A	10.5 V		1 Ah
16 CHA	10 A 14.4 V	130 %Ah 15 h 18		1 Ah

If the step time reaches 15 hours before 130 %Ah of the preceding discharge step are reached, step 16 will be terminated and the program continued with step 18.

17 CYC	150 * 1			

When all 150 cycles were executed successfully, the program will start again with step 1.

F1: Up	F3: Left	F5: Del. line	F7: Calculate	F9: Terminate
F2: Right	F4: Down	F6: Ins. line	F8: Head	F0: Save

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13.8.3.5 Special Features of the Skip Function

If a cycle is discontinued due to a step skip, the present value of the cycle counter will be memorized and counting will be continued when the program goes back into this cycle.

If a cycle is discontinued before the **CYC** step is reached, the cycle counter will not continue counting.

If the regular program run reaches a step with **BEG** or if the program skips to a step with **BEG**, the respective cycle counter will be set to "1".

13.8.4 Ramp Function with Nominal Power Values

In the same way as the programming of a ramp function with increasing or decreasing current, it is also possible to enter nominal power values. See example in chapter 13.4.

Operator	Nominal Value	Termination	Abort	Registration
CHA	500 - 1500 Watt	1 h		5 sec
PAU		30 min		0.1 V
DCH	1000 - 200 Watt	20 min		0.1 V

Please note that it is not allowed to abbreviate the unit "Watt" and that it must be entered after the **second** numerical value.

Options

Option: Cooling Water Control System

The formation rectifiers are prepared with two temperature probes T1 and T2 for each power unit cabinet via water bath controlling.

Using the **SET** operator within a formation program, a potential-free relay output can be activated. This switching output remains activated during a program interruption (BRK or INT) and after an error ER1 to ER5 or BRK occurs.

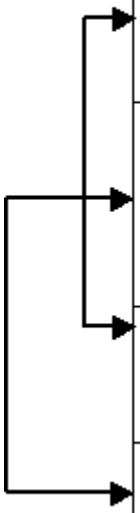
Depending on the entry, the **relay cooling water** or **relay delta temperature** is set.

The following picture shows some examples for activating potential-free relay outputs. If several entries are made together with one **SET** operator (see program step 3), the order of entries is irrelevant.

Additional entries in the **Abort** or **Registration** column are not allowed. For more details on the **Nominal Value** column, see **Option: Temperature Regulation**.

Step Number	Operator	Nominal Value	Termination	Abort	Registration
1	SET		> 80.0 C		
2	SET		< 70.0 C		
3	SET		> 80.0 C < 70.0 C		
4	SET		12.0 C		
5	STO				

Overview: Operators for Temperature Control



Operator	Termination	Action
SET	**> XX.X C**	**Relay Cooling Water Temperature** activated when T1>XX.X°C **or** T2>XX.X°C **Relay Cooling Water Temperature** deactivated when T1<XX.X°C**-2°C and** T2<XX.X°C**-2°C** Due to the switching hysteresis, the threshold moves by 2°C.
SET	**< XX.X C**	**Relay Cooling Water Temperature** deactivated when T1<XX.X°C **and** T2<XX.X°C **Relay Cooling Water Temperature** activated when T1>XX.X°C**+2°C or** T2>XX.X°C**+2°C** Due to the switching hysteresis, the threshold moves by 2°C.
SET	**> XX.X C** **< YY.Y C**	**Relay Cooling Water Temperature** activated when T1>XX.X°C **or** T2>XX.X°C **Relay Cooling Water Temperature** deactivated when T1<YY.Y°C **and** T2<YY.Y°C
SET	**XX.X C**	**Relay delta Temperature** activated when |T1-T2|>XX.X°C **Relay delta Temperature** deactivated when |T1-T2|<XX.X°C

The automatic switching hysteresis of 2°C is only valid when using the operator for the first time. If parameter changes are made afterwards through other **SET** commands in the program, upper and lower value always have to be redefined.

Option: Temperature Regulation

Besides the common current regulation it is also possible to use the measured temperature as regulating parameter.

The next table shows how the regulating parameters for the temperature regulation are entered into the **Nominal Value** column. Quantities are entered with the ending "C". Additionally, the **SET** operator has to be written into the **Operator** column.

Operator	Nominal Value	Action
SET	**XX.X C**	Nominal temperature value in degree centigrade
SET	**< XX.X C**	Temperature threshold value in degree centigrade: when exceeding this value, the on-load current is abruptly reduced by the with **= 0.X C** adjustable part of the present nominal current.
SET	**= 0.X C**	Reduction factor for the present nominal current when exceeding the threshold value < XX.X C. The new, reduced nominal value is calculated by: $I_{red} = I_{soll}-(I_{soll}*0.X)$. Adjustable range 0.10<X<0.99
SET	**> 0.X C**	Adjustable I-portion of the temperature regulator: Increasing the value: faster tracking Decreasing the value: slower tracking Adjustable range 0.01<X<0.99

All entered values are valid for the entire program starting with the belonging **SET** operator, but can be overwritten by a new **SET** command.

The **SET** command can also be used together with a time x that has to be entered in the **Termination** column. The relay **cooling water** is then set for the time x and entirely independent from the actual temperature and other parameters.

Parameters can also be written en bloc with only one **SET** operator:

Step Number	Operator	Nominal Value	Termination	Abort	Registration
1	SET	77.0 C < 75.0 C > 0.3 C = 0.1 C			
2	CHA	60 A	3:00:00 h		
3	STO				

1 Starting Phase

During the starting phase after the program start, charging is done constantly with the preset current value. If the measured temperature exceeds the temperature threshold value **<XX.X C**, the current is abruptly decreased by the reduction factor **= 0.X C** (see the following pictures).

2 Monitoring Phase

During the monitoring phase the current is kept constant. It is checked whether

- the temperature increase becomes negative

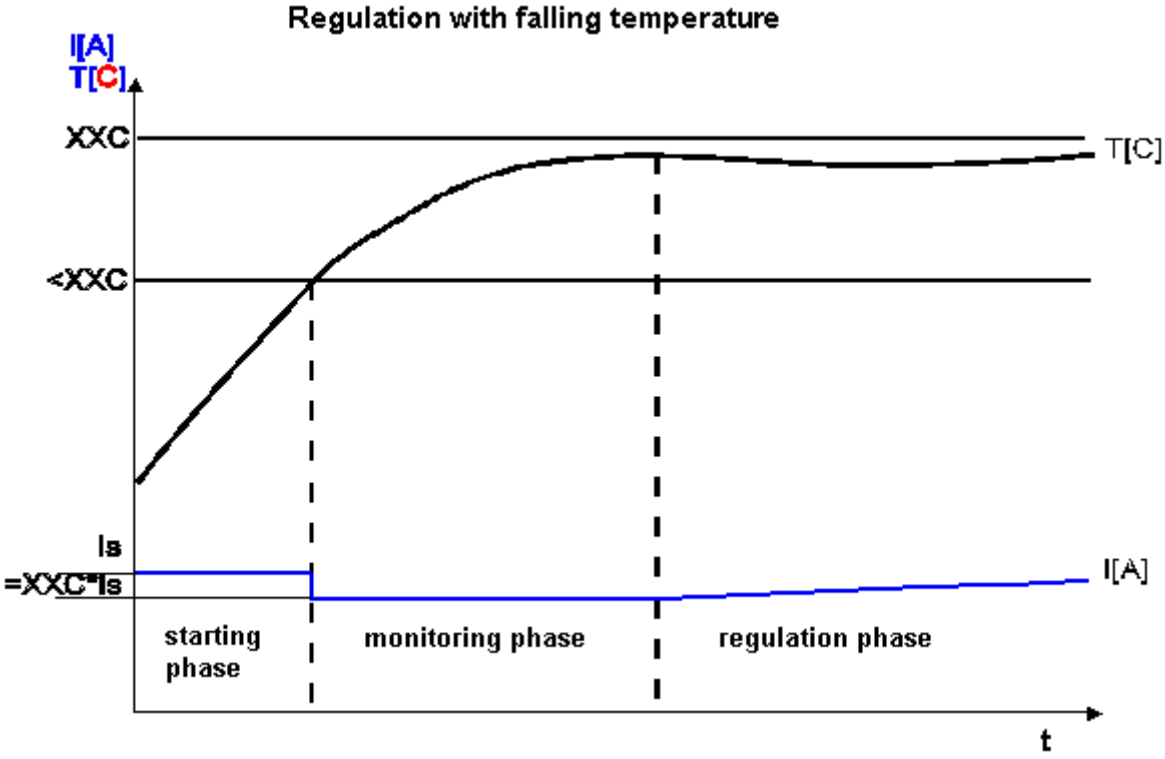
- the nominal temperature value is exceeded

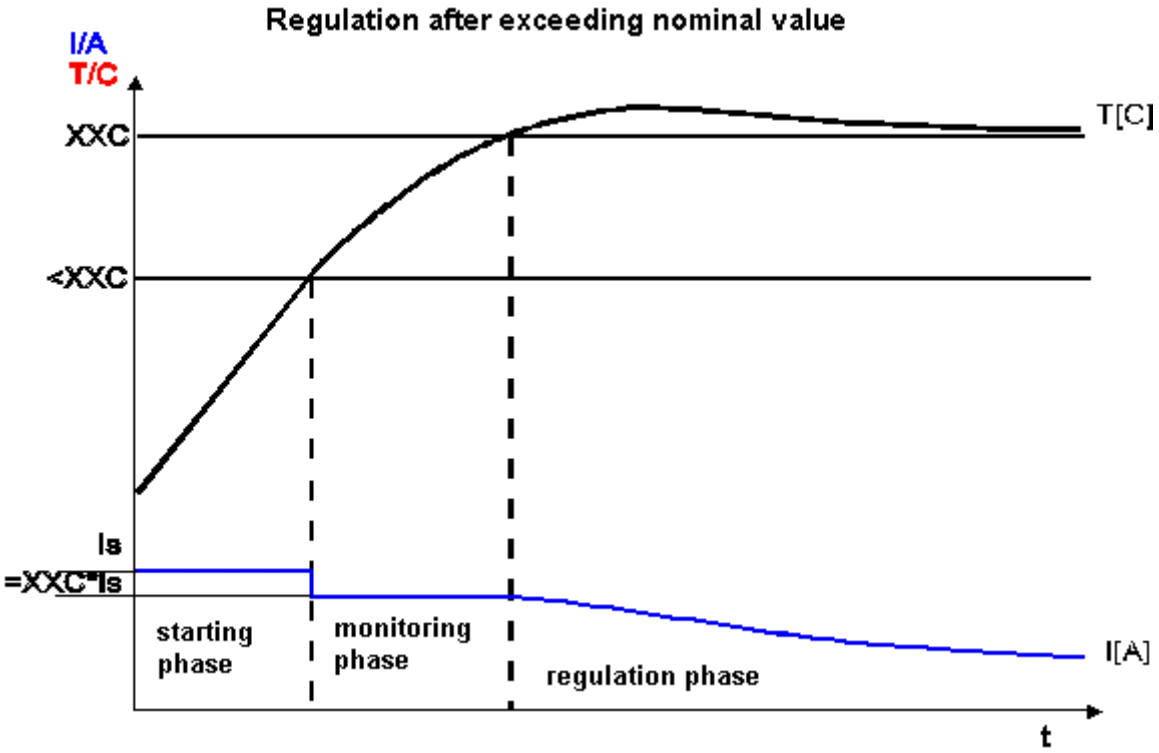
Should one of these conditions be fulfilled, the actual temperature regulation phase begins.

3 Regulation Phase

Now the temperature is regulated to its nominal value. The adjustable I-part of the temperature controller **> XX.X C** influences how strong the regulator interferes into the process. The influence is limited to three times the jump value of the entered reduction factor **= 0.X C** during starting phase, but not greater than 75% of the present nominal current.

If the temperature descents **0.5 °C** below the temperature threshold value **<XX.X C**, the full nominal current is released.





This allows to run charge and discharge pulses within the full scale current range with minimum pulse widths of one second. In addition there is a special software algorithm to detect the "State of Charge (SOC)" of the battery string during a certain pulse formation period.

Option: SOC detection

With this option there is a special software algorithm to detect the "State of Charge (SOC)" of the battery string during a certain pulse formation period.

1 SOC by delta I

1.1 Theory of the "SOC delta I detection" algorithm

Running constant resistance discharge pulses in-between the initial charge period, the current amplitude will continuously increase by each discharge pulse due to the increasing SOC of the battery string. The SOC detection algorithm indicates if the increase of the actual current amplitude compared to the previous one is less than a predefined difference e.g. 0.2 A. This condition has to be proved for a free programmable time period of e.g. 90 min.

1.2 The "SOC delta I" in Battery Manager

Three additional parameters are necessary to set up the **SOC detection**. The parameters are **dIsec**, **dIa** and **dIb**.

Parameter	Description
dIsec	**d**elta **I** minimum in **sec**onds
dIa	**d**elta **I** of current between two pulses, function **a**
dIb	**d**elta **I** of current between first pulse which was inside **dIa** value and last measured pulse, function **b**

The parameter **dIsec** defines the time in seconds where the increase of the current amplitude compared to the previous one should be less than a predefined difference **dIa**.

The parameters **dIsec** and **dIa** are necessary to run the **SOC detection**.

For safety reasons parameter **dIb** can be used in addition to terminate the time period **dIsec** in case of unexpected current increase.

Parameter **dIb** evaluates the difference between the actual measured discharge amplitude and the first measured discharge amplitude when the **dIa** condition became true.

The following example shows how to implement pulse charge/discharge steps in a formation schedule including the **SOC detection**. Please note that the formation schedule does not reflect a real battery formation with respect to the programmed time. But it yields sufficient information to understand the **SOC detection**.

The following test is made at one circuit of the MCDR 30-360-12 with 13 batteries connected.

1.3 Pulse formation schedule with "SOC"

Step Number	Operator	Nominal Value	Termination	Abort	Registration
1	SET	60 dlsec			
2	DCH	5 A	2 min		0.1 A 0.5 V
3	CHA	5 A	2 min		0.1 A 0.5 V
4	BEG				
5	CHA	20 A 187 V	3 sec		0.1 A 0.5 V
6	CHA	0 A	2 sec		0.1 A 0.5 V
7	CYC	10 *			
8	BEG				
9	CHA	20 A 187 V	2 sec		0.1 A 0.5 V
10	DCH	10 Ohm 30 A	1 sec < 0.5 dla > 1.0 dlb		
11	CYC	1000 *			
12	BEG				
13	CHA	20 A 187 V	20 sec		0.1 A 0.5 V
14	DCH	10 A	5 sec		0.1 A 0.5 V
15	CYC	10 *			
16	DCH	20 A	20 sec		0.1 A 0.5 V
17	CHA	20 A	20 sec		0.1 A 0.5 V
18	DCH	30 A	1 min		0.1 A 0.5 V
19	STO				

1.4 Explanation of program steps

Step 1

1	SET	60 dlsec			

Minimum time for delta I detection is set to 60 seconds.

Steps 2 and 3

2	DCH	5 A	2 min		0.1 A 0.5 V
3	CHA	5 A	2 min		0.1 A 0.5 V

Standard discharge and charge.

Steps 4 to 7

4	BEG				
5	CHA	20 A 187 V	3 sec		0.1 A 0.5 V
6	CHA	0 A	2 sec		0.1 A 0.5 V
7	CYC	10 *			

Pulse charge 3 seconds with 2 seconds pause.

Steps 8 to 11

8	BEG				
9	CHA	20 A 187 V	2 sec		0.1 A 0.5 V
10	DCH	10 Ohm 30 A	1 sec < 0.5 dla > 1.0 dlb		
11	CYC	1000 *			

Pulse charge 2 seconds with 1 second constant resistance discharge pulse at 10 Ohm.
The cycle will be terminated by **SOC detection**. The cycle will be terminated when the current difference (delta I) between two discharge pulses is less than 0.5 A and when this condition was due for 60 seconds, or the current difference (delta I) between the first pulse within 0.5 A and any other discharge pulse will be greater than 1.0 A.

Steps 12 to 15

12	BEG				
13	CHA	20 A 187 V	20 sec		0.1 A 0.5 V
14	DCH	10 A	5 sec		0.1 A 0.5 V
15	CYC	10 *			

10 times pulse charging with 20 seconds charge and 5 seconds discharge.

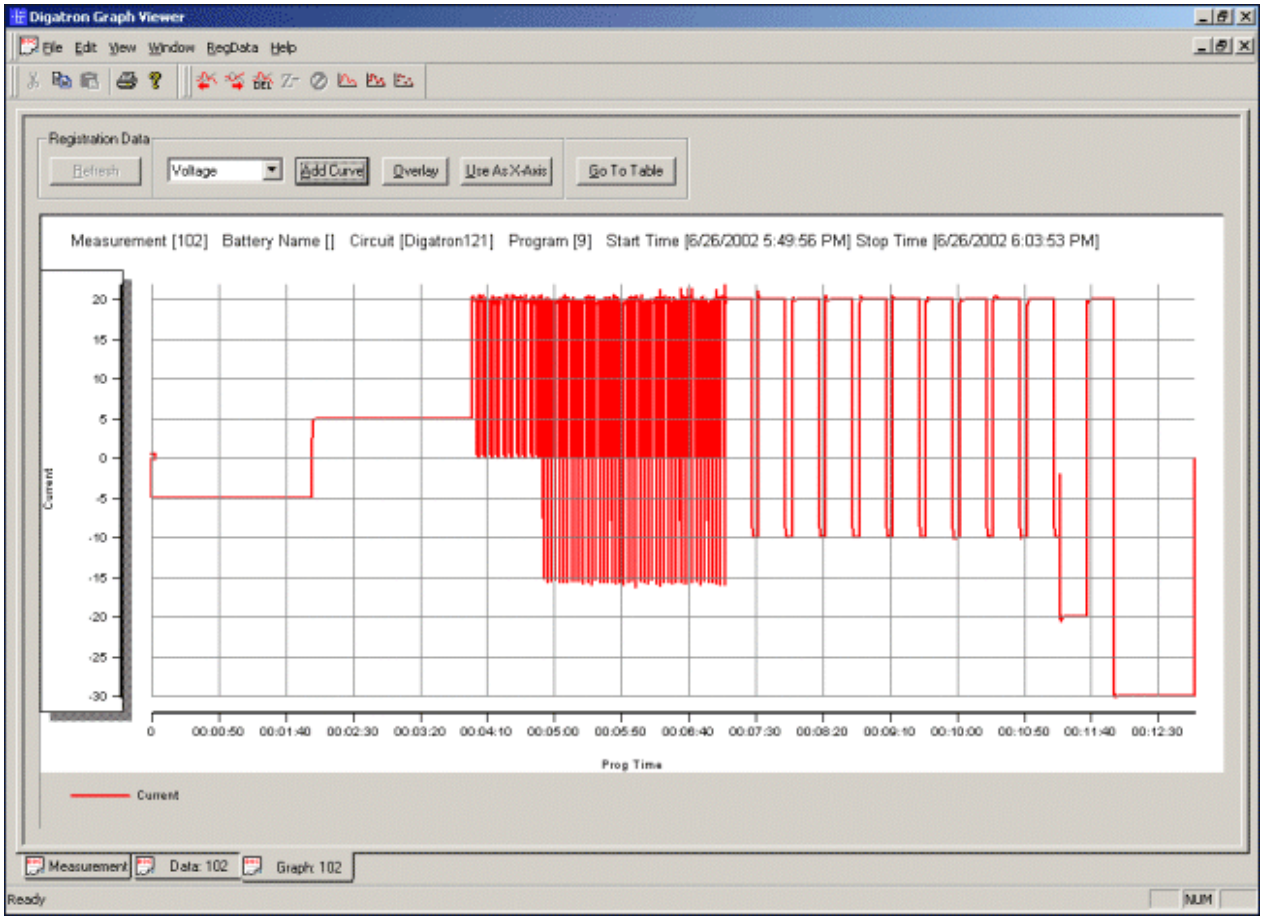
Steps 16 to 18

16	DCH	20 A	20 sec		0.1 A 0.5 V
17	CHA	20 A	20 sec		0.1 A 0.5 V
18	DCH	30 A	1 min		0.1 A 0.5 V

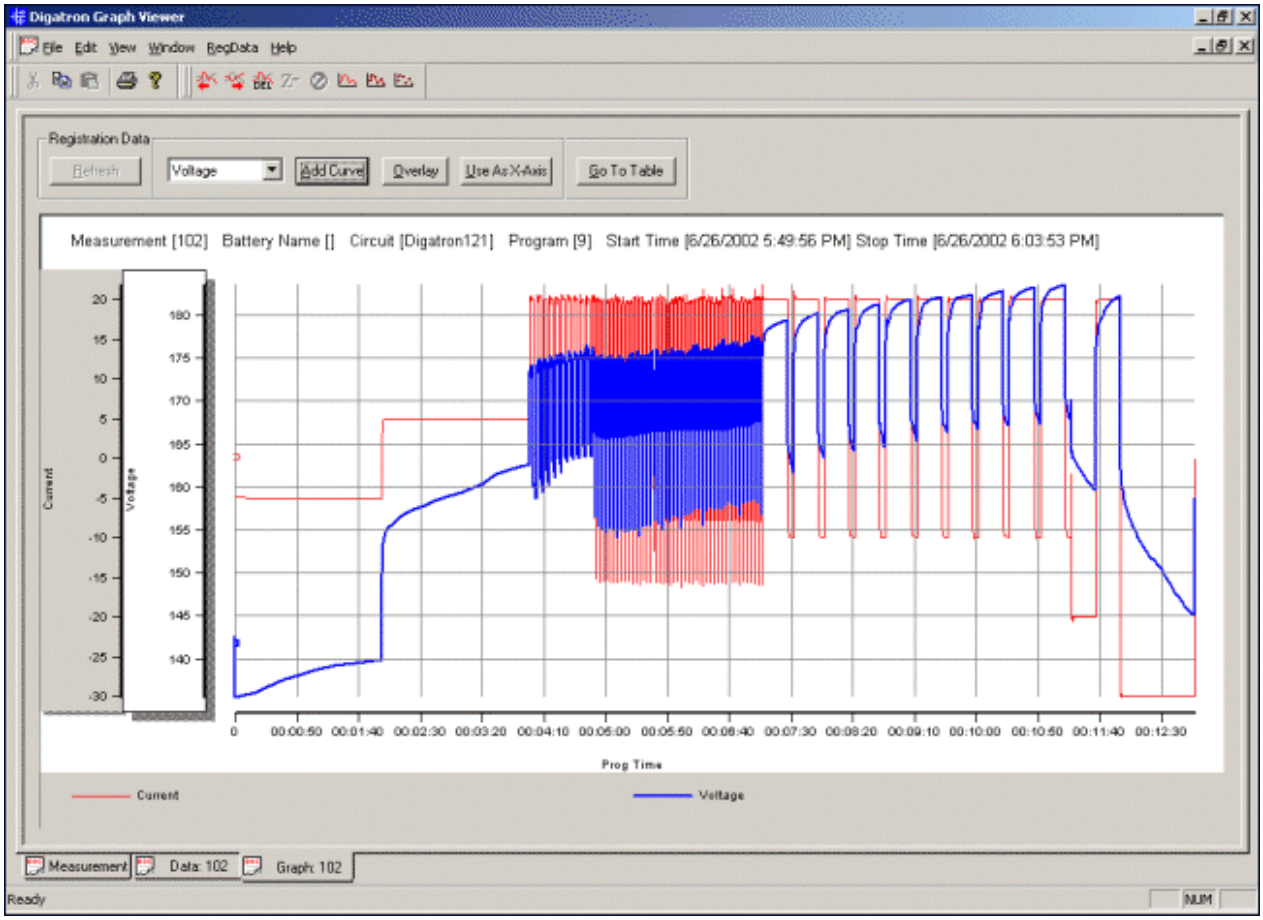
Standard charge and discharge steps.

1.5 Results

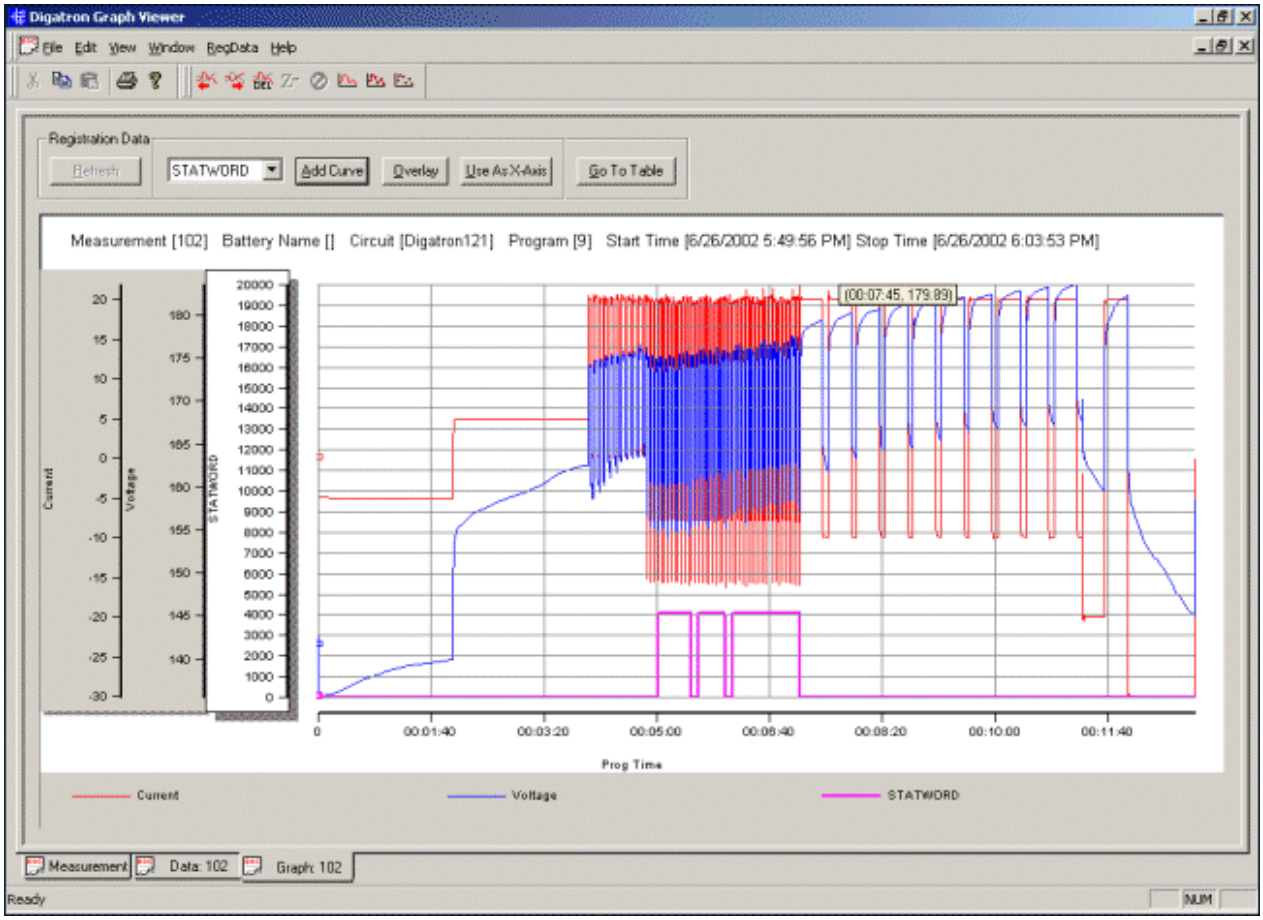
Graph, showing the current



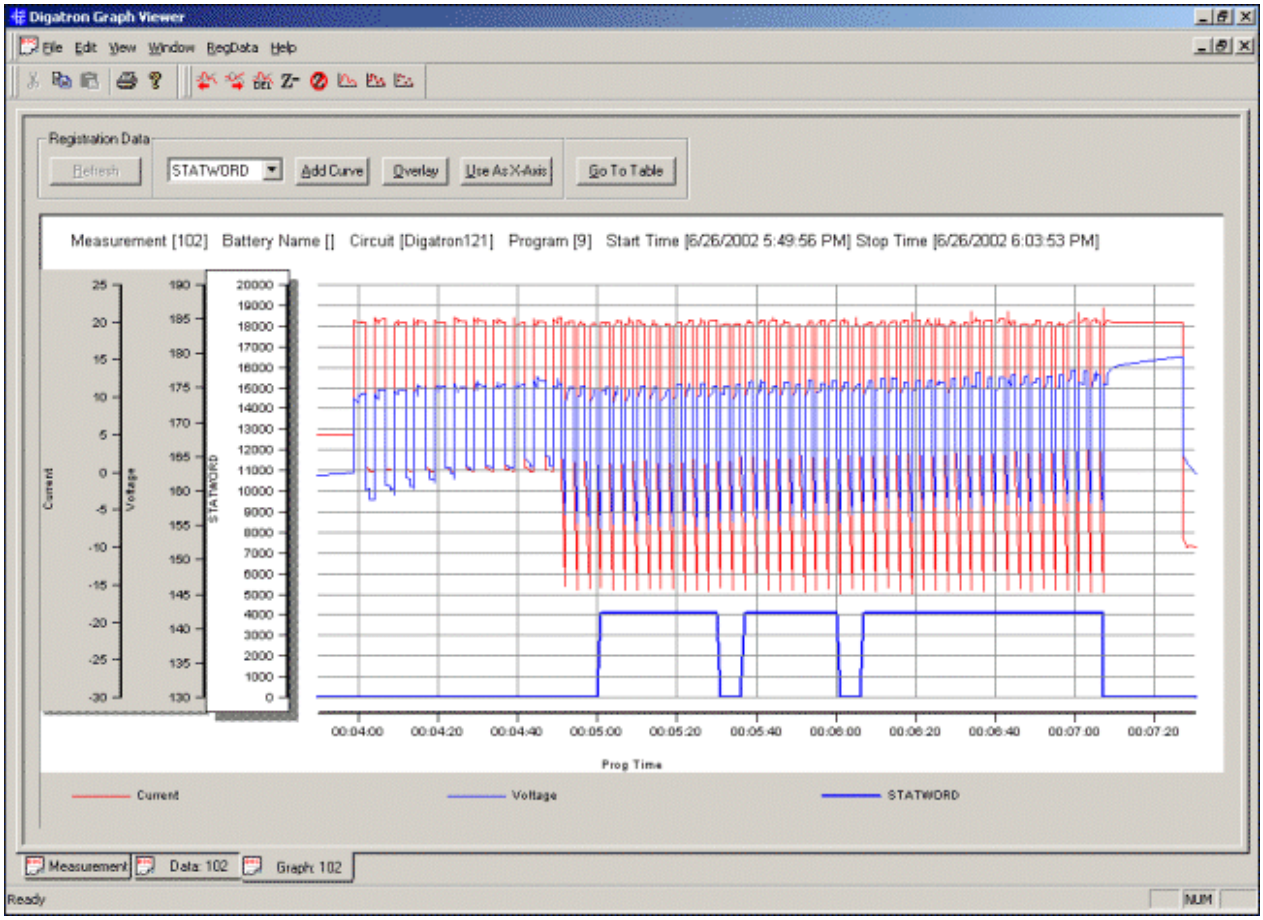
Graph, showing current and voltage



Graph, showing current and voltage with timer

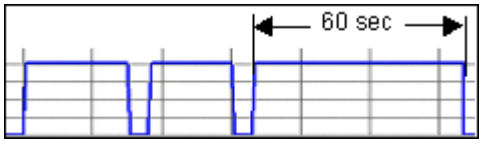


Graph, showing current and voltage with timer (zoomed)

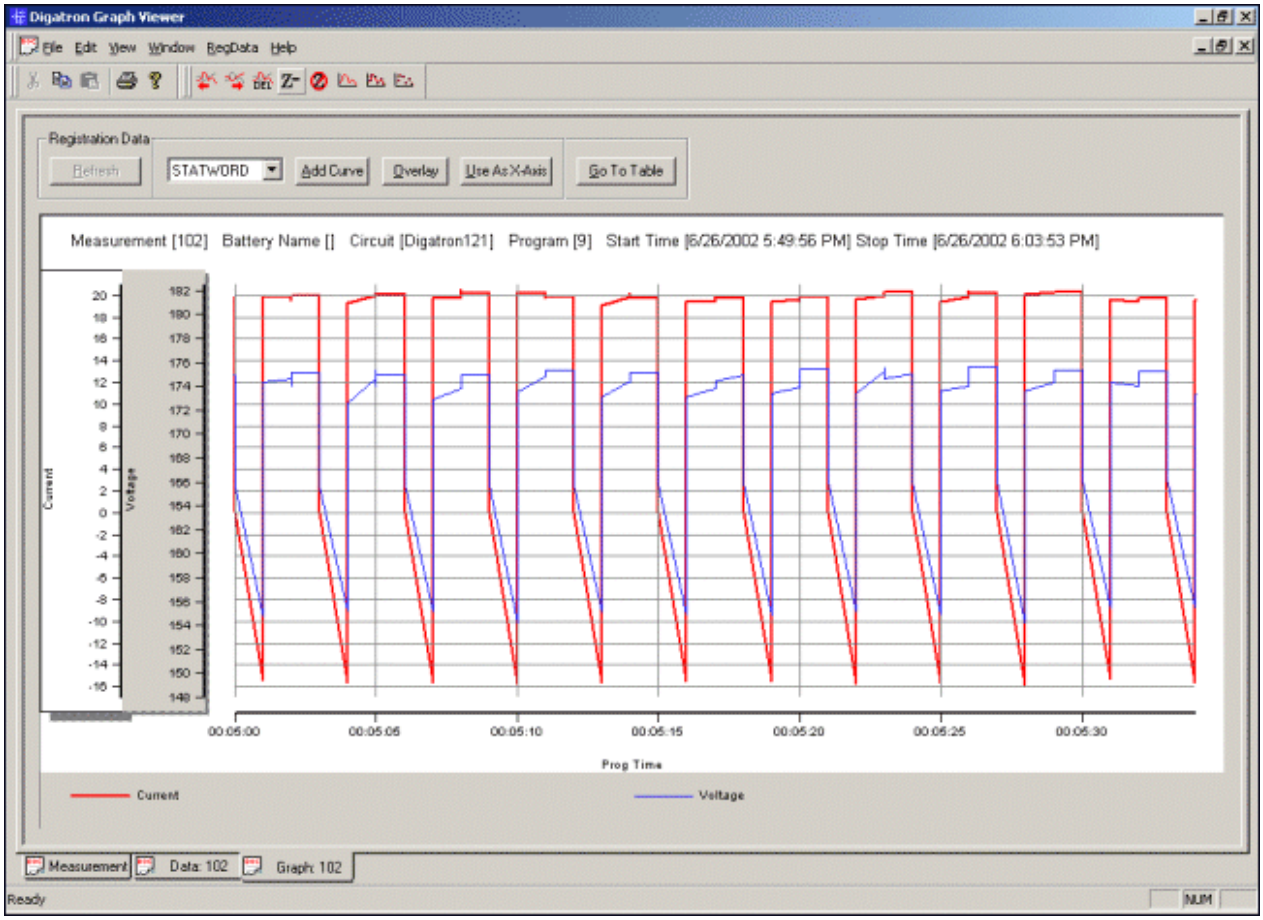


The zoomed graphic points out the part where the batteries are pulsed and the **SOC detection** takes effect.

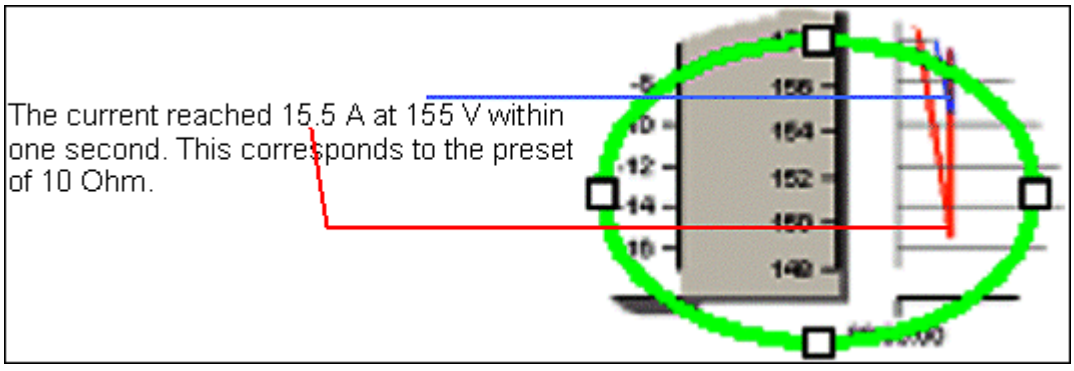
The blue colored **STATWORD** signal indicates the timer operation. The timer had to be started three times before the dIa condition was stable for 60 seconds to terminate the cycle.



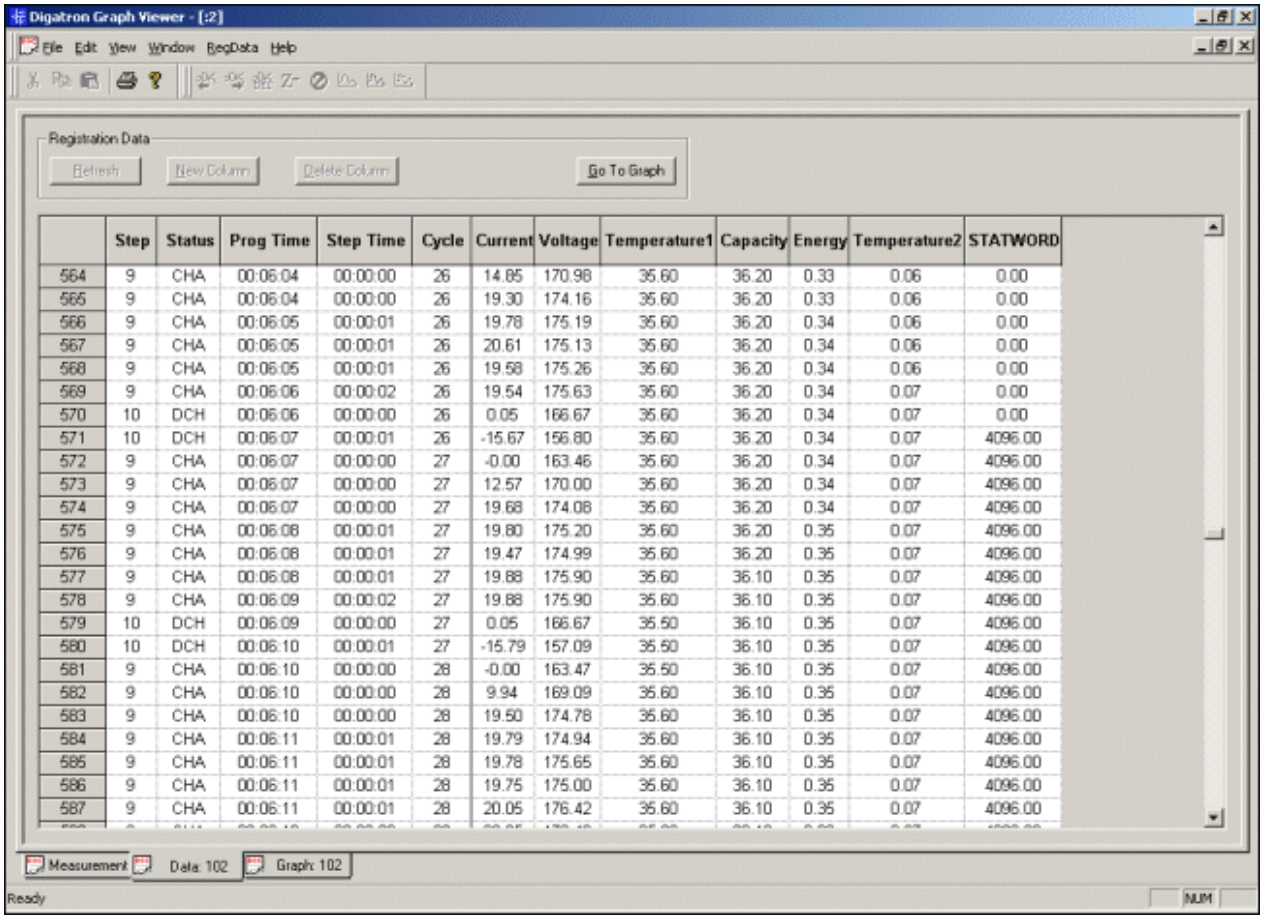
Graph, showing current and voltage (multiple zoom)



These graphs confirm that the MCDR unit can establish a one second constant resistance discharge pulse. The discharge current is calculated and set to keep a constant discharge at 10 Ohm resistance.



Numeric data display



	Step	Status	Prog Time	Step Time	Cycle	Current	Voltage	Temperature1	Capacity	Energy	Temperature2	STATWORD
564	9	CHA	00:06:04	00:00:00	26	14.85	170.98	35.60	36.20	0.33	0.06	0.00
565	9	CHA	00:06:04	00:00:00	26	19.30	174.16	35.60	36.20	0.33	0.06	0.00
566	9	CHA	00:06:05	00:00:01	26	19.78	175.19	35.60	36.20	0.34	0.06	0.00
567	9	CHA	00:06:05	00:00:01	26	20.61	175.13	35.60	36.20	0.34	0.06	0.00
568	9	CHA	00:06:05	00:00:01	26	19.58	175.26	35.60	36.20	0.34	0.06	0.00
569	9	CHA	00:06:06	00:00:02	26	19.54	175.63	35.60	36.20	0.34	0.07	0.00
570	10	DCH	00:06:06	00:00:00	26	0.05	166.67	35.60	36.20	0.34	0.07	0.00
571	10	DCH	00:06:07	00:00:01	26	-15.67	156.80	35.60	36.20	0.34	0.07	4096.00
572	9	CHA	00:06:07	00:00:00	27	-0.00	163.46	35.60	36.20	0.34	0.07	4096.00
573	9	CHA	00:06:07	00:00:00	27	12.57	170.00	35.60	36.20	0.34	0.07	4096.00
574	9	CHA	00:06:07	00:00:00	27	19.68	174.08	35.60	36.20	0.34	0.07	4096.00
575	9	CHA	00:06:08	00:00:01	27	19.80	175.20	35.60	36.20	0.35	0.07	4096.00
576	9	CHA	00:06:08	00:00:01	27	19.47	174.99	35.60	36.20	0.35	0.07	4096.00
577	9	CHA	00:06:08	00:00:01	27	19.88	175.90	35.60	36.10	0.35	0.07	4096.00
578	9	CHA	00:06:09	00:00:02	27	19.88	175.90	35.60	36.10	0.35	0.07	4096.00
579	10	DCH	00:06:09	00:00:00	27	0.05	166.67	35.50	36.10	0.35	0.07	4096.00
580	10	DCH	00:06:10	00:00:01	27	-15.79	157.09	35.50	36.10	0.35	0.07	4096.00
581	9	CHA	00:06:10	00:00:00	28	-0.00	163.47	35.50	36.10	0.35	0.07	4096.00
582	9	CHA	00:06:10	00:00:00	28	9.94	169.09	35.60	36.10	0.35	0.07	4096.00
583	9	CHA	00:06:10	00:00:00	28	19.50	174.78	35.60	36.10	0.35	0.07	4096.00
584	9	CHA	00:06:11	00:00:01	28	19.79	174.94	35.60	36.10	0.35	0.07	4096.00
585	9	CHA	00:06:11	00:00:01	28	19.78	175.65	35.60	36.10	0.35	0.07	4096.00
586	9	CHA	00:06:11	00:00:01	28	19.75	175.00	35.60	36.10	0.35	0.07	4096.00
587	9	CHA	00:06:11	00:00:01	28	20.05	176.42	35.60	36.10	0.35	0.07	4096.00

The numeric shows the registration data of the formation. The **STATWORD** channel is used to indicate several situations during the program. The value **4096** indicates that the **dIa** value is reached and the timer is started. 60 seconds later the **STATWORD** will change to 0 and the cycle will be terminated. If the time is not reached but the **STATWORD** indicates **12288** the termination was done by **dIb** condition.

2 SOC by delta U

2.1 Theory of the "SOC delta U detection" algorithm

SOC algorithm should be started when set voltage value (Ux) on battery terminals in pauses between charging impulses is achieved.

Counting of **delta T (dUsec)** time starts only if requirement **0 ≤ ΔU ≤ ΔUx** is fulfilled. If this is not true during set period of time **delta T** (i.e. density of electrolyte in batteries is not stabilized), the count of time stops and time value is to be zeroed. Time counting re-starts when requirement **0 ≤ ΔU ≤ ΔUx** is fulfilled anew.

SOC algorithm should be stopped and the formation program should jump to the following step when time value delta T is achieved, or the requirement **U < 0** is fulfilled that indicates commencement of batteries re-charging

2.2 The "SOC delta U" in Battery Manager

Two additional parameters are necessary to set up the SOC detection. The parameters are called: **dUsec**, **dU**.

In detail:

Parameter	Description
dUsec	**d**elta **U** minimum time in **sec**onds
dU	**d**elta **U** of voltage between first value of step and last value of step

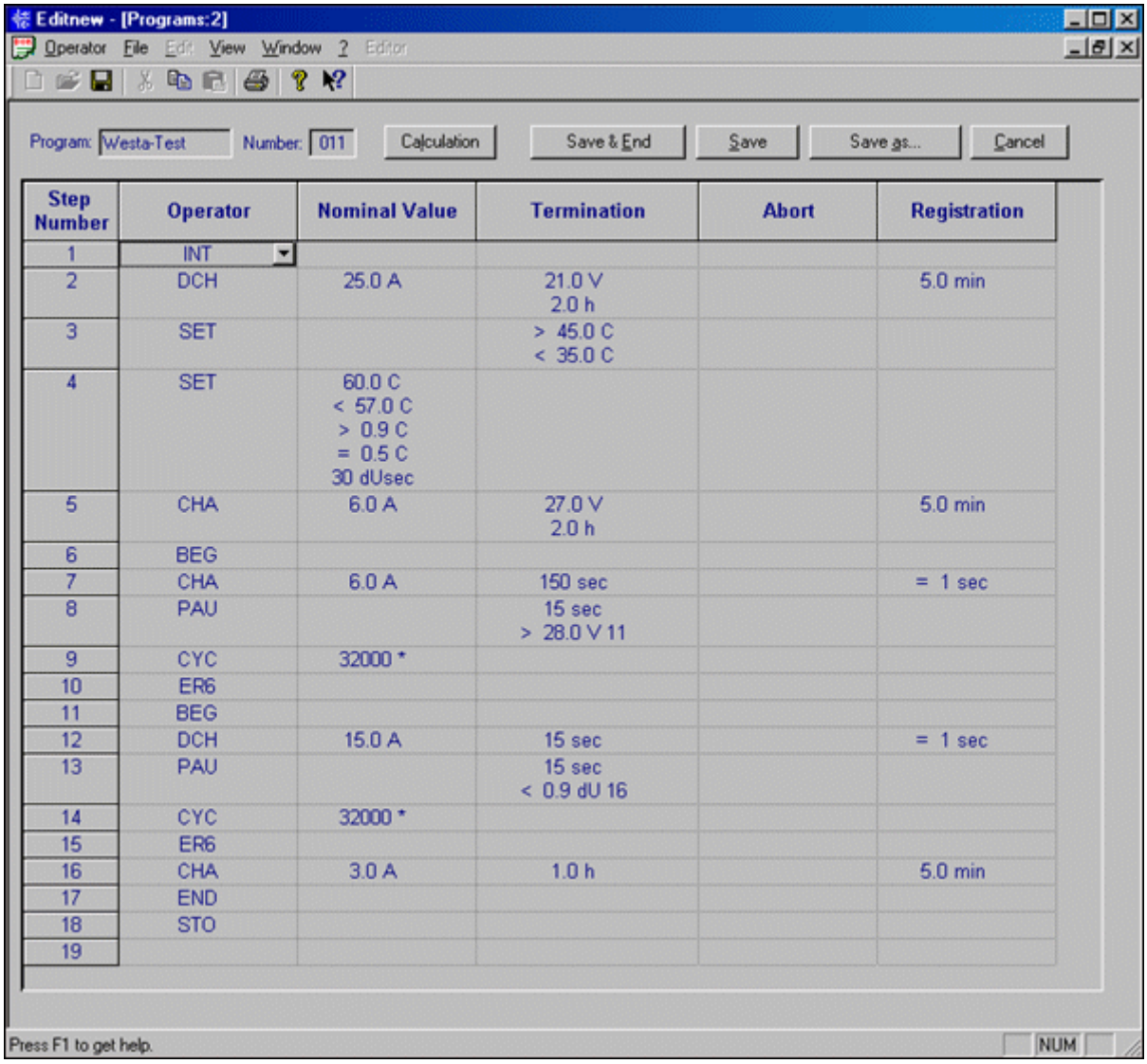
The parameter **dUsec** defines the time in seconds where the ΔU of a step must be less than a preset value (ΔUx).

The parameters **dUsec** and **dU** are necessary to run the **SOC detection**.

The following example shows how to implement pulse charge/discharge steps in a formation schedule including the **SOC detection**. Please note that the formation schedule does not reflect a real battery formation with respect to the programmed time. But it yields sufficient information to understand the **SOC detection**.

The following test is made at one circuit of the MCDR 30-360-12 with 2 batteries connected.

2.3 Pulse formation schedule with "SOC deltaU"



Step Number	Operator	Nominal Value	Termination	Abort	Registration
1	INT				
2	DCH	25.0 A	21.0 V 2.0 h		5.0 min
3	SET		> 45.0 C < 35.0 C		
4	SET	60.0 C < 57.0 C > 0.9 C = 0.5 C 30 dUsec			
5	CHA	6.0 A	27.0 V 2.0 h		5.0 min
6	BEG				
7	CHA	6.0 A	150 sec		= 1 sec
8	PAU		15 sec > 28.0 V 11		
9	CYC	32000 *			
10	ER6				
11	BEG				
12	DCH	15.0 A	15 sec		= 1 sec
13	PAU		15 sec < 0.9 dU 16		
14	CYC	32000 *			
15	ER6				
16	CHA	3.0 A	1.0 h		5.0 min
17	END				
18	STO				
19					

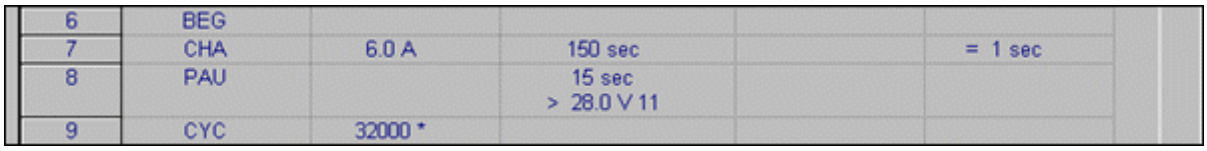
2.4 Explanation of program steps

Step 4



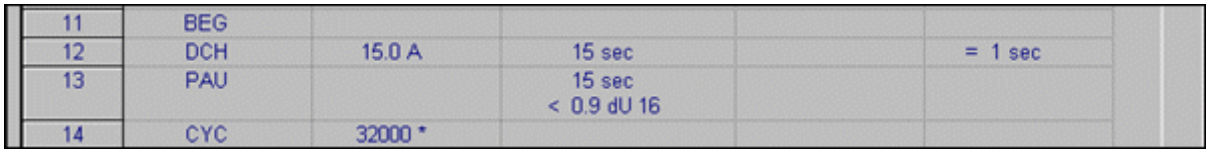
Minimum time for delta U detection is set to 30 seconds.

Steps 6 to 9



Pulse charge 150 seconds with 15 seconds pause. If voltage reaches 28.0 V during pause step the cycle is terminated and the program continues with step 11.

Steps 11 to 14



Pulse discharge 15 seconds with 15 seconds pause. The cycle will be terminated by **SOC detection**.

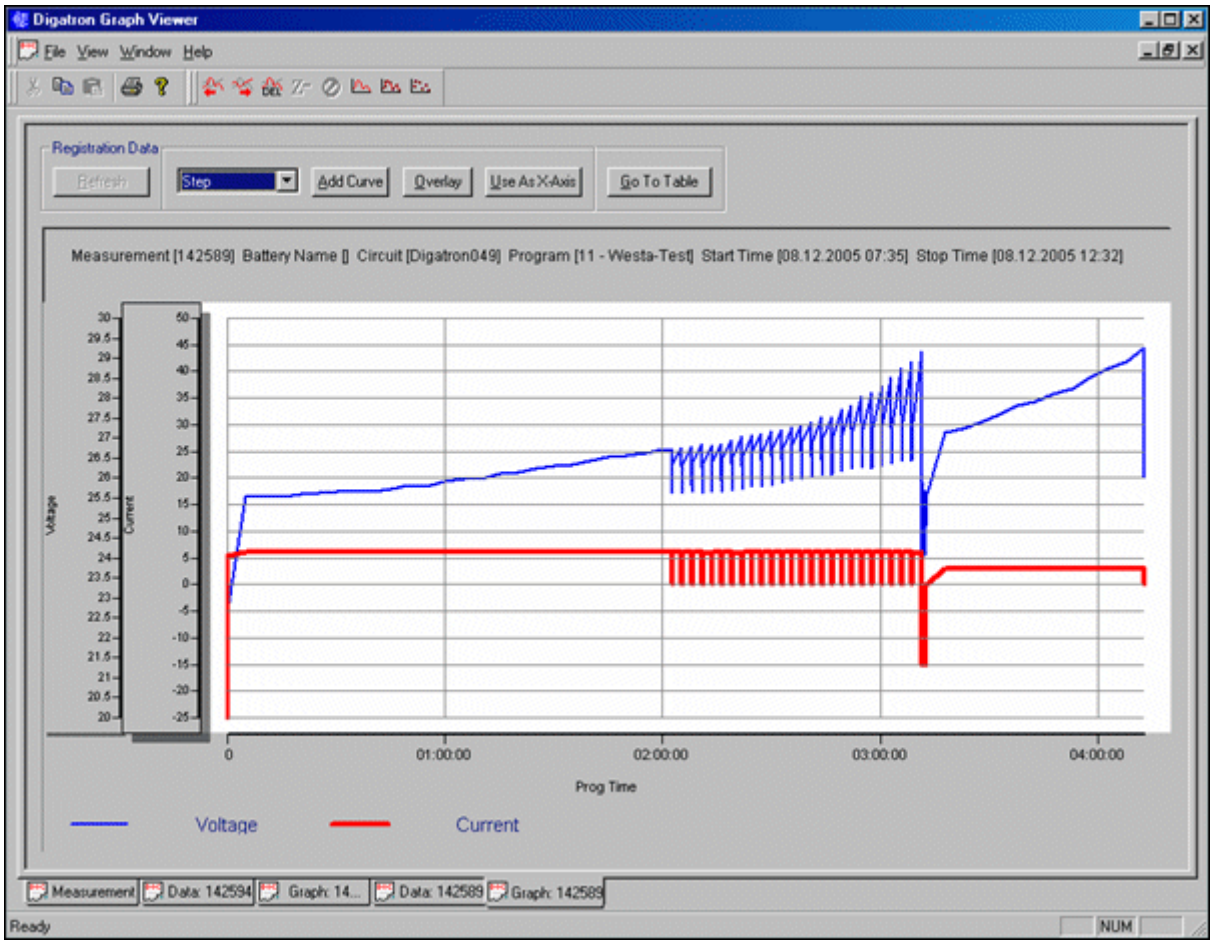
Delta U (dU) is calculated with last step voltage minus first step voltage. If this value is less than 0.9 and this condition was due for 30 seconds (30 dUsec see step 4) the cycle will be terminated and program will continue with step 16.

If the calculated delta U (dU) is less than 0.0 Volts the cycle will be terminated immediately and the program will continue with step 16.

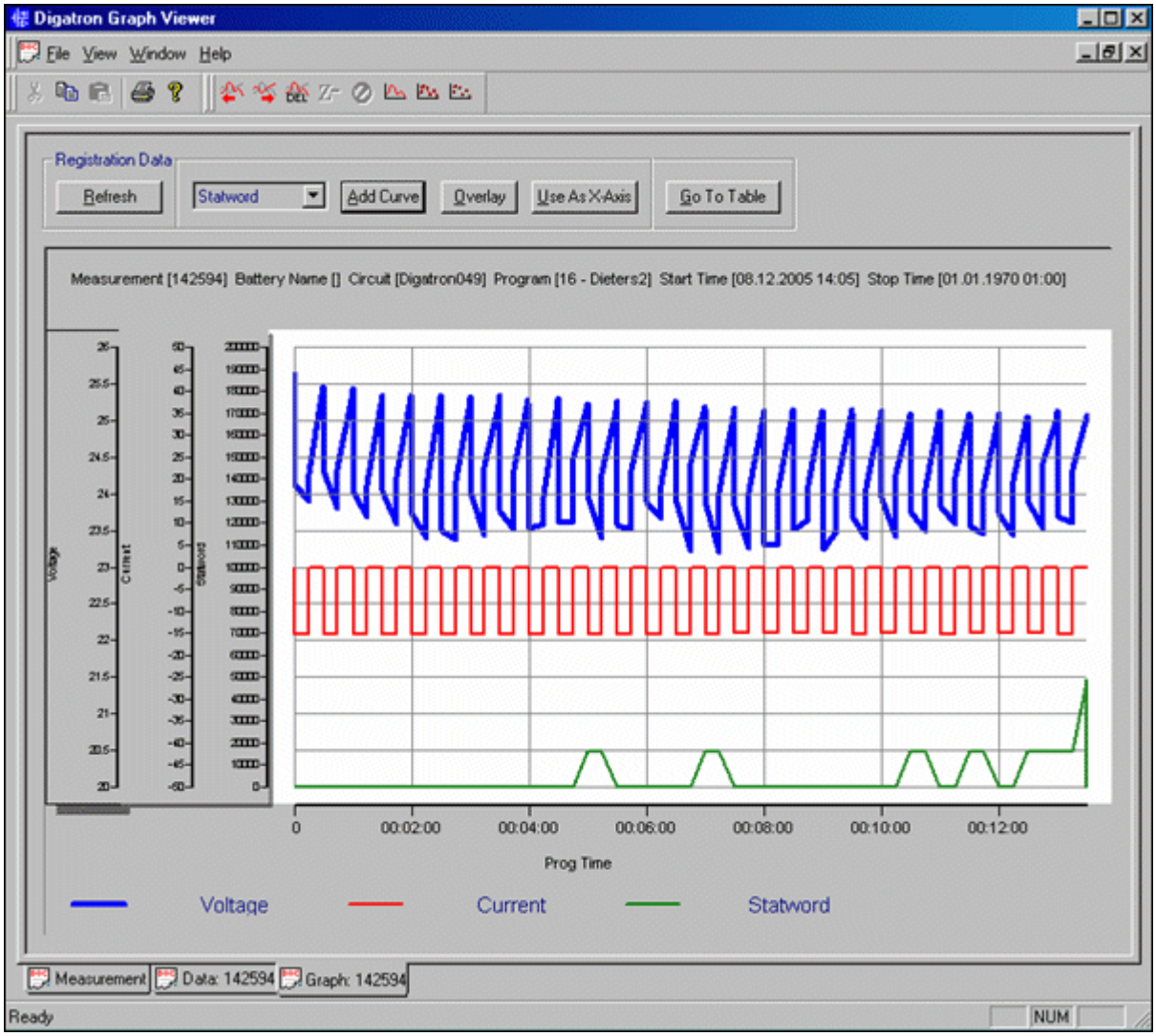
2.5 Results

Graph, showing current and voltage

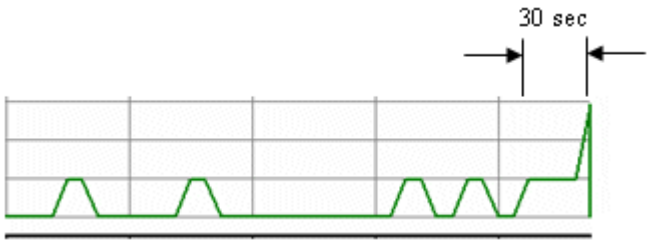
The time duration of the SOC is very small because of the used batteries it wasn't possible to get a better result. Important is the functionality of the algorithm.



Graph, showing current and voltage with timer (zoomed)

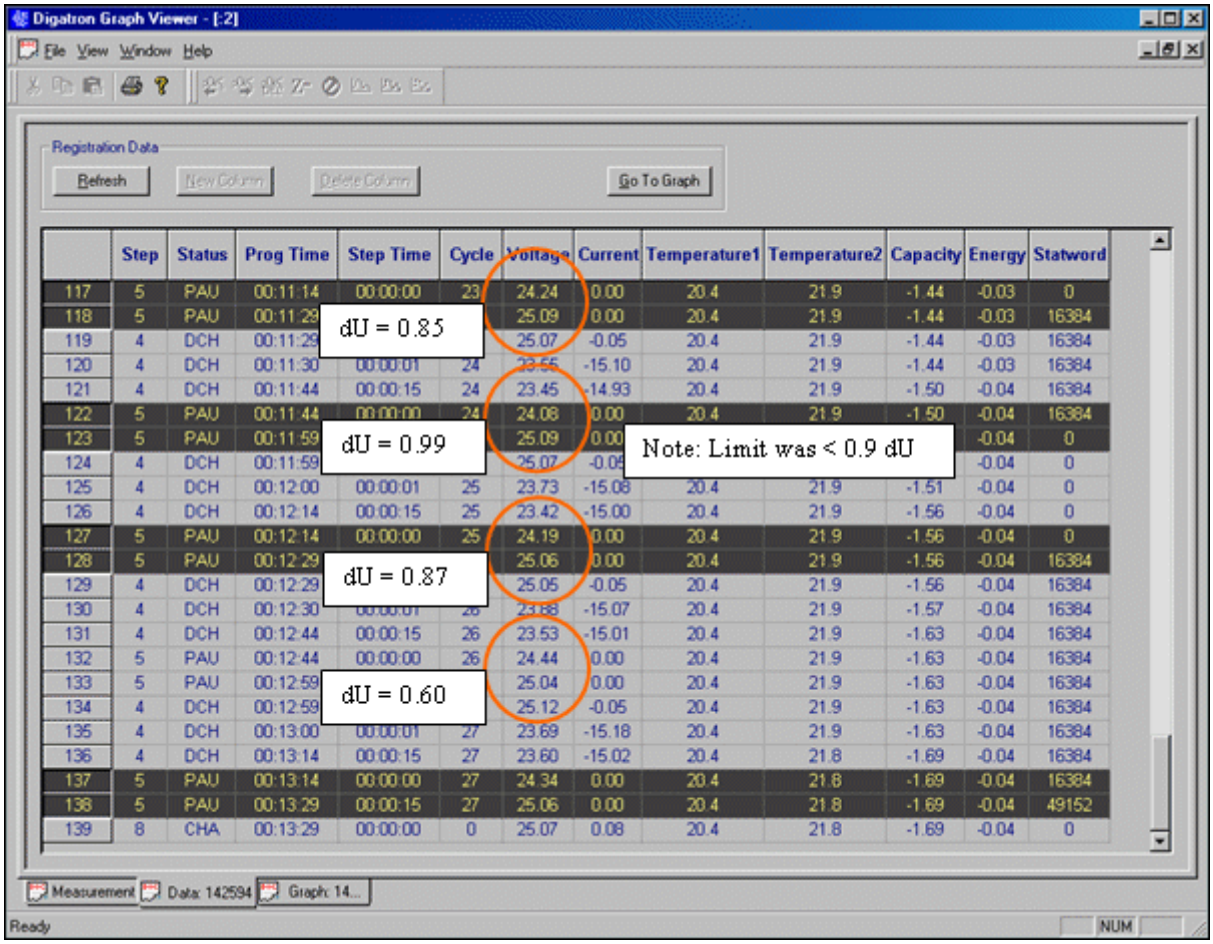


The zoomed graphic points out the part where the batteries are pulsed and the **SOC detection** takes effect. (Steps 11 to 14 with 15 A discharge.)



The **STATWORD** signal indicates the delta T (dUsec) timer operation. The timer had to be started five times before the **dU** condition was stable for 30 seconds to terminate the cycle.

Numeric data display



	Step	Status	Prog Time	Step Time	Cycle	Voltage	Current	Temperature1	Temperature2	Capacity	Energy	Statword
117	5	PAU	00:11:14	00:00:00	23	24.24	0.00	20.4	21.9	-1.44	-0.03	0
118	5	PAU	00:11:29			25.09	0.00	20.4	21.9	-1.44	-0.03	16384
119	4	DCH	00:11:29			25.07	-0.05	20.4	21.9	-1.44	-0.03	16384
120	4	DCH	00:11:30				-15.10	20.4	21.9	-1.44	-0.03	16384
121	4	DCH	00:11:44	00:00:15	24	23.45	-14.93	20.4	21.9	-1.50	-0.04	16384
122	5	PAU	00:11:44			24.08	0.00	20.4	21.9	-1.50	-0.04	16384
123	5	PAU	00:11:59			25.09	0.00				-0.04	0
124	4	DCH	00:11:59			25.07	-0.05				-0.04	0
125	4	DCH	00:12:00	00:00:01	25	23.73	-15.08	20.4	21.9	-1.51	-0.04	0
126	4	DCH	00:12:14	00:00:15	25	23.42	-15.00	20.4	21.9	-1.56	-0.04	0
127	5	PAU	00:12:14	00:00:00	25	24.19	0.00	20.4	21.9	-1.56	-0.04	0
128	5	PAU	00:12:29			25.06	0.00	20.4	21.9	-1.56	-0.04	16384
129	4	DCH	00:12:29			25.05	-0.05	20.4	21.9	-1.56	-0.04	16384
130	4	DCH	00:12:30				-15.07	20.4	21.9	-1.57	-0.04	16384
131	4	DCH	00:12:44	00:00:15	26	23.53	-15.01	20.4	21.9	-1.63	-0.04	16384
132	5	PAU	00:12:44	00:00:00	26	24.44	0.00	20.4	21.9	-1.63	-0.04	16384
133	5	PAU	00:12:59			25.04	0.00	20.4	21.9	-1.63	-0.04	16384
134	4	DCH	00:12:59			25.12	-0.05	20.4	21.9	-1.63	-0.04	16384
135	4	DCH	00:13:00				-15.18	20.4	21.9	-1.63	-0.04	16384
136	4	DCH	00:13:14	00:00:15	27	23.60	-15.02	20.4	21.8	-1.69	-0.04	16384
137	5	PAU	00:13:14	00:00:00	27	24.34	0.00	20.4	21.8	-1.69	-0.04	16384
138	5	PAU	00:13:29	00:00:15	27	25.06	0.00	20.4	21.8	-1.69	-0.04	49152
139	8	CHA	00:13:29	00:00:00	0	25.07	0.08	20.4	21.8	-1.69	-0.04	0

The numeric shows the registration data of the formation. The **STATWORD** channel is used to indicate several situations during the program. The value **16384** indicates that the **dU** value was reached and the timer was started.

The last timer difference is higher than 30 seconds because the deltaU condition is checked at the end of pause step only.

3 Table of STATWORD channel

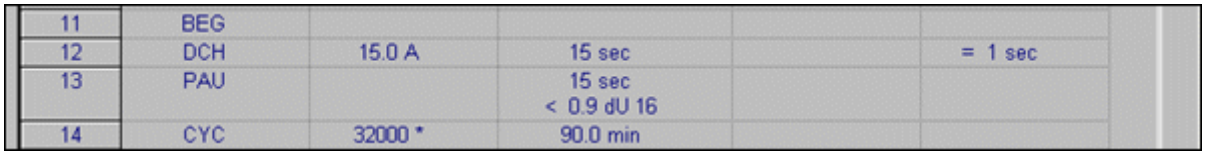
Value	Description
1	valve for water bath is switched on
2	temperature difference between the two sensors reached limit
4	during start, the program analysis went wrong, programming not allowed
8	error at communication to the water bath box
256	program was interrupted by user from PC
512	program was interrupted by user from water bath box
4096	timer for **SOC detection** started, **dIa** value reached
12288	**dIb** limit reached during **dIa**
16384	timer for **SOC delta U detection** started, **dU** value reached
49152	**dU** limit reached, **dU** permanent fulfilled during **dUsec**

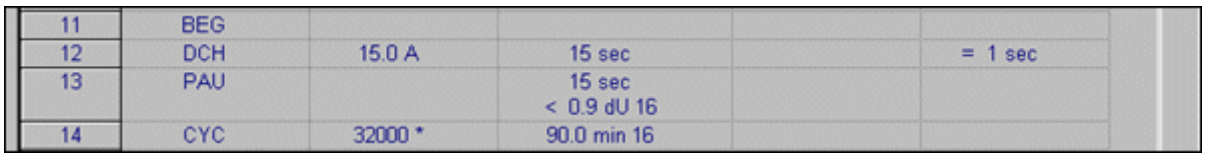
4 Additional Features

Because of the program complexity it could be useful to terminate a cycle by time or to set an error point inside the program which is not reached within the normal operation.

A time limit could be set within a cycle line with or without a jump parameter.

Examples:





Setting an error point inside the program with ER6:



This step is reached if the cycle is terminated by counts which should not happen during normal operation.

Option: Load Management (LOM)

Usage

The Load Management (LOM) allows the reduction of the programmed current by a specified percentage to reduce the power consumption in a formation during peak load times. If the actual power consumption of the plant is close to a defined limit, LOM can be activated to reduce the power consumption of the formation rectifiers.

The current reduction can be applied for all program steps with the operators **CHA**, **RCH** and **DCH**. Just add a program line with the operator **LOM**. After inserting a program step with the **LOM** operator, the **LOM** factor entered in the belonging INI file (see below) will be valid for the nominal current values of **all subsequent** program steps until the **LOM** operator is entered again into the program. Thus, it is possible to include single program steps into the current reduction or to exclude them. If, for instance, the current reduction shall only apply for program steps with the operator **CHA = "Charge"**, you have to set **LOM** operators **before and after** these program steps.

Example:

Step Number	Operator	Nominal Value	Termination	Abort	Registration
1	LOM				
2	CHA	1000 A	2 h	250 V	
3	LOM				
4	CHA	100 A	15 min	250 V	
5	STO				

Fig. 1: LOM operator to limit the current for a single program step

Alarm Server

The **AlarmServer.exe** needs to be running to make use of the LOM operator. The start command for the Alarm Server must be as follows:

AlarmServer.exe /i=IniFileName

The INI file is thereby present in the same directory as the **AlarmServer.exe.** If its name is e.g. **AlarmServer.ini,** the **Target** in the **Properties** of the shortcut to **AlarmServer.exe** must be defined as

[Path\]AlarmServer.exe /i=AlarmServer.ini

Alarm Server GUI

The interface of the alarm server shows the current state of the process and gives some hints on error situations.

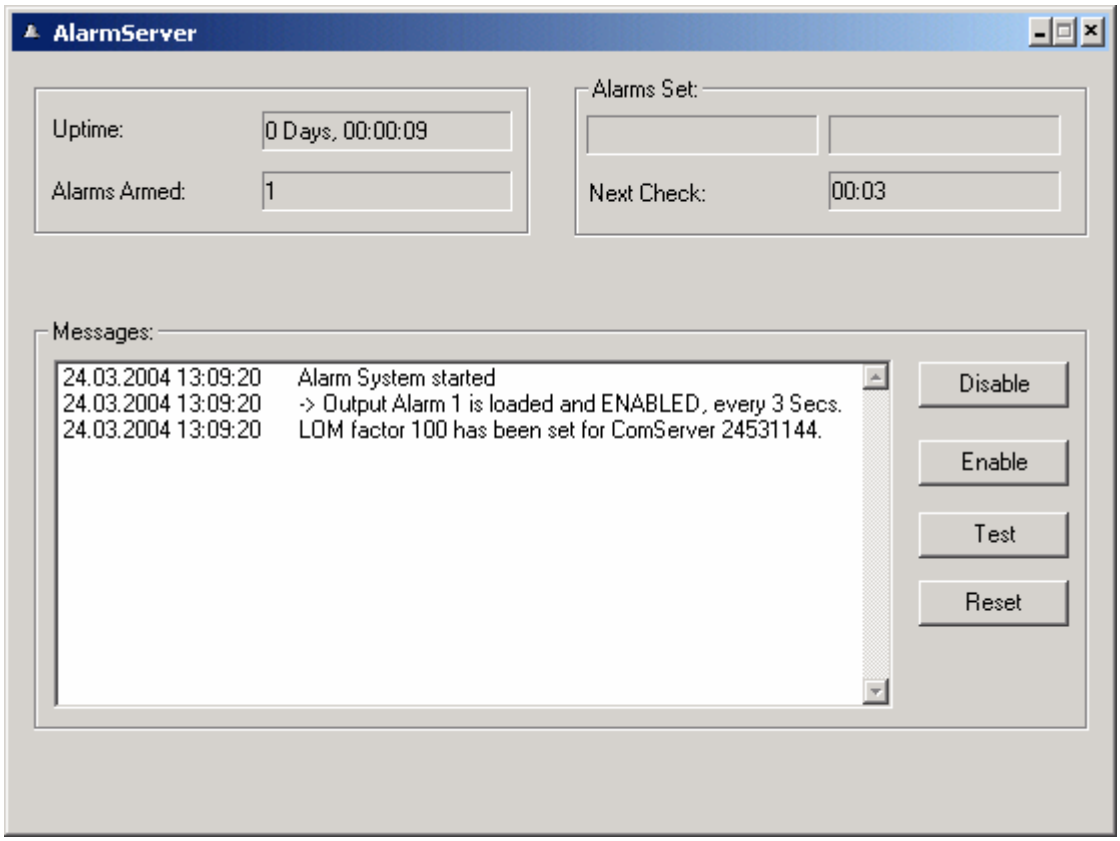


Fig. 2: Alarm Server Interface

The buttons allow the following actions:

Disable
Select a line and click the **Disable** button to terminate the alarm.

Enable
Select a line and click the **Enable** button to activate the alarm.

Test
Select a line and click the **Test** button. The corresponding output is set.

Reset
With the **Reset** button the output is cleared.

Contents of the INI file

The INI file that you define as part of the shortcut to **AlarmServer.exe** allows mainly the setting of the LOM factor, e.g. the reduction of the current. There are some other settings that have to agree with your system:

```
[Global]
Total_Channels=8            ; total number of LOM channels to set
ShowMode=1                  ; 1 = show GUI, 0 = hide GUI
Port=4799                   ; port number where the BM3Service can connect

[Channel 1]                 ; definitions for Channel 1
QueryType=LOM               ; kind of check (Service,Server,Input,LOM)
QueryDesc=ServiceName       ; description (name) of the type, e.g. Lom123
Enabled=1                   ; 1=channel is enabled, 0=channel is disabled
Period_in_Seconds=10        ; check every n sec
True_is_Energize=1          ; should always be 1

[ServiceName]
Name=BM3 Service      ; name of service
InLine=0              ; input line to check
State=1               ; value for which the LOM FactorOn value is used (1 or 0)
FactorOn=77           ; LOM factor (reduced current value in %)
FactorOff=100         ; must not be changed!
```

Example:

```
[System]
Language=English

[Global]
Total_Channels=1
Port=4799

[Channel 1]
QueryType=LOM
QueryDesc=Lom1
Enabled=1
Period_in_Seconds=3
True_is_Energize=1

[Lom1]
Name=LOM1
InLine=0
State=1
FactorOn=77
FactorOff=100
Server=localhost:4712
```

this value is assigned here

Adjust the INI file to your circumstances. To change the LOM factor, modify the **`FactorOn`** value. To reduce the current with LOM e.g. by 20%, enter **`FactorOn=80`**.